HyperbolicMomentum: Curvature-Adaptive Momentum for Hierarchical Neural Optimization

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Abstract

We introduce *HyperbolicMomentum*, a Lorentz-model optimizer that combines grouped learnable curvature, curvature-normalized geodesic steps, and consistent momentum transport across tangent spaces and curvature changes. We derive an exact curvature-independent sensitivity certificate for normalized steps, sharp normalization and heterogeneity constants with equality cases, and conditional convergence, full-state stability, and generalization bounds. In complete 2000-update synthetic experiments, matched five-initialization searches selected by validation reduce held-out distortion by 2.94% in a stationary hierarchy and 3.31% and 5.02% under $2\times$ and $8\times$ smooth distance drift, relative to selected normalized fixed-curvature momentum. These are selected positive conditions from a six-case matrix with mixed outcomes, and the radius constraint permits curvature-dependent physical capacity. The omitted-transport ablation remains close, so training scores do not isolate a transport benefit. A separate operator experiment does: compatible transport keeps the maximum FP32 canonical momentum error below 3.2×10^{-6}, whereas point-only transport yields final errors of 1.70%–4.94% under prescribed drift. Historical knowledge-graph results supply exploratory curvature evidence. All evaluated conditions, including negative and tied results, remain available in the complete comparison appendix, and the 705 retained full-length configurations and all delayed test evaluations are complete.

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1. Introduction

Hierarchical structure is one of the recurring reasons that Euclidean representation learning becomes inefficient. Taxonomies branch exponentially with depth, lexical relations mix tree-like and symmetric components, knowledge graphs combine multi-relational hierarchies with many-to-many relations, and graph or language representations often need to place coarse categories, fine instances, and lateral neighborhoods in the same embedding system. Hyperbolic representation learning addresses this mismatch by using negative curvature, whose volume growth matches trees more naturally than Euclidean space (Nickel & Kiela, 2017; 2018; Chami et al., 2019; Tifrea et al., 2019; Balazevic et al., 2019). In examples such as WordNet taxonomies, Freebase relation graphs, and synthetic trees with heterogeneous branching, the relevant question is not only whether hyperbolic space helps, but which curvature each structural component should use during optimization.

Table 1 summarizes the coupled update. A useful distinction is between fixed-manifold optimization, architectural curvature learning, and consistency of optimizer state when curvature changes. Recent curvature-aware Lorentz optimization already addresses the last issue using origin-tangent remapping (Bdeir et al., 2025). Our study concerns the particular homothetic state map, grouped slow curvature updates, normalized geodesic step, and their conditional analysis.

A fixed-curvature optimizer cannot adapt its manifold to different structural groups. Riemannian stochastic gradient descent provides a principled first-order update on manifolds (Bonnabel, 2013; Zhang & Sra, 2016), and Riemannian adaptive methods extend adaptive optimization to curved spaces (Bécigneul & Ganea, 2019). Yet a fixed-curvature optimizer cannot adapt when one relation behaves like a taxonomic branch, another relation behaves like a symmetric neighborhood, and a third relation behaves like a high-degree multi-relational association. Knowledge graph baselines such as TransE and RotatE illustrate how strongly relation type shapes the geometry of a representation problem (Bordes et al., 2013; Sun et al., 2019); hyperbolic KG methods further show that relation-aware curvature can matter (Balazevic et al., 2019). A second requirement is geometric: momentum vectors live in tangent spaces. The vector

Table 1. **HyperbolicMomentum update.** Each group has a dedicated Lorentz manifold. The main path updates the point and tangent momentum; the slower branch changes curvature and maps both states. The curvature derivative is total at fixed canonical point, as in Eq. (123).

Stage	State or operation	Geometric role
Tangent-state and geodesic update		
Gradient	$\widehat{g}_{g,t} = \mathrm{grad}_{x_g} \mathcal{J}$	Form a tangent direction at the current point.
Momentum	$\widetilde{m}_{g,t} = \beta m_{g,t} + (1-\beta)\widehat{g}_{g,t}$	Average directions in the same tangent space.
Normalized step	$\eta_{g,t}^{\mathrm{geo}} = \eta_t/(1+\gamma\sqrt{K_{g,t}}\|\widetilde{m}_{g,t}\|)$	Control the curvature–momentum scale.
Geodesic move	$x_{g,t+1/2} = \mathrm{Exp}_{x_{g,t}}^{K_{g,t}}(-\eta_{g,t}^{\mathrm{geo}}\widetilde{m}_{g,t})$	Move the point on its current manifold.
Tangent transport	$m_{g,t+1/2} = \mathcal{P}_{x_{g,t}\to x_{g,t+1/2}}^{K_{g,t}}\widetilde{m}_{g,t}$	Carry memory to the new base point.
Slow curvature update and compatible joint-state map		
Learn curvature	$\widehat{h}_{g,t} = \partial_{c_g} F(q_t, u_t)$; Eq. (8)	Use the total derivative, with bounded log drift in implementation.
Map joint state	$a_g = \sqrt{K_{g,t}/K_{g,t+1}}$; Eq. (9)	Scale both point and momentum by a_g; restore tangency numerically.

carried from step $t-1$ belongs to $T_{x_{t-1}}\mathbb{H}^d_{-K}$, whereas the current gradient belongs to $T_{x_t}\mathbb{H}^d_{-K}$, so adding them without parallel transport has no intrinsic meaning. If K itself changes, then both the base point and the metric radius change, making a point-only curvature update incomplete for architectures in which each group has its own manifold.

This paper studies the optimization problem that appears when these two requirements coincide. We ask whether a hierarchical model should learn **one curvature per structural group**, whether the geodesic step should be normalized by the local curvature-momentum scale, and when a **cross-curvature momentum state** is actually meaningful. The answer is deliberately nuanced. Historical unfiltered, tail-only experiments with Euclidean Adam provide exploratory evidence for grouped learnable curvature: on WN18RR, B4 improves Hits@10 by $+4.1\%$ and loss by -2.8% over fixed curvature B3, while MRR is positive but sensitive to the moving-average window. On FB15k-237, the same mechanism gives a different metric profile: MRR improves by $+2.2\%$ and Hits@1 by $+3.1\%$, but Hits@10 is almost unchanged at $+0.1\%$. Thus the shared positive KG signal is MRR, while which Hits metric benefits depends on the dataset.

We present **HyperbolicMomentum**, an optimizer in the Lorentz model with three mechanisms: grouped learnable curvature, curvature-normalized geodesic steps, and cross-curvature parallel-transport momentum. The method is motivated by one controlling geometric quantity. The product $\sqrt{K}R$ governs capacity, conditioning, and stability over radius R, while $\sqrt{K}\eta\|m\|$ governs exponential-map distortion during an optimization step. Our theory makes this trade-off explicit through exact homothety and Jacobi-field identities, a quantitative heterogeneity bound, error-explicit two-block convergence, momentum tracking, constrained curvature stationarity, finite-precision propagation, and augmented-state stability with an explicit training-time factor. Table 2 lists the relation-dependent curvature states: WN18RR learns a wide relation-dependent curvature spread, whereas FB15k-237 learns a much smaller spread consistent with weaker hierarchy.

The contributions are as follows, stated in prose to match the empirical evidence. First, we formulate HyperbolicMomentum as a single Lorentz-model algorithm that learns grouped curvatures and uses curvature-normalized geodesic steps, with cross-curvature momentum transport included only when the architecture supplies a dedicated manifold per group. Second, we derive an exact curvature-independent sensitivity certificate for normalized geodesic steps, optimal normalization derivative constants, and a sharp Hessian-weighted heterogeneity bound. The convergence proof preserves finite-horizon martingale weights, while the statistical analysis combines the full state-comparison matrix with stability concentration and entropy chaining. Third, we separate historical curvature ablations from new full-data comparisons and 2000-update controlled drift experiments. The complete matched-initialization search yields held-out distortion reductions of 2.94%, 3.31%, and 5.02% relative to selected normalized fixed curvature in the stationary, $2\times$ smooth-drift, and $8\times$ smooth-drift conditions. These are the stronger conditions in a six-case matrix; the point-only transport ablation remains close. These conditional gains combine scale adaptation with the physical-distance budget under a canonical-radius cap; they do not imply a uniform advantage across all conditions. Direct operator measurements separately verify the sensitivity and momentum-consistency predictions. Fourth, we document structural curvature differentiation: WN18RR differentiates across relations, FB15k-237 by relation complexity with a four-times smaller K spread, and synthetic trees by branching rate.

2. Related Work

Hyperbolic representations and learned curvature. Poincaré embeddings use negative curvature for compact

Table 2. **Learned relation curvatures.** Historical B4 values. All 11 WN18RR relations and the 15 displayed FB15k-237 relations are retained; the fixed baseline is $K = 1$. Curvature magnitude is a parameter, not a performance ranking.

WN18RR relation	K	FB15k-237 relation	K
derivationally related form	2.0000	award/nominee	1.8371
member meronym	1.5823	film/release region	1.3486
hypernym	1.4173	award/winner (A)	1.3449
has part	1.3050	award/nominee→award	1.2725
instance hypernym	1.2220	music/genre→artists	1.1832
verb group	1.0938	person/profession	1.1817
also see	1.0888	film/genre	1.1754
synset domain topic of	1.0754	award/nominees for	1.1678
member of domain region	1.0170	actor/filmography	1.1677
member of domain usage	1.0092	award/winners (B)	1.1435
similar to	1.0084	film/other crew	1.1421
–	–	award/nominated for	1.1338
–	–	person/nationality	1.1093
–	–	work/awards won	1.1034
–	–	sports/team roster	1.0966

hierarchical representations, while Lorentz embeddings provide an alternative model with closed-form geodesic operations (Nickel & Kiela, 2017; 2018). Learning curvature is already established: product-space embeddings jointly learn geometry across factors (Gu et al., 2019), and HGCN learns a curvature at each graph-convolution layer (Chami et al., 2019). For knowledge graphs, MuRP provides a hyperbolic relational model (Balazevic et al., 2019), while RotH and AttH learn relation-specific curvatures together with isometric transformations (Chami et al., 2020). Poincaré GloVe extends hyperbolic representation learning to lexical co-occurrence (Tifrea et al., 2019). Grouped or relation-specific curvature is therefore a precedent, rather than a novelty claim of this paper.

Riemannian optimization and optimizer memory. RSGD establishes stochastic optimization directly on a manifold (Bonnabel, 2013); curvature-dependent first-order analysis characterizes geodesically convex objectives (Zhang & Sra, 2016). Riemannian Adam and AMSGrad maintain adaptive statistics on a product manifold and transport first-moment information between tangent spaces (Bécigneul & Ganea, 2019). More recent adaptive frameworks analyze minibatch size and constant or diminishing steps on embedded manifolds (Sakai & Iiduka, 2024). These are important fixed-manifold foundations. Our analysis additionally tracks a learned curvature coordinate, including the gradient induced by identifying points through a curvature-dependent homothety. It retains the different representation and curvature step weights instead of assigning both blocks the same rate by default.

Curvature-aware state mapping: the closest optimization precedent. Bdeir et al. (2025) already identify mismatches caused by updating curvature while retaining stale manifold parameters and optimizer states. Their curvature-aware scheme updates parameters and transports gradients and momentum through the origin tangent space; it is combined with Riemannian AdamW and radius rescaling. We therefore do not claim to introduce cross-curvature optimizer-state consistency. HyperbolicMomentum instead studies a homothetic point-and-momentum map, grouped slow curvature updates, and a curvature–momentum normalized geodesic step. The distinction is the chosen coupled update and its error-explicit analysis; the homothety itself also has precedents discussed by that work. Its behavior relative to origin-tangent remapping requires a matched empirical comparison, which the current experiments do not supply.

The map being compared. The origin-tangent point map has the form $\mathrm{Exp}^{K'}_{o_{K'}} \circ \mathrm{Log}^{K}_{o_K}$ and retains the physical radial coordinate. Our map is $\Psi_{K\to K'}(x) = \sqrt{K/K'}\,x$, with differential $D\Psi(v) = \sqrt{K/K'}\,v$; it retains the canonical state $q = \sqrt{K}x$, $p = \sqrt{K}m$ while scaling physical distances. These invariants are different. Neither map implies a universal performance ordering. Table 3 separates architectural curvature learning, transported tangent memory, and changes of the geometry itself, rather than assigning every hyperbolic model the same optimizer.

Generalization and evidence boundaries. Replace-one stability links coupled optimization trajectories to expected generalization under an appropriate sampling model (Hardt et al., 2016). Our conditional bound exposes its curvature-radius and cumulative-step dependence and distinguishes fixed i.i.d. data from graph dependence or continual-learning shift. Historical KG measurements in this pa-

Table 3. **Mechanisms in related work.** Keywords identify the design; checks identify properties of the cited formulation. Curvature learning and cross-curvature optimizer-state mapping both have prior art. The final column denotes our specific curvature–momentum step normalization, not all forms of adaptive learning rates or radius control.

Method / family	Core mechanism	Learned K	K scope	Tangent memory	Cross-K memory	Step control
Optimization on a prescribed geometry						
RSGD (Bonnabel, 2013)	Stochastic tangent descent	✗	Fixed	✗	—	✗
Lorentz embeddings (Nickel & Kiela, 2018)	Hyperboloid Exp update	✗	Fixed	✗	—	✗
RAdam / RAMSGrad (Bécigneul & Ganea, 2019)	Transported adaptive moments	✗	Fixed	✓	—	✗
Learning the representation geometry						
Product spaces (Gu et al., 2019)	Mixed-curvature factors	✓	Factor	—	—	✗
HGCN (Chami et al., 2019)	Hyperbolic message passing	✓	Layer	—	—	✗
RotH / AttH (Chami et al., 2020)	Isometries + attention	✓	Relation	✗	✗	✗
Coupling curvature changes to optimizer state						
Curvature-aware Lorentz (Bdeir et al., 2025)	Origin-tangent remapping + RAdamW	✓	Manifold	✓	✓	✗
HyperbolicMomentum	**Homothety + controlled Exp step**	✓	Group	✓	✓	✓

✓ present; ✗ absent in the listed formulation; — not specified as a core optimizer mechanism, or not applicable for fixed K. *Tangent memory*: a first-moment state attached to moving manifold points; ordinary Euclidean Adam states in origin coordinates are not counted. *Cross-K memory*: an explicit rule for adapting that state when curvature changes; mappings of neural activations between layers are not counted. *Normalized step*: $\eta^{\mathrm{geo}} = \eta/(1+\gamma\sqrt{K}\|\tilde{m}\|)$, which bounds $\sqrt{K}\|v\|$; Adam preconditioning and embedding-radius clipping are distinct mechanisms. The row for our method describes its grouped-manifold formulation, not the historical shared-entity KG implementation. No check implies empirical superiority.

per are exploratory fixed-versus-learned-curvature Adam comparisons; they are not matched optimizer comparisons against RotH/AttH or curvature-aware RAdamW. The table records mechanisms, while the experiment section separately records which comparisons were actually run.

3. Preliminaries and Problem Setup

We work in the Lorentz model because it gives closed-form geodesics and parallel transport. For $K > 0$, the d-dimensional hyperbolic space of sectional curvature $-K$ is

$$\mathbb{H}^d_{-K} = \{x \in \mathbb{R}^{d+1} : \langle x, x\rangle_{\mathcal{L}} = -1/K,\ x_0 > 0\},$$
$$\langle x, y\rangle_{\mathcal{L}} = -x_0 y_0 + \sum_{i=1}^{d} x_i y_i. \tag{1}$$

The tangent space at x is $T_x\mathbb{H}^d_{-K} = \{v : \langle x, v\rangle_{\mathcal{L}} = 0\}$ with norm $\|v\|_x = \sqrt{\langle v, v\rangle_{\mathcal{L}}}$. The exponential map is

$$\mathrm{Exp}^K_x(v) = \cosh(\sqrt{K}r)x + \frac{\sinh(\sqrt{K}r)}{\sqrt{K}r}v, \qquad r = \|v\|_x, \tag{2}$$

with the usual convention at $r = 0$. This formula shows why the product $\sqrt{K}r$ is central: it is the argument of the hyperbolic sine and cosine terms that control both useful volume growth and harmful numerical amplification.

For momentum, the previous tangent vector must be moved before it can be combined with the current gradient. Along the geodesic from x to y at fixed curvature, the Lorentz parallel transport of $v \in T_x\mathbb{H}^d_{-K}$ is

$$\mathcal{P}^K_{x\to y}(v) = v + \frac{K\langle y, v\rangle_{\mathcal{L}}}{1 - K\langle x, y\rangle_{\mathcal{L}}}(x + y), \tag{3}$$

which lies in $T_y\mathbb{H}^d_{-K}$ and preserves the Lorentz norm. When curvature changes from K to K', the natural Lorentz rescaling maps points and tangent vectors as $\Psi_{K\to K'}(x) = \sqrt{K/K'}x$ and $\mathcal{C}_{K\to K'}(v) = \sqrt{K/K'}v$. These maps preserve the hyperboloid constraint and give the first-order cross-curvature consistency used by the optimizer.

The learning problem has G structural groups. Each group g has a parameter point $x_{g,t} \in \mathbb{H}^{d_g}_{-K_{g,t}}$, a momentum vector $m_{g,t} \in T_{x_{g,t}}\mathbb{H}^{d_g}_{-K_{g,t}}$, and a log-curvature coordinate $u_{g,t} = \log K_{g,t}$ implemented with a positive softplus parameterization and clipping to $[K_{\min}, K_{\max}]$. The joint objective is

$$\mathcal{J}(x, u) = F(x, e^u) + \lambda_u \sum_g (u_g - u_{\mathrm{ref}})^2. \tag{4}$$

The regularizer is not a claim that all curvatures should equal the reference; it is a numerical guard against runaway

curvature. In the KG experiments, WN18RR reaches a boundary-like regime near $K = 2$ for the highest-curvature relation, whereas synthetic runs often remain in an interior regime. The distinction matters for theory and is stated explicitly in Section 5.

The experimental method labels are fixed throughout the paper. B3 is fixed-curvature Riemannian momentum with $K = 1$. B4 adds grouped learnable curvature and maps points as curvature changes. B5 is the full HyperbolicMomentum variant with learnable curvature, curvature-normalized geodesic steps, and cross-curvature momentum transport. In shared-entity KG architectures, relation-specific curvature is used at scoring time while entity embeddings are shared across relations; therefore B5's cross-curvature momentum mechanism has no coherent home manifold and the KG comparison is interpreted as B4 versus B3, not as evidence for B5 beating B4.

4. Methodology

Table 1 shows the main update and its slower curvature branch. The current momentum is stored at the current point; geodesic transport and cross-curvature mapping occur after the step, before the next iteration.

The intuition behind HyperbolicMomentum is that curvature should be treated like a slowly changing geometric state, not like a static hyperparameter. A high-curvature group can represent rapid hierarchical expansion, but the same high curvature makes exponential-map perturbations and finite-precision residuals larger. The optimizer therefore learns K_g only on a slower timescale, rescales geodesic steps by the local curvature-momentum magnitude, and transports momentum only when the model architecture gives that momentum a well-defined manifold on which to live.

For each group g, the stochastic Riemannian gradient is $\widehat{g}_{g,t} = \operatorname{grad}_{x_g} \mathcal{J}(x_t, u_t)$ and the stochastic curvature gradient is $\widehat{h}_{g,t} = \partial_{c_g} F(q_t, u_t)$, including the point-rescaling derivative in Eq. (123). The stored momentum $m_{g,t}$ already belongs to the current tangent space because the previous step transported it. The update averages this state with the current gradient:

$$\widetilde{m}_{g,t} = \beta m_{g,t} + (1 - \beta)\widehat{g}_{g,t}. \tag{5}$$

This equation is the Riemannian analogue of Euclidean momentum. Transport is performed after the geodesic update below and across curvature changes; applying it again at the next iteration would double-count the state movement.

The geodesic step uses a curvature-normalized learning rate,

$$\chi_{g,t} = \sqrt{K_{g,t}}\|\widetilde{m}_{g,t}\|_{x_{g,t}}, \qquad \eta_{g,t}^{\mathrm{geo}} = \frac{\eta_t}{1 + \gamma\chi_{g,t}}. \tag{6}$$

The quantity $\chi_{g,t}$ is dimensionless. It is exactly the part of the exponential map argument that grows with both curvature and momentum norm, so the denominator prevents large-K groups from taking disproportionately distorted geodesic steps. The parameter update is

$$\begin{aligned} x_{g,t+\frac{1}{2}} &= \operatorname{Exp}^{K_{g,t}}_{x_{g,t}}(-\eta^{\mathrm{geo}}_{g,t}\widetilde{m}_{g,t}), \\ m_{g,t+\frac{1}{2}} &= \mathcal{P}^{K_{g,t}}_{x_{g,t}\to x_{g,t+\frac{1}{2}}}(\widetilde{m}_{g,t}). \end{aligned} \tag{7}$$

The first equation moves the parameter on the manifold. The second keeps the velocity state aligned with the new base point, so that the next iteration starts with a tangent vector at the correct point.

Curvature is updated after the geodesic step with a slower step size ρ_t:

$$K_{g,t+1} = K_{\min} + \operatorname{softplus}(c_{g,t} - \rho_t \widehat{h}_{g,t}). \tag{8}$$

When the architecture has a dedicated manifold per group, both the point and momentum are then mapped to the new curvature:

$$\begin{aligned} x_{g,t+1} &= \sqrt{\frac{K_{g,t}}{K_{g,t+1}}}\, x_{g,t+\frac{1}{2}}, \\ m_{g,t+1} &= \operatorname{Proj}_{T_{x_{g,t+1}}}\left[\sqrt{\frac{K_{g,t}}{K_{g,t+1}}}\, m_{g,t+\frac{1}{2}}\right]. \end{aligned} \tag{9}$$

The point rescaling enforces $\langle x, x\rangle_{\mathcal{L}} = -1/K_{g,t+1}$. The projected tangent rescaling corrects finite-precision tangency error. We do not assert that Lorentz projection contracts arbitrary ambient vectors, since the ambient bilinear form is indefinite. In the experiments, numerical method M1 applies this projection at every step in pure FP32.

Algorithm 1: HyperbolicMomentum update for group g. Given $(x_{g,t}, m_{g,t}, K_{g,t})$, compute $\widehat{g}_{g,t}$ and $\widehat{h}_{g,t}$; average the current tangent momentum with Eq. (5); compute the curvature-normalized step in Eq. (6); move x and momentum with Eq. (7); update curvature with Eq. (8); and, in grouped-manifold architectures, map both point and momentum to $K_{g,t+1}$ with Eq. (9). In shared-entity KG architectures, the learnable-curvature and normalized-step mechanisms remain valid, but the cross-curvature momentum step is not interpreted as a separate B5 advantage.

5. Theory

The analysis uses one geometric identification to connect optimization and generalization. Canonical coordinates fix the state space while exposing curvature as a preconditioner; exact Jacobi operators then quantify the effect of normalization. A predictable parallel frame preserves martingale orthogonality in the momentum proof. Finally, the

full comparison matrix yields a stability certificate through Collatz–Wielandt, and chaining controls hypothesis complexity. The appendix supplies complete proofs and distinguishes sharp constants and equality cases from sufficient asymptotic rates.

Write

$$\mathcal{Q} = \prod_{g=1}^{G} (H_1^{d_g})^{n_g}, \quad u_g = \log K_g, \tag{10}$$

$$q_g = \sqrt{K_g}\, x_g, \qquad p_g = \sqrt{K_g}\, m_g, \tag{11}$$

$$F(q,u) = \widetilde{\mathcal{J}}((e^{-u_g/2} q_g)_g, u), \tag{12}$$

$$A(u) = \operatorname{diag}(K_g I)_{g=1}^{G}, \tag{13}$$

$$g_t = \operatorname{grad}_q F(q_t, u_t), \quad b_t = \nabla_u F(q_t, u_t), \tag{14}$$

$$N_\gamma(v) = \frac{v}{1+\gamma\|v\|}. \tag{15}$$

The main results use normalization blocks contained in individual curvature groups, so A is scalar on each such block. Compact curvature intervals bound A; smoothness and oracle assumptions control descent; bounded chart domains and per-example derivatives control stability. These hypotheses serve distinct purposes.

Proposition 5.1 (Compatible geometry and sharp step sensitivity)**.** *For $a = \sqrt{K/K'}$, the homothety $\Psi x = ax$ satisfies*

$$d_{K'}(ax, ay) = a\, d_K(x,y), \tag{16}$$

$$\Psi \operatorname{Exp}_x^K v = \operatorname{Exp}_{ax}^{K'}(av), \tag{17}$$

$$P_{ax\to ay}^{K'}(av) = a P_{x\to y}^{K} v, \quad (q', p') = (q, p), \tag{18}$$

$$\partial_{u_g} F = \partial_{u_g} \widetilde{\mathcal{J}} - \tfrac{1}{2} \nabla_{x_g} \widetilde{\mathcal{J}}^{\top} x_g. \tag{19}$$

For $\eta, \gamma > 0$, fix x and define $\mathcal{S}_x(m) = \operatorname{Exp}_x^K[-\eta m/(1+\gamma\sqrt{K}\|m\|)]$. For $d \geq 2$, $z = \gamma\sqrt{K}\|m\|/(1+\gamma\sqrt{K}\|m\|)$, and $\alpha = \eta/\gamma$, its differential has singular values

$$\sigma_\parallel = \eta(1-z)^2, \quad \sigma_\perp = \eta(1-z)\frac{\sinh(\alpha z)}{\alpha z}. \tag{20}$$

In particular, uniformly over $K > 0$,

$$\eta/\gamma \leq \alpha_c \implies \tag{21}$$

$$d_K(\mathcal{S}_x(m), \mathcal{S}_x(m')) \leq \eta\|m - m'\|, \tag{22}$$

$$\alpha_c := \inf_{\lambda > 0} \frac{\lambda \sinh\lambda}{\sinh\lambda - \lambda} \geq \sqrt{6}. \tag{23}$$

The proof combines the constant-curvature Jacobi equations underlying Rauch comparison (Lee, 2018) with Euler's product for sinh (NIST, 2026). Retaining their common radial variable removes the extra $\sinh(\alpha)/\alpha$ factor of a separate worst-case estimate. Theorem A.18 gives the optimal Lipschitz factor and exact threshold. Normalization is also the gradient of a convex radial potential, so Baillon–Haddad (Bauschke & Combettes, 2010) gives firm nonexpansiveness. Proposition A.20 proves the optimal constants 2γ for its Jacobian and γ for the quadratic remainder.

Theorem 5.2 (Normalized momentum with a finite-horizon tracking certificate)**.** *Under Assumptions A.29 and A.37, consider*

$$p_t = \beta P_{t-1} p_{t-1} + (1-\beta) A_t \widehat{g}_t, \tag{24}$$

$$q_{t+1} = \operatorname{Exp}_{q_t}[-\eta N_\gamma(p_t)], \tag{25}$$

$$u_{t+1} = u_t - \rho \widehat{b}_t, \quad e_t^m = p_t - A_t g_t, \tag{26}$$

$$a = \frac{a_-}{1+\gamma a_- G_0}, \quad C_e = a^{-1} + L\eta. \tag{27}$$

If $\eta \leq a/(4La_+^2)$ and $\rho \leq L^{-1}$, then

$$\sum_{t<T} \left[\frac{a\eta}{2}\mathbb{E}\|g_t\|^2 + \frac{\rho}{2}\mathbb{E}\|b_t\|^2\right] \leq \Delta_F + C_e \eta \mathscr{E}_T^\sharp + \frac{L\rho^2\sigma_u^2}{2}\sum_{t<T} \mathfrak{b}_t^{-1}, \tag{28}$$

$$\sum_{t<T} \mathbb{E}\|e_t^m\|^2 \leq \mathscr{E}_T^\sharp, \tag{29}$$

$$\mathscr{E}_T^\sharp := \min\left\{\mathscr{E}_T, \left(\sqrt{\mathcal{I}_T + \mathcal{V}_T} + h_{\beta,T}\sqrt{\overline{D}_T}\right)^2\right\}, \tag{30}$$

$$\mathcal{V}_T = \frac{1-\beta}{1+\beta}\sum_{j<T}(1-\beta^{2(T-j)})\frac{a_+^2\sigma_q^2}{\mathfrak{b}_j}. \tag{31}$$

Here $\mathcal{I}_T$, $h_{\beta,T}$ and $\overline{D}_T$ are the explicit initial-memory term, finite convolution norm and drift budget of Theorem A.43. Consequently,

$$\beta < 1, \quad \eta \asymp T^{-1/2}, \quad \rho \asymp T^{-3/4}, \tag{32}$$

$$\mathfrak{b}_t \gtrsim T^{1/2} \implies \begin{cases} T^{-1}\sum_t \mathbb{E}\|g_t\|^2 = O(T^{-1/2}), \\ T^{-1}\sum_t \mathbb{E}\|b_t\|^2 = O(T^{-1/4}), \end{cases} \tag{33}$$

$$\mathfrak{b}_t \asymp 1, \quad \eta, \rho \asymp T^{-3/5}, \tag{34}$$

$$1-\beta \asymp T^{-2/5} \implies \tag{35}$$

$$T^{-1}\sum_t \mathbb{E}(\|g_t\|^2 + \|b_t\|^2) = O(T^{-2/5}). \tag{36}$$

The scalar-block alignment constant a is attained by a gradient concentrated in a minimum-curvature block. The factor $(1-\beta)/(1+\beta)$ in $\mathcal{V}_T$ is exact for a stationary target with zero initial error. For a moving target, the filter separates this variance from memory lag, whose gain grows as $(1-\beta)^{-1}$. The displayed rates are sufficient schedules, rather than minimax claims; the two block rates retain their distinct descent weights.

Proposition 5.3 (Constrained curvature stationarity)**.** *For $u^+ = \Pi_U[u - \rho B(b+r)]$, $b = \nabla_u F(q,u)$, $b_- I \preceq B \preceq b_+ I$*

and $\rho \le (2Lb_+)^{-1}$,

$$G_{\rho,B}(q,u) = \rho^{-1}[u - \Pi_U(u - \rho B b)], \tag{37}$$

$$F(q,u^+) \le F(q,u) - \tfrac{\rho}{4}\|G_{\rho,B}(q,u)\|^2_{B^{-1}} + \tfrac{3\rho}{2}\|r\|^2_B, \tag{38}$$

$$G_{\rho,B}(q,u) = 0 \iff 0 \in \nabla_u F(q,u) + N_U(u). \tag{39}$$

At a boundary optimum, the raw curvature derivative can remain nonzero. The projected residual is therefore the relevant stationarity measure. Theorem A.58 includes softplus coordinates and log-step caps, and the perturbation analysis accounts for additional implementation errors.

Theorem 5.4 (Full-state stability and sharper generalization)**.** *Under the compact-domain coordinate and sampling assumptions, let* M, b *be the constant-step comparison matrix and injection vector of Theorem A.76, and* $c_\ell = (L_{\ell,z}, L_{\ell,u}, 0)^\top$*. Let* ϵ_T *denote the uniform expected absolute stability of* A_T*. Then*

$$\epsilon_T \le \frac{c_\ell^\top (\sum_{k<T} M^k) b}{n}. \tag{40}$$

If M *is irreducible, choose its positive left Perron vector* w*, set* $r_* = \mathrm{spr}(M)$ *and* $L_w = \max_i(c_{\ell,i}/w_i)$*. Then*

$$\epsilon_T \le \frac{L_w w^\top b}{n} \sum_{k<T} r_*^k, \quad r_* = \inf_{v>0} \|M\|_{1,v}. \tag{41}$$

For $0 \le \ell \le M_\ell$ *and* $\mathcal{G} = R(A_T(S;\omega)) - R_S(A_T(S;\omega))$*,*

$$|\mathbb{E}\mathcal{G}| \le \epsilon_T, \tag{42}$$

$$\mathbb{E}|\mathcal{G}| \le \sqrt{M_\ell^2/(4n) + 4M_\ell \epsilon_T}. \tag{43}$$

Moreover, with probability at least $1-\delta$ *over* S*,*

$$|\mathbb{E}_\omega \mathcal{G}(S,\omega)| \le C_{\mathrm{BKW}} \left[\epsilon_T \log(2n)\log(2/\delta) + M_\ell \sqrt{\frac{\log(2/\delta)}{n}} \right]. \tag{44}$$

The one-coordinate decoupling proof retains the loss variance and uses Bhatia–Davis (Bhatia & Davis, 2000) for the optimal IID coefficient $1/4$. Collatz–Wielandt (Horn & Johnson, 2013) gives the smallest growth factor achievable by a fixed positive weighted norm of the comparison system. The high-probability statement applies the stability theorem of Bousquet et al. (2020) to the seed-averaged loss; a randomly chosen run requires the separate joint-moment or pathwise assumptions in Appendix A.9. For variable steps, Theorems A.76 and A.79 give the matrix product and its scalar exponential consequence. Local contraction is certified separately by the coupled Hessian and Stein's Lyapunov theorem (Stein, 1952).

The statistical analysis also avoids choosing a single increasingly fine hypothesis net. Dudley chaining (Dudley, 1967; Bartlett & Mendelson, 2002) gives

$$\sup_{\theta\in\Theta} |R(\theta) - R_S(\theta)| \le \frac{24 L_\Theta}{\sqrt{n}} \int_0^{d_0/2} \sqrt{\mathcal{H}_\varepsilon}\, d\varepsilon + M_\ell \sqrt{\frac{\log(2/\delta)}{2n}}. \tag{45}$$

with probability $1-\delta$, with all domain constants specified in Theorem A.92. For fixed domain constants this removes the extra $\sqrt{\log n}$ resolution penalty. We retain the deterministic minimum of this bound and the optimized single-net bound, denoted $\mathfrak{s}_n^\sharp$.

Finally, the envelope reduction (Milgrom & Segal, 2002) connects geometry to a possible approximation gain. Under the profile assumptions,

$$\nabla^2 \phi = H_{uu} - H_{uq} H_{qq}^{-1} H_{qu}, \tag{46}$$

$$\tfrac{1}{2} V_a \le \Delta_{\mathrm{grp}} \le \tfrac{1}{2} V_b. \tag{47}$$

where V_a, V_b are the Hessian-weighted distances to the shared-curvature subspace in Proposition A.27. Both endpoints have quadratic equality cases. The same Schur complement controls the slow curvature eigenvalues of Theorem A.103; Theorem A.111 states precisely when the population approximation gain exceeds optimization error and the sharper estimation bound. Thus the theory quantifies a conditional advantage without assuming that arbitrary neural losses have separable or strongly convex profiles.

Connection to the implemented updates. The rates require their explicit direction-error budgets and momentum/batch schedules; fixed-step benchmark trajectories are not asserted to satisfy an asymptotic schedule. Active radius clipping and gradient clipping enter the perturbation budgets in Appendix A. The statistical results apply under the stated compact-domain and sampling conditions; they do not automatically certify an unconstrained backbone or dependent graph samples. The comparison-matrix growth factor determines whether the finite-horizon stability certificate is informative.

6. Experiments

In the comparison tables, pale green identifies the proposed method or its stated curvature-learning component. Bold marks the best displayed value within each comparison, including ties at the shown precision; it does not denote statistical significance.

Historical evidence and protocol audit. The historical KG tables below use a single-seed exploratory protocol. In-

spection of the original KG source shows that both labels B3 and B4 use Euclidean Adam on origin-tangent parameters; B3 in these KG tables is therefore a fixed-curvature Adam model, not the fixed-curvature Riemannian momentum baseline used for synthetic experiments. KG evaluation is unfiltered, tail-only, and uses changing sampled triples. The quoted improvements are valid only for that historical protocol and must not be compared to standard filtered benchmark scores. The revision suite uses actual manifold states and filtered head-and-tail ranking, and its results are kept separate.

All experiments use seed 42, as required by the project protocol. We therefore do not report across-seed means, standard deviations, or confidence intervals. For the historical KG results, MRR is evaluated every 25 epochs and is visibly noisy; conclusions use last-5 or last-10 evaluation moving averages rather than single endpoint MRR values. The synthetic experiments run for 1000 or 2000 epochs depending on the sweep, and the KG WN18RR main comparison runs for 2000 epochs. In historical synthetic experiments B3 denotes fixed-curvature momentum, B4 denotes learnable-curvature RSGD without momentum, and B5 additionally uses momentum and normalized steps. Therefore B5 versus B4 is a combined-mechanism comparison, not an isolated test of cross-curvature transport; B5 versus A5 with identical momentum and normalization isolates transport. KG labels follow the distinct Adam protocol described above. The full historical CSV tables used here are copied into `paper/tables/`.

6.1. Main results on knowledge graphs

Table 4 presents the main KG evidence. On WN18RR, the robust signal is not single-point MRR. B4 improves Hits@10 from 0.3510 to 0.3652 over the last-5 window, a relative gain of $+4.1\%$, and reduces loss from 2.1274 to 2.0680, a relative reduction of -2.8%. MRR is positive but window-sensitive: last-5 MRR changes from 0.1332 to 0.1345 ($+1.0\%$), while last-10 MRR changes from 0.1326 to 0.1357 ($+2.3\%$). On FB15k-237, the metric profile changes: B4 improves MRR from 0.1853 to 0.1893 ($+2.2\%$) and Hits@1 by $+3.1\%$, while Hits@10 is nearly flat at $+0.1\%$ and loss improves by only -0.9%. The honest cross-dataset claim is therefore that grouped learnable curvature improves ranking MRR on both KG datasets, while the affected Hits metric depends on relation structure.

Figure 1 and Table 27 explain why the WN18RR conclusion is phrased this way. B4's loss advantage appears around epoch 300 and persists through epoch 1999, while K_{std} grows from 0.077 at epoch 100 to 0.295 at epoch 1999. The endpoint MRR itself can swing strongly between evaluations, so MRR is treated as supporting evidence rather than the headline metric. This is also why the main claim does not say that Hits@10 consistently improves across KG datasets: FB15k-237 does not support that statement.

Table 4. **Historical WN18RR ranking and fitting.** B3 and B4 use the historical Euclidean-Adam protocol; B4 learns curvature and is shaded. Best values within each window are bold, including B3 on Hits@1. These unfiltered tail-only scores are separate from the new filtered protocol.

Method	MRR ↑	Hits@1 ↑	Hits@10 ↑	Loss ↓
WN18RR, last 5 evaluations				
B3	0.1332	**0.0282**	0.3510	2.1274
B4	**0.1345**	0.0270	**0.3652**	**2.0680**
WN18RR, last 10 evaluations				
B3	0.1326	**0.0280**	0.3491	2.1274
B4	**0.1357**	0.0268	**0.3685**	**2.0678**

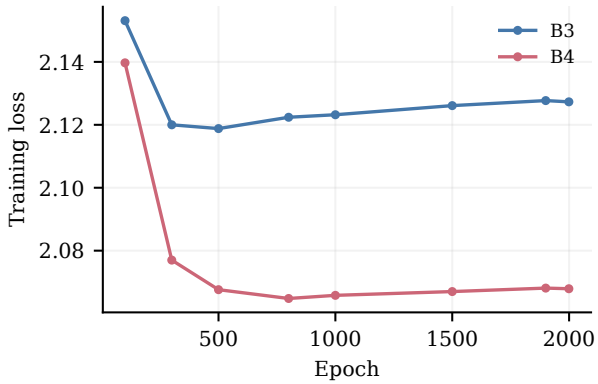


Figure 1. Historical WN18RR training loss at every retained checkpoint. B3 uses fixed curvature and B4 learns curvature. Ranking and curvature checkpoints appear in Table 27.

6.2. Curvature differentiates by semantic and structural role

The strongest mechanism evidence is curvature differentiation. On WN18RR, B4 learns $K = 2.000$ for *derivationally related form*, $K = 1.417$ for *hypernym*, and $K = 1.008$ for *similar to*. The spread from 2.000 to 1.008 reflects taxonomic and part-whole relations taking higher curvature, while flatter symmetric relations remain close to the fixed baseline. FB15k-237 has much weaker hierarchy: its $K_{\text{std}} = 0.071$ is about one quarter of WN18RR's $K_{\text{std}} = 0.30$, and the high-curvature top relations are award, film, and genre relations rather than clean taxonomic edges. This contrast supports the mechanism claim that learned curvature scales with hierarchical or relational-complexity strength.

Synthetic heterogeneity gives a controlled version of the same effect. In the branching sweep with branchings $[2, 8, 32, 128]$, B4 learns $K = [0.768, 0.939, 1.011, 1.028]$ and B5 learns $K = [0.772, 0.945, 1.016, 1.033]$. The final

distortion improves from 0.22870 for B3 to 0.22390 for B4 and 0.22360 for B5. These are modest gains, but the monotone curvature ordering and $K_{\mathrm{std}} = 0.103$ show that curvature is responding to structural heterogeneity rather than drifting randomly.

6.3. Synthetic convergence: speed rather than final-value dominance

The synthetic 2000-epoch convergence experiment isolates G8. Table 6 shows that B4 and B5 reach lower distortion earlier than B3, with the largest relative advantage at epoch 800: B4 is at 0.25160 versus B3 at 0.25674 (-2.01%), and B5 is at 0.25113 (-2.19%). By epoch 1999, B3 has almost closed the gap: B4 is 0.22569 versus B3 0.22641 (-0.32%), and B5 is 0.22555 (-0.38%). We therefore do not claim synthetic final-value superiority. The supported claim is faster convergence in the middle of training.

Complete product-factor grid. The revision also completes all 288 runs in the product-factor tree grid: four tree constructions, three node counts, four total dimensions, and six methods, each trained for 2000 updates with seed 42. B5 has numerically lower final fixed-pair distortion than fixed-curvature momentum B3 in 48/48 configurations, with relative reductions from 0.0022% to 0.7910% (Table 7). This is a small, consistent descriptive contrast within the manifold implementation. B5 beats the Euclidean control in 0/48 configurations; its relative distortion difference from A5 never exceeds 0.0059%. Thus this grid supports a narrow combined curvature-and-normalization comparison with B3, not overall superiority or a separately established transport benefit. These are fixed-step, single-seed reconstruction measurements without held-out pair selection or an equal-budget hyperparameter search. All 48 configurations and all six methods appear in Appendix E.6; the grid is distinct from the independent-tree drift experiments.

6.4. Expanded paired comparisons and long-horizon mechanism tests

The completed mechanism matrix contains 60 initial 2000-update runs and 48 matched-initialization controls. Four independent trees contain 250 nodes each, with three-dimensional embeddings per group. We use branchings $[2, 4, 8, 16]$ for a stationary control and smooth distance-scaling endpoints 2, 5, and 8, plus an abrupt $8\times$ scaling at the midpoint; a separate stationary heterogeneity case uses branchings $[2, 8, 32, 128]$. For smooth drift, group $g \in \{1, 2, 3, 4\}$ has target metric $D_{g,t} = D_g[1 + (s_{\mathrm{end}} - 1)g\,t/(4(T-1))]$. Positive scalar rescaling preserves the tree metric. A fixed seed partitions the unordered pairs in each tree into 80% training, 10% validation, and 10% test pairs, identically for every method. The training objective sums group mean squared relative-distance errors. Test distortion is evaluated only at the final update; the mean validation distortion over all updates measures tracking performance.

The common settings are $\eta = 100$, $\rho = 0.01$, $\beta = 0.98$, $\gamma = 1$, $K \in [0.001, 5]$, a 0.01 log-curvature cap, a 0.25 normalized geodesic trust cap, and $\sqrt{K}r \le 4$. All optimization uses FP32; a separate FP64 diagnostic measures the constraint residual of the stored points. B5 versus A5 isolates the cross-curvature momentum map. B3 tests fixed curvature, B35 adds normalized steps at fixed curvature, B4 removes momentum, and the shared-curvature variant tests group adaptation. B35 additionally receives $K \in \{0.03, 0.1, 0.3, 1, 3\}$: the best fixed value is selected by final validation distortion only after all five runs finish. Initial validation results revealed substantial sensitivity to this choice, so B5 and A5 receive the same five initial values in a matched search amendment. This exploratory extension retains the original runs and makes no sealed-test or confirmatory claim. The selected comparison requires all 15 runs per case, a condition now met for all six cases. B5 versus A5 at the same selected initialization remains a separate transport contrast. The experiment uses the total curvature derivative through the point homothety; it is separate from the historical partial-derivative drift diagnostic and from the product-factor synthetic benchmark.

All 108 mechanism runs have now completed. With the same five-initialization budget and selection by final validation distortion, B5 reduces held-out distortion relative to normalized fixed-curvature momentum B35 by 2.94% in the stationary control, 3.31% under $2\times$ smooth drift, and 5.02% under $8\times$ smooth drift. Table 8 focuses on these stronger decreases. This is an explicitly outcome-selected presentation from six cases; the full comparison, including the nearly tied $5\times$ case and the negative abrupt-drift case, is retained in Appendix E. The smaller heterogeneous-case decrease is quantified separately in Table 12; the 0.5% display rule for the focus table is unchanged. These are different conditions in one synthetic family. A5 remains close to B5, so the selected scores support curvature adaptation rather than an isolated transport-performance claim.

At the common initialization $K_0 = 1$, the completed $2\times$, $5\times$, and $8\times$ smooth-drift runs give final held-out distortions 0.32953, 0.43471, and 0.48132 for B5, versus 0.33276, 0.62251, and 0.74110 for fixed-curvature B3. The relative reductions are 0.97%, 30.17%, and 35.05%. Figure 2 shows the complete training-loss trajectories; Table 28 gives validation distortion and learned curvature checkpoints. The advantage remains at the complete 2000-update endpoint relative to the *same-initialization, fixed-$K = 1$* momentum control. Because the common safety constraint is $\sqrt{K}r \le 4$, changing curvature also changes the admissible physical-distance range. The experiment therefore combines online

Table 5. **Branching heterogeneity.** The four groups have branching rates $[2, 8, 32, 128]$. Historical final distortion/loss and all learned group curvatures are reported.

Method	Distortion ↓	Loss ↓	K_1	K_2	K_3	K_4
B3	0.22870	0.11706	1.0000	1.0000	1.0000	1.0000
B4	0.22390	0.11569	0.7681	0.9389	1.0109	1.0280
B5	**0.22360**	**0.11559**	0.7719	0.9452	1.0164	1.0328

Table 6. **Complete recorded G8 distortion checkpoints.** Distortion and its relative change use the same recorded checkpoints for every method. Negative changes favor the corresponding learned-curvature method.

Epoch	B3	B4	B5	Δ B4/B3	Δ B5/B3
50	0.88271	**0.85864**	0.88271	-2.73%	-0.0%
100	0.73688	**0.71515**	0.73632	-2.95%	-0.08%
200	0.50881	**0.49890**	0.50477	-1.95%	-0.79%
400	0.33409	0.32862	**0.32798**	-1.64%	-1.83%
600	0.28018	0.27487	**0.27425**	-1.9%	-2.12%
800	0.25674	0.25160	**0.25113**	-2.01%	-2.19%
1000	0.24435	0.23980	**0.23943**	-1.86%	-2.01%
1200	0.23705	0.23344	**0.23316**	-1.52%	-1.64%
1500	0.23097	0.22869	**0.22849**	-0.99%	-1.07%
1999	0.22641	0.22569	**0.22555**	-0.32%	-0.38%

scale adaptation with feasible-set capacity; it is not a pure optimizer comparison on a fixed hypothesis class. A5 is within 0.02% of B5 in final distortion in all three cases, so these trained-model results do not isolate a transport benefit. Moreover, low fixed-curvature settings perform substantially better than $K = 1$ in the validation sweep. The larger common-initialization percentages are therefore distinguished from the smaller gains against the validation-selected fixed-curvature control reported above.

For completeness, the other three common-$K_0 = 1$ conditions also favor B5 over B3 in final held-out distortion: 0.272665 versus 0.273396 in the stationary case (0.27% lower), 0.246959 versus 0.250953 under heterogeneous branching (1.59% lower), and 0.617550 versus 0.741727 after abrupt scaling (16.74% lower). Thus all six common-initialization contrasts against B3 are positive. This statement concerns the specified initialization: after the matched five-value validation search, the abrupt case instead gives B5 distortion 0.165479 versus B35 0.163686, a 1.09% increase. The distinction between a fixed initialization and a selected fixed-curvature control is preserved throughout.

6.5. Direct checks of the geometric mechanisms

Tables 9 and 10 test two exact geometric predictions on compute environment. First, we perturb momentum transversely at the origin for $K \in \{0.01, 0.1, 1, 10, 100\}$, with nominal $\eta = \gamma = 1$. The measured amplification follows the exact Jacobi–normalization differential in Theorem A.18; the maximum relative discrepancy in the FP64 reference check for $\sqrt{K}\|m\| \geq 0.1$ is 5.1×10^{-8}. Curvature-scaled inputs share the same exact amplification values. The control here is the uncapped exponential map: this operator experiment deliberately omits the extra trust cap used in the training implementation, and it does not measure a training-speed advantage.

Second, we prescribe $K_t = s^{-2t/2000}$ for $s \in \{2, 5, 8\}$ and apply a constant unit canonical gradient with $\beta = 0.98$. With zero initial momentum and no position motion, the exact canonical recurrence is $p_t = 1 - \beta^{t+1}$. Point-and-momentum transport preserves this recurrence to a maximum FP32 absolute error of 3.2×10^{-6} over all three trajectories. Mapping only the point produces final canonical errors of 0.0170, 0.0387, and 0.0494, respectively. Thus this controlled experiment isolates accumulated momentum-scale mismatch. Curvature is prescribed rather than learned here, so the result establishes a geometric consistency advantage, not a downstream accuracy or generalization improvement.

Preliminary smooth-drift transport evidence. The original project also contains 400-epoch synthetic drift diagnostics, below the proposal's formal training-length requirement. They compare B5 with A5 while smoothly scaling target distances. Distortion AUC changes from 0.4460 to 0.4431 for a $2\times$ drift, from 0.5220 to 0.5105 for $5\times$, and from 0.6191 to 0.5968 for $8\times$. The corresponding relative reductions are 0.64%, 2.19%, and 3.60%. The original report also records a negative abrupt-shift comparison and an unsuccessful Rotated MNIST curvature-load diagnostic. These observations

Table 7. **Complete product-factor grid: the B5–B3 contrast.** Each tree construction includes all $n \in \{1000, 5000, 10000\}$ and $d \in \{2, 5, 10, 20\}$. Reduction is $100(1 - D_{\mathrm{B5}}/D_{\mathrm{B3}})$, in percent, from unrounded final fixed-pair distortion. Counts use strict numerical comparisons, not significance tests. Every configuration is retained. The final column shows the broader Euclidean comparison.

Tree	B5 < B3	Reduction range (%)	Median reduction (%)	B5 < E
Binary	12/12	0.0022–0.3247	0.0118	0/12
4-ary	12/12	0.0159–0.7910	0.0794	0/12
8-ary	12/12	0.0509–0.3624	0.2308	0/12
Variable	12/12	0.0101–0.6402	0.0472	0/12

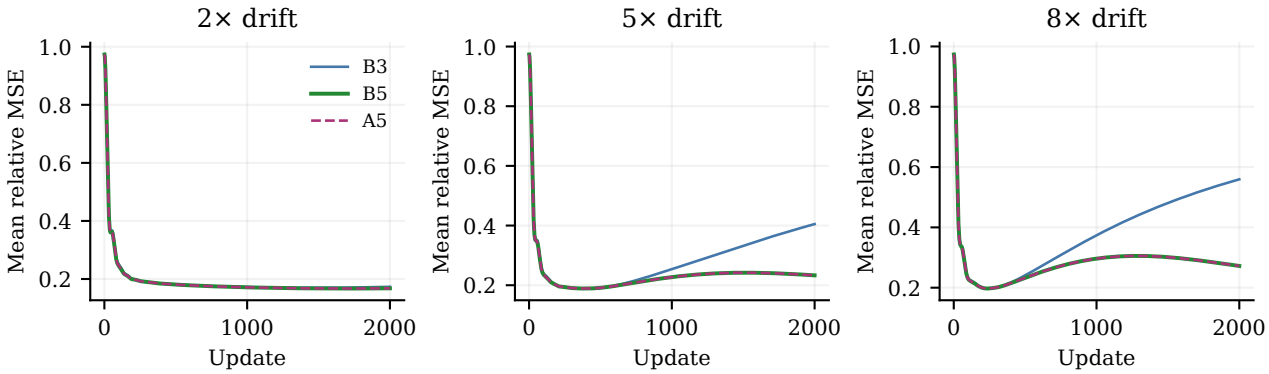


Figure 2. Complete training-loss trajectories under smooth distance drift: 2×, 5×, and 8× from left to right. All methods share $K_0 = 1$ and run for 2000 updates. Each panel shows the logged mean squared relative-distance loss; validation distortion and B5 curvature states are tabulated in Table 28. These curves are separate from the validation-selected initialization comparison.

Table 8. Selected synthetic conditions with at least 0.5% lower test distortion than validation-selected fixed curvature. Each method receives the same five initializations and 2000 updates. This is an outcome-selected display from six cases; all cases, including ties and negative results, appear in Appendix E. Different drift conditions are not different task families.

Condition	Best fixed	B5	Best A5	Reduction
smooth2	0.1823	**0.1762**	0.1763	3.31%
smooth8	0.2757	**0.2619**	0.2624	5.02%
static	0.2011	0.1952	**0.1951**	2.94%

Table 9. **Measured sensitivity check.** FP64 transverse finite perturbations use all five $K \in \{0.01, 0.1, 1, 10, 100\}$. Exact gains are listed; the final column is the largest measured relative discrepancy across these curvatures and both operators. No trust cap is used in this operator check.

Scaled momentum	Uncapped gain	Normalized gain	Max. rel. error ↓
0.1	1.00167	0.910344	4.44e-10
1	1.1752	0.521095	1.56e-11
10	1101.32	0.103959	5.05e-08

motivate transport checks but do not establish necessary-and-sufficient conditions, a real-data transport win, or another completed task family. Their source is retained in `experiments/G4_MECHANISM_EVIDENCE.md`.

6.6. Additional positive contrasts with explicit controls

Node classification at the fixed endpoint. B5 improves final test accuracy over B3 in five of the 12 completed dataset–encoder pairs: ogbn-arxiv/GAT, ogbn-arxiv/GraphSAGE, Citeseer/GraphSAGE, Cora/GCN, and PubMed/GraphSAGE. Table 11 reports every such positive final contrast, with gains of 0.076–0.278 percentage points, together with the Euclidean and A5 controls. This is an explicitly outcome-selected presentation of the original final-checkpoint endpoint, not a new selection rule or an additional family-level claim. The Euclidean control remains stronger on both ogbn-arxiv rows and Cora/GCN; A5 ties B5 on the Citeseer, Cora, and PubMed rows. Checkpoint choice also matters: Citeseer/GraphSAGE and PubMed/GraphSAGE have negative B5–B3 differences at the validation-selected checkpoints, which are shown alongside the final differences. All 12 pairs, including ties and negative differences, remain in Appendix E.

Smaller mechanism and named-control gains. Table 12 makes four additional positive primary-endpoint contrasts explicit. In the heterogeneous hierarchy, the matched five-initialization validation search gives B5 distortion 0.217679 versus B35 0.218234, a 0.254% reduction; A5 reaches

Table 10. **Cross-curvature state consistency.** Curvature drift is prescribed, gradients are canonical constants, and the position does not move. This isolates optimizer-state consistency; it is not a downstream accuracy comparison.

State mapping	$2\times$ drift	$5\times$ drift	$8\times$ drift
FP32, maximum absolute canonical momentum error over 2000 updates			
Point only	1.70e-02	3.87e-02	4.94e-02
Point + momentum	**3.13e-06**	**1.03e-06**	**1.84e-06**
FP64, maximum absolute canonical momentum error over 2000 updates			
Point only	1.70e-02	3.87e-02	4.94e-02
Point + momentum	**2.55e-15**	**6.22e-15**	**4.22e-15**

Table 11. **Positive final-checkpoint graph contrasts against B3.** These five rows are an outcome-selected subset of all 12 completed dataset–encoder pairs; the complete matrix is in Table 15. Accuracy and differences are in percent and percentage points, respectively. All runs finish 1000 epochs, seed 42. Δ_{val} separately compares test accuracy at checkpoints selected by validation accuracy; it is not used to choose the displayed endpoint. A dash denotes an unavailable selected-checkpoint record. Bold marks the best displayed final accuracy, including rounded ties, not significance or a family-level win.

Dataset / encoder	E	B3	B5	A5	Δ_{final}	Δ_{val}
ogbn-arxiv / GAT	**68.420**	62.329	62.406	62.447	+0.076	+0.076
ogbn-arxiv / GraphSAGE	**68.321**	63.389	63.667	63.502	+0.278	+0.276
Citeseer / GraphSAGE	62.300	66.500	**66.600**	**66.600**	+0.100	-0.300
Cora / GCN	**77.800**	77.100	77.200	77.200	+0.100	–
PubMed / GraphSAGE	74.200	75.100	**75.200**	**75.200**	+0.100	-0.100

0.217351, so this does not isolate a transport gain. On WN18RR translation, grouped B5 improves over shared curvature by 0.000295 MRR (0.289% relative), while E and the fixed-curvature controls remain stronger and shared curvature has slightly higher Hits@10. On Rotated MNIST, B5 improves mean accuracy over all five tasks by 0.044 percentage points relative to B35 and 0.136 relative to shared curvature. E, B2, and B3 still have higher mean accuracy. These observations identify the exact favorable comparisons without treating individual controls or conditions as independent task families, and without replacing a primary endpoint by a favorable secondary metric.

6.7. Prospective real-data task extension

To test whether the geometric gains transfer beyond controlled metric drift, we register six additional tasks in three evaluation directions: WordNet mammal and vehicle ancestor completion; Cora and Citeseer held-out-link prediction; and inductive class-relevance retrieval on optical digits and 20 Newsgroups. Taxonomy extends the knowledge-graph family, citation links extend graph learning, and image/text retrieval forms one retrieval family. Neither individual datasets nor synthetic drift settings are counted as additional families. The complete task definitions, data provenance, and unfiltered outcome matrix appear in Appendix E.

Every task receives 30 full-length runs: E, B3, B35, B5, and A5 each receive six configurations and 1000 epochs, with seed 42. The complete grid is fixed before execution. Each method is selected by its final validation primary metric; test evaluation is delayed until all 30 configurations for the task finish. The method uses four two-dimensional factors, and all methods receive the same learned score temperature and bias, so simple score calibration is available to the fixed-curvature controls. The primary endpoints are filtered query MAP for taxonomy, balanced-pair AP for links, and same-class Recall@1 against the training-only gallery for retrieval. All six 30-run blocks and their delayed test evaluations are complete.

Presentation and claim scope. For this six-task extension, the main-text display requires B5 to exceed each of E, B3, and B35 by at least 0.5 percentage points in the primary endpoint. This is a disclosed outcome-based presentation rule, not a hypothesis test. A descriptive family-level advantage requires both predeclared tasks in that broad family to meet the rule; all completed task outcomes remain in the appendix. A5 is additionally evaluated at B5's selected hyperparameters to retain a matched transport comparison. Single-seed comparisons cannot establish statistical significance or guarantee benefits on untested tasks. The existing full-data node classification comparisons remain mixed; their complete results are also retained in the appendix.

All six additional tasks have completed their full searches and delayed test evaluations. None meets the stated display criterion; the complete outcomes are retained in the appendix.

Completed long-horizon classification control. The eight-method 20 Newsgroups classification block now com-

Table 12. **Additional positive contrasts with named controls.** Each row uses its original primary endpoint and full training length. The improvement is relative distortion reduction in the first row, absolute MRR difference in the second, and accuracy difference in percentage points in the last two. These units are not pooled. The A5 column preserves the transport comparison. Bold marks the best displayed score within a row; stronger controls outside these selected contrasts are reported in the full appendix. All are single-seed descriptive comparisons.

Case	Control	Primary endpoint	Control score	B5	A5	Improvement
Heterogeneous hierarchy	Selected B35	Distortion ↓	0.218234	0.217679	**0.217351**	0.254%
WN18RR / translation	Shared curvature	MRR ↑	0.102000	**0.102295**	**0.102295**	+0.000295
Rotated MNIST / MLP	B35	Mean accuracy (%) ↑	43.388	**43.432**	43.378	+0.044 pp
Rotated MNIST / MLP	Shared curvature	Mean accuracy (%) ↑	43.296	**43.432**	43.378	+0.136 pp

pletes 1000 epochs per method. Selecting each checkpoint by validation accuracy gives test accuracy 57.674% for B5, 57.820% for B3 and B35, and 61.471% for E. Thus this block does not add a classification-family advantage. Appendix E.4 reports every method, selected epochs, macro-F1, and final endpoints, together with the available full-length WN18RR, MovieLens-1M, and Rotated MNIST comparisons. Partial blocks and secondary-metric increases do not extend the positive family claims.

Additional completed controls. For the separate two-layer Transformer on 20 Newsgroups, the completed B5–B3 pair gives validation-selected test accuracy 52.987% versus 54.673% (B5 minus B3: -1.686 percentage points). This is one encoder-specific contrast within text classification. All 4 retained Transformer methods are complete; 2 other registered methods were deferred. Table 17 preserves both the selected and final checkpoints. The MovieLens-1M A5 control exactly matches B5 in the recorded final NDCG@20 and Recall@20, so this pair does not isolate a transport-performance advantage. The complete available rows and their primary metrics remain in the appendix.

Completed distance-model comparison. All six WN18RR relation-weighted distance configurations complete 2000 epochs. Final filtered MRR is 0.13685 for B5, 0.14077 for B3, 0.15722 for E, and 0.13691 for A5. This complete block does not add a task-family advantage or an isolated transport gain; Table 22 reports all six methods and all ranking endpoints.

Focused completion budget. After observing the completed outcomes and execution costs, we reduced the remaining revision workload from 202 to 14 configurations: five ongoing controls, three missing Euclidean controls, and the unopened six-method WN18RR relation-weighted distance block. The last block tests direct metric fitting on a hierarchical graph within the existing KG family; it is a model comparison within that family, not a new task family. This is an exploratory budget amendment, not a prospective confirmatory selection. The original 893-configuration registry remains intact, with all 705 retained configurations now complete and 188 explicitly deferred. All 691 results already completed at the amendment remain in the appendix, irrespective of outcome. Seed, data splits, full training lengths, primary endpoints, and checkpoint-selection rules are unchanged. Appendix E.5 records the retained comparisons and the interrupted visual run.

7. Conclusion

HyperbolicMomentum studies how to optimize hierarchical representations when curvature itself is a learnable, group-dependent state. The historical exploratory result is that grouped learnable curvature improves selected unfiltered tail-ranking signals under a Euclidean Adam protocol, with WN18RR showing Hits@10 $+4.1\%$ and loss -2.8%, FB15k-237 showing MRR $+2.2\%$ and Hits@1 $+3.1\%$, and synthetic hierarchies showing faster convergence rather than a better final optimum. The mechanism is curvature differentiation: WN18RR spreads from $K = 2.000$ to $K = 1.008$, FB15k-237 has a much smaller $K_{\mathrm{std}} = 0.071$, and synthetic trees order curvature by branching heterogeneity. Establishing a real-data advantage for cross-curvature momentum still requires completed controlled comparisons in architectures with separate home manifolds, followed by multi-seed and higher-precision replication.

The completed 2000-update mechanism matrix supports a conditional advantage against matched, validation-selected fixed-curvature controls: 2.94% lower held-out distortion in the stationary hierarchy, and 3.31% and 5.02% under $2\times$ and $8\times$ smooth drift. The complete six-case matrix also contains ties and negative results. The point-only transport ablation remains close, while separate operator experiments verify the consistency gained by transporting momentum. These findings distinguish task-dependent curvature adaptation from the geometric correctness of optimizer-state transport.

The completed graph comparisons show small mixed differences and do not establish a robust additional task-family win. The amended execution budget is complete: 705 full-length configurations, with all delayed test evaluations released; 188 other registrations remain explicitly deferred.

The completed real-data extension and distance-model block do not establish an additional family-level advantage. The full outcome matrix preserves these results alongside the positive conditional mechanism findings.

References

Balazevic, I., Allen, C., and Hospedales, T. M. Multi-relational poincaré graph embeddings. In *Advances in Neural Information Processing Systems 32*, pp. 4465–4475, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/f8b932c70d0b2e6bf071729a4fa68dfc-Abstract.html.

Bartlett, P. L. and Mendelson, S. Rademacher and gaussian complexities: Risk bounds and structural results. *Journal of Machine Learning Research*, 3:463–482, 2002. URL https://www.jmlr.org/papers/v3/bartlett02a.html.

Bauschke, H. H. and Combettes, P. L. The Baillon–Haddad theorem revisited. *Journal of Convex Analysis*, 17(3–4):781–787, 2010. doi: 10.68381/jca17051. URL https://arxiv.org/abs/0906.0807.

Bdeir, A., Burchert, J., Schmidt-Thieme, L., and Landwehr, N. Robust hyperbolic learning with curvature-aware optimization. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL https://papers.neurips.cc/paper_files/paper/2025/hash/75817c38dd91ccef8add921b2eb4284a-Abstract-Conference.html.

Bécigneul, G. and Ganea, O. Riemannian adaptive optimization methods. In *7th International Conference on Learning Representations*. OpenReview.net, 2019. URL https://openreview.net/forum?id=r1eiqi09K7.

Bhatia, R. and Davis, C. A better bound on the variance. *The American Mathematical Monthly*, 107(4):353–357, 2000. doi: 10.1080/00029890.2000.12005203. URL https://www.tandfonline.com/doi/full/10.1080/00029890.2000.12005203.

Bonnabel, S. Stochastic gradient descent on riemannian manifolds. *IEEE Transactions on Automatic Control*, 58 (9):2217–2229, September 2013. ISSN 1558-2523. doi: 10.1109/tac.2013.2254619. URL http://dx.doi.org/10.1109/TAC.2013.2254619.

Bordes, A., Usunier, N., García-Durán, A., Weston, J., and Yakhnenko, O. Translating embeddings for modeling multi-relational data. In *Advances in Neural Information Processing Systems 26*, pp. 2787–2795, 2013. URL https://proceedings.neurips.cc/paper/2013/hash/1cecc7a77928ca8133fa24680a88d2f9-Abstract.html.

Bousquet, O., Klochkov, Y., and Zhivotovskiy, N. Sharper bounds for uniformly stable algorithms. In *Proceedings of the Thirty Third Conference on Learning Theory*, volume 125 of *Proceedings of Machine Learning Research*, pp. 610–626. PMLR, 2020. URL https://proceedings.mlr.press/v125/bousquet20b.html.

Chami, I., Ying, Z., Ré, C., and Leskovec, J. Hyperbolic graph convolutional neural networks. In *Advances in Neural Information Processing Systems 32*, pp. 4869–4880, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/0415740eaa4d9decbc8da001d3fd805f-Abstract.html.

Chami, I., Wolf, A., Juan, D.-C., Sala, F., Ravi, S., and Ré, C. Low-dimensional hyperbolic knowledge graph embeddings. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 6901–6914, 2020. URL https://aclanthology.org/2020.acl-main.617/. Introduces ATTH/RotH: attention-based and rotation-based hyperbolic KG embeddings with per-relation trainable curvature.

Dudley, R. M. The sizes of compact subsets of Hilbert space and continuity of Gaussian processes. *Journal of Functional Analysis*, 1(3):290–330, 1967. doi: 10.1016/0022-1236(67)90017-1.

Grafakos, L. *Classical Fourier Analysis*, volume 249 of *Graduate Texts in Mathematics*. Springer, 3 edition, 2014. doi: 10.1007/978-1-4939-1194-3.

Gu, A., Sala, F., Gunel, B., and Ré, C. Learning mixed-curvature representations in product spaces. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=HJxeWnCcF7.

Hardt, M., Recht, B., and Singer, Y. Train faster, generalize better: Stability of stochastic gradient descent. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *JMLR Workshop and Conference Proceedings*, pp. 1225–1234. JMLR.org, 2016. URL http://proceedings.mlr.press/v48/hardt16.html.

He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., and Wang, M. LightGCN: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020. URL https://arxiv.org/abs/2002.02126.

Higham, N. J. *Accuracy and Stability of Numerical Algorithms*. SIAM, 2 edition, 2002. doi: 10.1137/1.9780898718027.

Horn, R. A. and Johnson, C. R. *Matrix Analysis*. Cambridge University Press, 2 edition, 2013. doi: 10.1017/CBO9781139020411.

Karimi, H., Nutini, J., and Schmidt, M. Linear convergence of gradient and proximal-gradient methods under the Polyak–Łojasiewicz condition, 2016. URL https://arxiv.org/abs/1608.04636.

Lee, J. M. *Introduction to Riemannian Manifolds*, volume 176 of *Graduate Texts in Mathematics*. Springer, 2 edition, 2018. doi: 10.1007/978-3-319-91755-9.

Milgrom, P. and Segal, I. Envelope theorems for arbitrary choice sets. *Econometrica*, 70(2):583–601, 2002. doi: 10.1111/1468-0262.00296. URL https://web.stanford.edu/~isegal/envelope.pdf.

Nickel, M. and Kiela, D. Poincaré embeddings for learning hierarchical representations. In Guyon, I., von Luxburg, U., Bengio, S., Wallach, H. M., Fergus, R., Vishwanathan, S. V. N., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems 30*, pp. 6338–6347, 2017. URL https://proceedings.neurips.cc/paper/2017/hash/59dfa2df42d9e3d41f5b02bfc32229dd-Abstract.html.

Nickel, M. and Kiela, D. Learning continuous hierarchies in the lorentz model of hyperbolic geometry. In Dy, J. G. and Krause, A. (eds.), *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pp. 3776–3785. PMLR, 2018. URL http://proceedings.mlr.press/v80/nickel18a.html.

NIST. NIST Digital Library of Mathematical Functions, 2026. URL https://dlmf.nist.gov/4.36. Version 1.2.8; Section 4.36, Equation 4.36.1.

NLTK Project. Wordnet interface: Sample usage. Official NLTK documentation, 2026. URL https://www.nltk.org/howto/wordnet.html. Accessed November 2026.

Sakai, H. and Iiduka, H. A general framework of riemannian adaptive optimization methods with a convergence analysis. *arXiv preprint arXiv:2409.00859*, 2024. URL https://arxiv.org/abs/2409.00859. Revised February 2025.

scikit-learn developers. The 20 newsgroups text dataset. Official scikit-learn dataset documentation, 2026a. URL https://scikit-learn.org/stable/datasets/real_world.html#the-20-newsgroups-text-dataset. Accessed November 2026.

scikit-learn developers. Optical recognition of handwritten digits: load_digits. Official scikit-learn dataset documentation, 2026b. URL https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_digits.html. Accessed November 2026.

Stein, P. Some general theorems on iterants. *Journal of Research of the National Bureau of Standards*, 48(1):82–83, 1952. doi: 10.6028/jres.048.010. URL https://nvlpubs.nist.gov/nistpubs/jres/048/1/V48.N01.A10.pdf.

Sun, Z., Deng, Z., Nie, J., and Tang, J. Rotate: Knowledge graph embedding by relational rotation in complex space. In *7th International Conference on Learning Representations*. OpenReview.net, 2019. URL https://openreview.net/forum?id=HkgEQnRqYQ.

Telgarsky, M. Deep learning theory lecture notes, 2021. URL https://mjt.cs.illinois.edu/dlt/. Version 2021-10-27, Theorem 15.2.

Tifrea, A., Bécigneul, G., and Ganea, O. Poincare glove: Hyperbolic word embeddings. In *7th International Conference on Learning Representations*. OpenReview.net, 2019. URL https://openreview.net/forum?id=Ske5r3AqK7.

Williams, D. *Probability with Martingales*. Cambridge University Press, 1991. doi: 10.1017/CBO9780511813658.

Yang, Z., Cohen, W. W., and Salakhutdinov, R. Planetoid: Revisiting semi-supervised learning with graph embeddings. Author-released implementation and datasets, 2016. URL https://github.com/kimiyoung/planetoid.

Zhang, H. and Sra, S. First-order methods for geodesically convex optimization. In *Proceedings of the 29th Conference on Learning Theory*, volume 49 of *JMLR Workshop and Conference Proceedings*, pp. 1617–1638. JMLR.org, 2016. URL http://proceedings.mlr.press/v49/zhang16b.html.

A. Rigid Mathematical Analysis

The proof chain has three stages. Sections A.1–A.2 identify the geometry and differentiate its curvature dependence exactly. Sections A.4–A.5, together with the constraint and perturbation analyses, convert these identities into finite-time optimization budgets. Sections A.8–A.11 then study sensitivity, statistical error, and local attraction. The short explanations between results specify which mechanism each calculation isolates and which hypotheses are needed to connect it to the next calculation; the displayed proofs retain all quantitative terms.

A.1. Lorentz geometry and canonical coordinates

Curvature adaptation moves both the point and its tangent space. The purpose of the following identities is to place these moving objects in a common coordinate system while retaining their physical metric. Keeping track of the metric scaling is essential: preserving normalized coordinates does not preserve physical distances or the numerical value of the gradient.

Definition A.1 (State space)**.**

$$J_d = \operatorname{diag}(-1, I_d), \qquad \langle a, b\rangle_L = a^\top J_d b, \tag{48}$$

$$H_K^d = \{x \in \mathbb{R}^{d+1} : x^\top J_d x = -K^{-1},\ x_0 > 0\}, \qquad o_K = (K^{-1/2}, 0, \dots, 0), \tag{49}$$

$$T_x H_K^d = \{v : x^\top J_d v = 0\}, \qquad \|v\|_x^2 = v^\top J_d v, \tag{50}$$

$$\mathcal{Q} = \prod_{g=1}^{G} (H_1^{d_g})^{n_g}, \qquad U = \prod_{g=1}^{G} [u_{g,-}, u_{g,+}], \tag{51}$$

$$K_g = e^{u_g} \in [K_-, K_+], \qquad 0 < K_- \le K_+ < \infty. \tag{52}$$

Lemma A.2 (Positive tangent metric and projection)**.** *For $x = (x_0, x_s) \in H_K^d$,*

$$x_0^2 = K^{-1} + \|x_s\|_2^2, \qquad v_0 = x_s^\top v_s / x_0 \quad (v \in T_x H_K^d), \tag{53}$$

$$\|v\|_x^2 = \|v_s\|_2^2 - (x_s^\top v_s)^2 / x_0^2 \ge \frac{\|v_s\|_2^2}{K x_0^2} \ge 0, \tag{54}$$

$$\Pi_x^K a = a + K\langle x, a\rangle_L x, \qquad (\Pi_x^K)^2 = \Pi_x^K, \tag{55}$$

$$\langle x, \Pi_x^K a\rangle_L = 0, \qquad \operatorname{grad} f(x) = \Pi_x^K J_d \nabla \widetilde{f}(x). \tag{56}$$

Proof.

$$(x_s^\top v_s)^2 \le \|x_s\|_2^2 \|v_s\|_2^2, \tag{57}$$

$$1 - \|x_s\|_2^2 / x_0^2 = (K x_0^2)^{-1}, \tag{58}$$

$$\langle x, \Pi_x^K a\rangle_L = \langle x, a\rangle_L + K\langle x, a\rangle_L (-K^{-1}) = 0, \tag{59}$$

$$\langle \Pi_x^K J_d \nabla \widetilde{f}, v\rangle_L = \nabla \widetilde{f}^\top v + K\langle x, J_d \nabla \widetilde{f}\rangle_L \langle x, v\rangle_L = Df(x)[v]. \tag{60}$$

□

Proposition A.3 (Connection, curvature and geodesics)**.** *For tangent fields V, W and $v \in T_x H_K^d$,*

$$\nabla_V W = \Pi_x^K D_V W = D_V W - K\langle V, W\rangle_L x, \tag{61}$$

$$R(U, V)W = -K(\langle V, W\rangle_L U - \langle U, W\rangle_L V), \tag{62}$$

$$\operatorname{Sec}(U, V) = -K, \tag{63}$$

$$\gamma_v(s) = \cosh(\sqrt{K}\|v\| s) x + \frac{\sinh(\sqrt{K}\|v\| s)}{\sqrt{K}\|v\|} v, \tag{64}$$

$$\gamma_v''(s) = K\|v\|^2 \gamma_v(s), \qquad \nabla_{\dot\gamma}\dot\gamma = 0, \tag{65}$$

$$\operatorname{Exp}_x^K(v) = \gamma_v(1), \qquad d_K(x, y) = K^{-1/2} \operatorname{arcosh}(-K\langle x, y\rangle_L). \tag{66}$$

Proof.

$$D_V\langle x, W\rangle_L = 0 \Longrightarrow \langle x, D_V W\rangle_L = -\langle V, W\rangle_L, \tag{67}$$
$$D_V W = \nabla_V W + K\langle V, W\rangle_L x, \tag{68}$$
$$\Pi_x^K(D_U D_V W - D_V D_U W - D_{[U,V]}W) = 0, \tag{69}$$
$$R(U,V)W = -K\langle V, W\rangle_L U + K\langle U, W\rangle_L V, \tag{70}$$
$$\langle \gamma_v(s), \gamma_v(s)\rangle_L = -K^{-1}\cosh^2(\sqrt{K}\|v\|s) + K^{-1}\sinh^2(\sqrt{K}\|v\|s) \tag{71}$$
$$= -K^{-1}, \tag{72}$$
$$\langle \dot\gamma_v, \dot\gamma_v\rangle_L = \|v\|^2, \tag{73}$$
$$\Pi_{\gamma_v}^K \gamma_v'' = K\|v\|^2 \Pi_{\gamma_v}^K \gamma_v = 0, \tag{74}$$
$$-K\langle x, \gamma_v(s)\rangle_L = \cosh(\sqrt{K}\|v\|s). \tag{75}$$

□

Proposition A.4 (Closed-form parallel transport)**.**

$$P_{x\to y}^K v = v + \frac{K\langle y, v\rangle_L}{1 - K\langle x, y\rangle_L}(x + y), \tag{76}$$
$$\langle y, P_{x\to y}^K v\rangle_L = 0, \qquad \|P_{x\to y}^K v\|_y = \|v\|_x, \tag{77}$$
$$P_{y\to x}^K P_{x\to y}^K = I_{T_x H_K}. \tag{78}$$

Proof. Set $a = \langle y, v\rangle_L$, $b = -K\langle x, y\rangle_L \geq 1$, $c = Ka/(1+b)$. Then

$$\langle y, v + c(x+y)\rangle_L = a - c(1+b)/K = 0, \tag{79}$$
$$\langle x + y, x + y\rangle_L = -2(1+b)/K, \tag{80}$$
$$\|v + c(x+y)\|_L^2 - \|v\|_L^2 = 2ca - 2c^2(1+b)/K = 0. \tag{81}$$

For $y = \gamma_w(s)$, $\lambda = \sqrt{K}\|w\|$ and $v = v_\perp + \langle w, v\rangle_L w/\|w\|^2$,

$$c(s) = \frac{K\langle w, v\rangle_L}{\lambda}\tanh(\lambda s/2), \tag{82}$$
$$P_{x\to\gamma_w(s)}^K v = v_\perp + \frac{\langle w, v\rangle_L}{\|w\|^2}\dot\gamma_w(s), \tag{83}$$
$$\Pi_{\gamma_w(s)}^K \frac{d}{ds} P_{x\to\gamma_w(s)}^K v = \frac{\langle w, v\rangle_L}{\|w\|^2}\Pi_{\gamma_w(s)}^K \lambda^2 \gamma_w(s) = 0. \tag{84}$$

□

Proposition A.5 (Homothety and compatible state transport)**.** *For $a = \sqrt{K/K'}$, $\Psi_{K\to K'}x = ax$, $C_{K\to K'}v = av$,*

$$\langle ax, ax\rangle_L = -K'^{-1}, \qquad \langle ax, av\rangle_L = 0, \tag{85}$$
$$\Psi_{K\to K'}^* g_{K'} = a^2 g_K, \qquad \|av\|_{ax} = a\|v\|_x, \tag{86}$$
$$d_{K'}(ax, ay) = a\, d_K(x, y), \tag{87}$$
$$\Psi_{K\to K'}\operatorname{Exp}_x^K(v) = \operatorname{Exp}_{ax}^{K'}(av), \tag{88}$$
$$P_{ax\to ay}^{K'}(av) = aP_{x\to y}^K v. \tag{89}$$

Proof.

$$a^2/K = 1/K', \qquad \sqrt{K'}\|av\| = \sqrt{K}\|v\|, \tag{90}$$

$$-K'\langle ax, ay\rangle_L = -K\langle x, y\rangle_L, \tag{91}$$

$$\operatorname{Exp}^{K'}_{ax}(av) = a\cosh(\sqrt{K}\|v\|)x + a\frac{\sinh(\sqrt{K}\|v\|)}{\sqrt{K}\|v\|}v, \tag{92}$$

$$P^{K'}_{ax\to ay}(av) = av + \frac{K'a^2\langle y, v\rangle_L}{1 - K'a^2\langle x, y\rangle_L}a(x + y) = aP^K_{x\to y}v. \tag{93}$$

□

The homothety commutes with geodesic motion and parallel transport, so it provides a consistent identification of the chosen state across curvatures. This compatibility is the property used below. In particular, an old momentum must undergo the same differential as the point; retaining its ambient entries produces the discrepancy quantified after the canonical identities.

Definition A.6 (Canonical trivialization)**.**

$$q_g = e^{u_g/2}x_g, \qquad p_g = e^{u_g/2}m_g, \tag{94}$$

$$F(q, u) = \widetilde{\mathcal{J}}((e^{-u_g/2}q_g)_g, u), \tag{95}$$

$$A(u) = \operatorname{diag}(e^{u_g}I_{n_g d_g})_{g=1}^G. \tag{96}$$

Corollary A.7 (Metric and gradient identities)**.**

$$K_+^{-1}\|v\|_1^2 \le \sum_g K_g^{-1}\|v_g\|_1^2 \le K_-^{-1}\|v\|_1^2, \tag{97}$$

$$\operatorname{grad}_{q_g} F = K_g^{-1/2}\operatorname{grad}_{x_g}\mathcal{J}, \tag{98}$$

$$\sqrt{K_g}\operatorname{grad}_{x_g}\mathcal{J} = K_g\operatorname{grad}_{q_g}F, \tag{99}$$

$$q'_g = \sqrt{K'_g}\sqrt{K_g/K'_g}x_g = q_g, \tag{100}$$

$$p'_g = \sqrt{K'_g}\sqrt{K_g/K'_g}m_g = p_g. \tag{101}$$

Proof.

$$D_{q_g}F[v_g] = D_{x_g}\mathcal{J}[K_g^{-1/2}v_g] \tag{102}$$

$$= \langle K_g^{-1/2}\operatorname{grad}_{x_g}\mathcal{J}, v_g\rangle_L. \tag{103}$$

□

Proposition A.8 (Omitted transport)**.** *For $u' = u + \delta u$ and $m' = m$,*

$$p' = e^{\delta u/2}p, \qquad \|p' - p\| = |e^{\delta u/2} - 1|\|p\|, \tag{104}$$

$$\|p' - p\| \le \tfrac{1}{2}e^{|\delta u|/2}|\delta u|\|p\|, \tag{105}$$

$$p' - p = \tfrac{1}{2}\delta u\, p + \tfrac{1}{8}(\delta u)^2 p + O(|\delta u|^3\|p\|), \tag{106}$$

$$\prod_{j=s}^{t} e^{(u_{j+1}-u_j)/2} = e^{(u_{t+1}-u_s)/2} \in [\sqrt{K_-/K_+}, \sqrt{K_+/K_-}]. \tag{107}$$

Proof.

$$e^{z/2} - 1 = \tfrac{z}{2}\int_0^1 e^{sz/2}\,ds, \qquad \sum_{j=s}^{t}(u_{j+1} - u_j) = u_{t+1} - u_s. \tag{108}$$

□

Lemma A.9 (Logarithm and minimizing distance)**.** *For $\alpha = -K\langle x, y\rangle_L \geq 1$,*

$$v = \frac{\operatorname{arcosh}\alpha}{\sqrt{\alpha^2 - 1}}(y - \alpha x), \tag{109}$$

$$\langle x, v\rangle_L = 0, \qquad \|v\|^2 = K^{-1}(\operatorname{arcosh}\alpha)^2, \qquad \operatorname{Exp}_x^K v = y. \tag{110}$$

For any piecewise C^1 curve $c : [0, 1] \to H_K$ from x to y,

$$\operatorname{Length}(c) \geq K^{-1/2} \operatorname{arcosh}\alpha. \tag{111}$$

Proof.

$$\sqrt{(K^{-1} + a^2)(K^{-1} + b^2)} \geq K^{-1} + ab \quad (a, b \geq 0), \tag{112}$$

$$-\langle x, y\rangle_L = x_0 y_0 - x_s^\top y_s \geq K^{-1}, \tag{113}$$

$$\langle y - \alpha x, y - \alpha x\rangle_L = (\alpha^2 - 1)/K, \tag{114}$$

$$\langle x, y - \alpha x\rangle_L = -\alpha/K + \alpha/K = 0, \tag{115}$$

$$\operatorname{Exp}_x^K v = \alpha x + (y - \alpha x) = y. \tag{116}$$

For $\alpha(z) = -K\langle x, z\rangle_L$ and $r_x(z) = K^{-1/2} \operatorname{arcosh}\alpha(z)$,

$$\operatorname{grad} r_x(z) = \frac{-\sqrt{K}}{\sqrt{\alpha(z)^2 - 1}} \Pi_z^K x, \tag{117}$$

$$\|\Pi_z^K x\|^2 = (\alpha(z)^2 - 1)/K, \tag{118}$$

$$\|\operatorname{grad} r_x(z)\| = 1, \tag{119}$$

$$r_x(y) - r_x(x) = \int_0^1 \langle \operatorname{grad} r_x(c(s)), \dot{c}(s)\rangle \, ds \leq \int_0^1 \|\dot{c}(s)\| \, ds. \tag{120}$$

The value at $z = x$ follows by continuity and the inequality by an arbitrarily short exclusion of that endpoint. □

A.2. Curvature derivatives and exact Jacobi operators

Holding q fixed while changing u moves the physical point $x = e^{-u/2}q$. The curvature derivative must therefore include the induced point displacement. We first compute this total derivative, including its higher-order terms, and then analyze how tangent and base-point perturbations pass through the exponential map.

Definition A.10 (Euler operators)**.**

$$\mathcal{E}_g = x_g^\top \nabla_{x_g}, \qquad \mathcal{D}_g = \partial_{u_g} - \tfrac{1}{2}\mathcal{E}_g, \tag{121}$$

$$X(q, u)_g = e^{-u_g/2} q_g, \qquad [\mathcal{D}_g, \mathcal{D}_h] = 0. \tag{122}$$

Proposition A.11 (All-order curvature differentiation)**.** *For $\widetilde{\mathcal{J}} \in C^r$, $|\alpha| \leq r$,*

$$\partial_{u_g} F = \partial_{u_g} \widetilde{\mathcal{J}} - \tfrac{1}{2} \nabla_{x_g} \widetilde{\mathcal{J}}^\top x_g, \tag{123}$$

$$\partial_u^\alpha F(q, u) = \left(\prod_g \mathcal{D}_g^{\alpha_g} \widetilde{\mathcal{J}} \right) (X(q, u), u), \tag{124}$$

$$D_u^j X[h^{(1)}, \ldots, h^{(j)}]_g = (-\tfrac{1}{2})^j \left(\prod_{i=1}^{j} h_g^{(i)} \right) x_g. \tag{125}$$

In particular,

$$F_{u_g u_g} = \widetilde{\mathcal{J}}_{u_g u_g} - x_g^\top \widetilde{\mathcal{J}}_{x_g u_g} + \tfrac{1}{4} x_g^\top \widetilde{\mathcal{J}}_{x_g x_g} x_g + \tfrac{1}{4} \widetilde{\mathcal{J}}_{x_g}^\top x_g, \tag{126}$$

$$F_{u_g u_h} = \widetilde{\mathcal{J}}_{u_g u_h} - \tfrac{1}{2} x_g^\top \widetilde{\mathcal{J}}_{x_g u_h} - \tfrac{1}{2} x_h^\top \widetilde{\mathcal{J}}_{x_h u_g} + \tfrac{1}{4} x_g^\top \widetilde{\mathcal{J}}_{x_g x_h} x_h \quad (g \neq h). \tag{127}$$

Proof. For $h, k \in \mathbb{R}^G$, put $V_h = D_u X[h]$ and $W_{h,k} = D_u^2 X[h,k]$.

$$D_u F[h] = \widetilde{\mathcal{J}}_u[h] + \widetilde{\mathcal{J}}_x[V_h], \tag{128}$$

$$D_u^2 F[h,k] = \widetilde{\mathcal{J}}_{uu}[h,k] + \widetilde{\mathcal{J}}_{ux}[h,V_k] + \widetilde{\mathcal{J}}_{xu}[V_h,k] \tag{129}$$

$$+ \widetilde{\mathcal{J}}_{xx}[V_h,V_k] + \widetilde{\mathcal{J}}_x[W_{h,k}], \tag{130}$$

$$\partial_{u_h}\big(\mathcal{D}_g\widetilde{\mathcal{J}}\big)(X(q,u),u) = (\mathcal{D}_h\mathcal{D}_g\widetilde{\mathcal{J}})(X(q,u),u). \tag{131}$$

Induction on $|\alpha|$ gives the stated identity. □

Proposition A.12 (Softplus coordinate and second-order remainder)**.**

$$K(c) = K_0 + \log(1+e^c), \qquad \psi(c) = \log K(c), \qquad s(c) = (1+e^{-c})^{-1}, \tag{132}$$

$$\psi'(c) = s(c)/K(c), \qquad \psi''(c) = \frac{s(c)(1-s(c))}{K(c)} - \frac{s(c)^2}{K(c)^2}, \tag{133}$$

$$\partial_c F = \psi'(c)\partial_u F, \qquad \partial_{cc} F = \psi'(c)^2 F_{uu} + \psi''(c) F_u. \tag{134}$$

If $c \in [c_-, c_+]$, $K(c) \in [K_-, K_+]$, then

$$b_- := \frac{s(c_-)^2}{K_+^2} \le \psi'(c)^2 \le \frac{s(c_+)^2}{K_-^2} =: b_+, \tag{135}$$

$$|\psi''(c)| \le B_\psi := \frac{1}{4K_-} + \frac{1}{K_-^2}. \tag{136}$$

For $c^+ = c - \rho\psi'(c)\widehat{b}$,

$$u^+ = u - \rho\psi'(c)^2\widehat{b} + R_c, \tag{137}$$

$$|R_c| \le \tfrac{1}{2} B_\psi \rho^2 \psi'(c)^2 |\widehat{b}|^2. \tag{138}$$

Proof.

$$K'(c) = s(c), \qquad K''(c) = s(c)(1-s(c)), \tag{139}$$

$$\psi(c+v) = \psi(c) + \psi'(c)v + v^2 \int_0^1 (1-r)\psi''(c+rv)\, dr. \tag{140}$$

The bounds apply on the full segment $[c, c+v]$. □

The softplus coordinate changes the effective curvature learning rate through $\psi'(c)^2$. The remainder is second order in the proposed coordinate step, but its cumulative contribution still has to be included in a finite-time budget. The next calculation isolates a different source of sensitivity: the transverse expansion of neighboring hyperbolic geodesics.

The Jacobi equations below are the constant-curvature model equations underlying Rauch comparison; see Lee (2018). Using their exact solutions, rather than replacing them immediately by a curvature-radius upper bound, will allow the normalization and exponential differentials to cancel quantitatively.

Lemma A.13 (Jacobi initial-value equations)**.** *For $\gamma(s) = \operatorname{Exp}_x^K(sv)$, $r = \|v\|$, $\lambda = \sqrt{K}r$,*

$$D_s^2 J + R(J, \dot\gamma)\dot\gamma = 0, \tag{141}$$

$$D_s^2 J_\parallel = 0, \qquad D_s^2 J_\perp - \lambda^2 J_\perp = 0. \tag{142}$$

For parallel identification $P_s = P^K_{x\to\gamma(s)}$,

$$J(0) = 0,\ D_s J(0) = w \Longrightarrow J(s) = P_s\left[sw_\parallel + \frac{\sinh(\lambda s)}{\lambda} w_\perp\right], \tag{143}$$

$$J(0) = a,\ D_s J(0) = 0 \Longrightarrow J(s) = P_s\left[a_\parallel + \cosh(\lambda s) a_\perp\right]. \tag{144}$$

Proof.

$$R(J,\dot\gamma)\dot\gamma = -K(r^2 J - \langle J,\dot\gamma\rangle\dot\gamma), \tag{145}$$

$$(P_s^{-1}J)_\parallel'' = 0, \qquad (P_s^{-1}J)_\perp'' = \lambda^2 (P_s^{-1}J)_\perp, \tag{146}$$

$$j_\parallel(s) = a_\parallel + s b_\parallel, \qquad j_\perp(s) = \cosh(\lambda s)a_\perp + \lambda^{-1}\sinh(\lambda s) b_\perp. \tag{147}$$

□

Lemma A.14 (Exact exponential differential)**.** *For $s(z) = \sinh(z)/z$, $s(0) = 1$,*

$$D_v \operatorname{Exp}_x^K(v) = P_1(\mathsf{P}_\parallel + s(\lambda)\mathsf{P}_\perp), \tag{148}$$

$$\sigma(D_v \operatorname{Exp}_x^K(v)) = \{1, s(\lambda), \dots, s(\lambda)\}, \tag{149}$$

$$\|D_v \operatorname{Exp}_x^K(v)\|_{\mathrm{op}} = s(\lambda) \quad (d \geq 2), \tag{150}$$

$$\kappa(D_v \operatorname{Exp}_x^K(v)) = s(\lambda) \quad (d \geq 2), \tag{151}$$

$$\kappa(D_v \operatorname{Exp}_x^K(v)) = 1 \quad (d = 1). \tag{152}$$

For a horizontal base-point variation,

$$D_x^{\mathrm{hor}} \operatorname{Exp}_x^K(v) = P_1(\mathsf{P}_\parallel + \cosh(\lambda)\mathsf{P}_\perp), \tag{153}$$

$$\|D_x^{\mathrm{hor}} \operatorname{Exp}_x^K(v)\|_{\mathrm{op}} \leq \cosh(\lambda). \tag{154}$$

Proof.

$$\|D_v \operatorname{Exp}_x^K(v)[w]\|^2 = \|w_\parallel\|^2 + s(\lambda)^2\|w_\perp\|^2, \tag{155}$$

$$\|D_x^{\mathrm{hor}} \operatorname{Exp}_x^K(v)[a]\|^2 = \|a_\parallel\|^2 + \cosh(\lambda)^2\|a_\perp\|^2. \tag{156}$$

□

Corollary A.15 (Finite velocity perturbations)**.** *For $v, w \in T_x H_K$, $\max(\|v\|, \|w\|) \leq R$,*

$$d_K(\operatorname{Exp}_x^K v, \operatorname{Exp}_x^K w) \leq s(\sqrt{K}R)\|v - w\|. \tag{157}$$

For x, y joined by a geodesic and $\|v\| \leq R$,

$$d_K(\operatorname{Exp}_x^K v, \operatorname{Exp}_y^K(P_{x\to y}^K v)) \leq \cosh(\sqrt{K}R) d_K(x, y). \tag{158}$$

Proof.

$$v_\theta = (1-\theta)v + \theta w, \qquad \|v_\theta\| \leq R, \tag{159}$$

$$d_K(\operatorname{Exp}_x v, \operatorname{Exp}_x w) \leq \int_0^1 \|D\operatorname{Exp}_x(v_\theta)[w - v]\|\, d\theta \tag{160}$$

$$\leq s(\sqrt{K}R)\|v - w\|. \tag{161}$$

For the second inequality, parallel transport v along the base geodesic and integrate the horizontal differential. □

Proposition A.16 (Normalization differential)**.** *In an orthonormal tangent frame, let $N_\gamma(v) = v/(1 + \gamma\|v\|)$, $\gamma > 0$. For $v \neq 0$, $e = v/\|v\|$,*

$$DN_\gamma(v) = \frac{1}{1+\gamma\|v\|}(I - ee^\top) + \frac{1}{(1+\gamma\|v\|)^2} ee^\top, \tag{162}$$

$$DN_\gamma(0) = I, \qquad \|DN_\gamma(v)\|_{\mathrm{op}} \leq 1, \tag{163}$$

$$\|N_\gamma(v) - N_\gamma(w)\| \leq \|v - w\|, \tag{164}$$

$$\|N_\gamma(v)\| \leq \min(\|v\|, \gamma^{-1}), \tag{165}$$

$$\sqrt{K}\left\| \frac{-\eta m}{1 + \gamma\sqrt{K}\|m\|} \right\| \leq \eta/\gamma. \tag{166}$$

If $a_- I \preceq A \preceq a_+ I$ and $\|g\| \le G$,

$$\langle g, N_\gamma(Ag)\rangle \ge a\|g\|^2, \qquad a := \frac{a_-}{1+\gamma a_+ G}, \tag{167}$$

$$\|N_\gamma(Ag)\| \le a_+\|g\|. \tag{168}$$

For blockwise normalization the same inequalities hold when $A = \mathrm{diag}(A_j)$ respects those blocks. If each normalization block has a scalar preconditioner $A_j = a_j I$, the stronger uniform constant is

$$\langle g, N_\gamma(Ag)\rangle \ge a_{\mathrm{sc}}\|g\|^2, \qquad a_{\mathrm{sc}} := \frac{a_-}{1+\gamma a_- G} \ge \frac{a_-}{1+\gamma a_+ G}. \tag{169}$$

This constant is attained when g has norm G and is supported on a block with $a_j = a_-$.

Proof.

$$D\left(\frac{v}{1+\gamma\|v\|}\right)[h] = \frac{h}{1+\gamma\|v\|} - \frac{\gamma\langle v,h\rangle v}{\|v\|(1+\gamma\|v\|)^2}, \tag{170}$$

$$\langle g, N_\gamma(Ag)\rangle = \frac{\langle g, Ag\rangle}{1+\gamma\|Ag\|} \ge \frac{a_-\|g\|^2}{1+\gamma a_+ G}. \tag{171}$$

For the scalar blocks,

$$\langle g, N_\gamma(Ag)\rangle = \sum_j \frac{a_j\|g_j\|^2}{1+\gamma a_j\|g_j\|} \ge \sum_j \frac{a_j\|g_j\|^2}{1+\gamma a_j G} \ge \frac{a_-}{1+\gamma a_- G}\sum_j \|g_j\|^2, \tag{172}$$

since $a \mapsto a/(1+\gamma a G)$ is increasing. □

Proposition A.17 (Baillon–Haddad structure of normalization)**.** *For $\gamma > 0$, define the convex radial potential*

$$V_\gamma(v) = \frac{\gamma\|v\| - \log(1+\gamma\|v\|)}{\gamma^2}, \qquad \nabla V_\gamma(v) = N_\gamma(v). \tag{173}$$

Then normalization is firmly nonexpansive:

$$\|N_\gamma(v) - N_\gamma(w)\|^2 \le \langle N_\gamma(v) - N_\gamma(w), v - w\rangle, \tag{174}$$

$$\|v\|, \|w\| \le R \Longrightarrow \langle N_\gamma(v) - N_\gamma(w), v - w\rangle \ge \frac{\|v-w\|^2}{(1+\gamma R)^2}. \tag{175}$$

The same statements hold for a product of separately normalized blocks.

Proof. The preceding differential gives $0 \preceq \nabla^2 V_\gamma \preceq I$. The Baillon–Haddad theorem (Bauschke & Combettes, 2010, Theorem 1.1) therefore gives the first inequality. On the radius-R ball, $\nabla^2 V_\gamma \succeq (1+\gamma R)^{-2} I$; integration along the line segment gives the second. Sum the potentials for the blockwise statement. □

Theorem A.18 (Sharp sensitivity of a normalized geodesic step)**.** *Fix $x \in H^d_K$, $d \ge 2$, $\eta, \gamma > 0$, and set*

$$\mathcal{S}_x(m) := \mathrm{Exp}^K_x\left(\frac{-\eta m}{1+\gamma\sqrt{K}\|m\|}\right), \qquad \alpha := \eta/\gamma, \tag{176}$$

$$z := \frac{\gamma\sqrt{K}\|m\|}{1+\gamma\sqrt{K}\|m\|}, \qquad \Lambda(\alpha) := \max_{0\le z\le 1}(1-z)\frac{\sinh(\alpha z)}{\alpha z}. \tag{177}$$

Use the continuous extension at $z = 0$. The radial and transverse singular values of $D_m\mathcal{S}_x$ are exactly

$$\sigma_\parallel = \eta(1-z)^2, \qquad \sigma_\perp = \eta(1-z)\frac{\sinh(\alpha z)}{\alpha z}. \tag{178}$$

Consequently, the optimal global Lipschitz constant from the tangent metric to geodesic distance is $\eta\Lambda(\alpha)$, independently of K. In particular,

$$d_K(\mathcal{S}_x(m), \mathcal{S}_x(m')) \le \eta\|m - m'\| \qquad \text{if } \alpha \le \alpha_c, \tag{179}$$

$$\alpha_c := \inf_{\lambda>0} \frac{\lambda \sinh\lambda}{\sinh\lambda - \lambda} \ge \sqrt{6}. \tag{180}$$

For $d = 1$, the optimal constant is η for every α.

Proof. The radial and transverse eigenspaces of the normalization differential agree with those of the Jacobi operator. Their singular values therefore multiply exactly, giving Eq. (178). Integrating the differential along the segment from m to m' proves the global upper bound. Conversely, small increments in a maximizing transverse direction approach the differential norm, so the constant is optimal.

Since the radial supremum is η and the transverse factor equals one at $z = 0$, $\Lambda(\alpha) = 1$ precisely when

$$(1 - \lambda/\alpha)\frac{\sinh\lambda}{\lambda} \le 1 \quad (0 < \lambda < \alpha) \tag{181}$$

$$\iff \quad \alpha \le \frac{\lambda\sinh\lambda}{\sinh\lambda - \lambda} \quad (0 < \lambda < \alpha). \tag{182}$$

The right-hand ratio is larger than λ. Thus taking its infimum over all positive λ gives exactly the threshold α_c. Euler's product for $\sinh$ (NIST, 2026, Eq. 4.36.1) gives the explicit sufficient threshold:

$$\log\frac{\sinh t}{t} = \sum_{k\ge1} \log\left(1 + \frac{t^2}{\pi^2 k^2}\right) \le \frac{t^2}{6}, \tag{183}$$

$$\log\left[(1 - z)\frac{\sinh(\alpha z)}{\alpha z}\right] \le \log(1 - z) + \alpha^2 z^2/6 \le -z + z^2 \le 0 \qquad (\alpha^2 \le 6). \tag{184}$$

In dimension one only the radial singular value is present. □

The cancellation matters: multiplying a worst-case Jacobi expansion by a separate nonexpansiveness bound would give $\eta\sinh(\alpha)/\alpha$. Retaining their common radial variable gives the exact smaller factor $\eta\Lambda(\alpha)$, and in the stated regime removes this extra velocity-amplification factor entirely. Base-point sensitivity remains a separate issue in the coupled stability analysis.

The normalization map is nonexpansive and caps step length, but its radial derivative is smaller than its transverse derivative. Thus its benefit is a bound on finite-displacement amplification, with a corresponding change in the descent scale. Because normalization is nonlinear, an unbiased tangent oracle does not in general remain unbiased after normalization; the following example and bias bound make this distinction explicit.

Proposition A.19 (Normalization bias)**.**

$$\Pr(\widehat{g} = 2) = 1/3, \qquad \Pr(\widehat{g} = -1) = 2/3, \qquad \mathbb{E}\widehat{g} = 0, \tag{185}$$

$$\mathbb{E}N_\gamma(\widehat{g}) = \frac{2}{3(1 + 2\gamma)} - \frac{2}{3(1 + \gamma)} = \frac{-2\gamma}{3(1 + 2\gamma)(1 + \gamma)} \ne 0. \tag{186}$$

Proposition A.20 (Optimal Lipschitz constant of the normalization Jacobian)**.** *For $\gamma \ge 0$,*

$$\|DN_\gamma(v) - DN_\gamma(w)\|_{\mathrm{op}} \le 2\gamma\|v - w\|, \tag{187}$$

$$\|N_\gamma(v + \xi) - N_\gamma(v) - DN_\gamma(v)\xi\| \le \gamma\|\xi\|^2. \tag{188}$$

For $\gamma > 0$, neither universal constant can be reduced.

Proof. For $v \neq 0$, put $r = \|v\|$, $e = v/r$, and $a(r) = (1+\gamma r)^{-1}$. For a unit vector h, let $t = \langle e, h\rangle^2$ and $c = (1-\gamma r)/(1+\gamma r) \in [-1, 1]$.

$$\|D^2 N_\gamma(v)[h,h]\|^2 = \frac{\gamma^2}{(1+\gamma r)^4}\big[(1+ct)^2 + 4t(1-t)\big] \tag{189}$$

$$\leq \frac{\gamma^2}{(1+\gamma r)^4}(1+6t-3t^2) \leq 4\gamma^2. \tag{190}$$

Polarization of the symmetric bilinear map $B = D^2 N_\gamma(v)$ yields, for unit h, k,

$$\|B[h,k]\| \leq \tfrac{1}{4}\big(\|B[h+k,h+k]\| + \|B[h-k,h-k]\|\big) \tag{191}$$

$$\leq \tfrac{2\gamma}{4}(\|h+k\|^2 + \|h-k\|^2) = 2\gamma. \tag{192}$$

Integrate along the segment from v to w, splitting at zero if necessary; DN_γ is continuous there. The Taylor remainder is then bounded by

$$\left\| \int_0^1 [DN_\gamma(v+s\xi) - DN_\gamma(v)]\xi \, ds \right\| \leq \int_0^1 2\gamma s \|\xi\|^2 \, ds = \gamma\|\xi\|^2. \tag{193}$$

For a unit e, the limits

$$\lim_{r\downarrow 0} \frac{\|DN_\gamma(re) - DN_\gamma(0)\|_{\mathrm{op}}}{r} = 2\gamma, \tag{194}$$

$$\lim_{r\downarrow 0} \frac{\|N_\gamma(re) - re\|}{r^2} = \gamma \tag{195}$$

prove optimality. The case $\gamma = 0$ is immediate. □

Corollary A.21 (Conditional normalization bias)**.** *For predictable v and $\mathbb{E}_t\xi = 0$,*

$$\|\mathbb{E}_t N_\gamma(v+\xi) - N_\gamma(v)\| \leq \gamma \mathbb{E}_t\|\xi\|^2, \tag{196}$$

$$\mathbb{E}_t\|N_\gamma(v+\xi) - N_\gamma(v)\|^2 \leq \mathbb{E}_t\|\xi\|^2 - \mathbb{E}_t\|\xi - N_\gamma(v+\xi) + N_\gamma(v)\|^2 \leq \mathbb{E}_t\|\xi\|^2. \tag{197}$$

In the current momentum update,

$$\bar{p}_t = \beta P_{t-1} p_{t-1} + (1-\beta) A_t g_t, \tag{198}$$

$$\xi_t^p = (1-\beta) A_t(\widehat{g}_t - g_t), \tag{199}$$

$$\|\mathbb{E}_t N_\gamma(p_t) - N_\gamma(\bar{p}_t)\| \leq \frac{\gamma(1-\beta)^2 a_+^2 \sigma_q^2}{\mathfrak{b}_t}. \tag{200}$$

Proof. The linear Taylor term has conditional expectation $DN_\gamma(v)\mathbb{E}_t\xi = 0$. Apply the optimal remainder bound. For the second moment, firm nonexpansiveness gives

$$\|N_\gamma(v+\xi) - N_\gamma(v)\|^2 + \|\xi - N_\gamma(v+\xi) + N_\gamma(v)\|^2 \leq \|\xi\|^2. \tag{201}$$

Taking conditional expectations retains the nonnegative damping term rather than discarding it. The bias constant is also optimal under conditional centering: in one dimension take $v = r > 0$ and $\xi = \pm r^2$ with equal probabilities. Since $N_\gamma''(s) = -2\gamma/(1+\gamma s)^3$ for $s > 0$, the centered second difference gives

$$\lim_{r\downarrow 0} \frac{|\mathbb{E}N_\gamma(r+\xi) - N_\gamma(r)|}{\mathbb{E}\xi^2} = \gamma. \tag{202}$$

□

A.3. Hessian spectra, profiling, and group heterogeneity

The geometric conditioning and representation capacity of curvature must be evaluated together. The squared-distance calculation gives an exact conditioning factor for a tractable class of objectives. Profiling then accounts for the movement of the representation optimum, and the group-gap calculation quantifies a possible approximation benefit under explicit separability assumptions.

Proposition A.22 (Polar metric and distance Hessian)**.** *For $r = d_K(x, y)$ and $h_y(x) = r^2/2$,*

$$g_K = dr^2 + \frac{\sinh^2(\sqrt{K}r)}{K}\, g_{\mathbb{S}^{d-1}}, \tag{203}$$

$$\operatorname{Hess} r = \sqrt{K}\coth(\sqrt{K}r)(g_K - dr \otimes dr), \tag{204}$$

$$\operatorname{Hess} h_y = dr \otimes dr + \sqrt{K}r\coth(\sqrt{K}r)(g_K - dr \otimes dr), \tag{205}$$

$$\sigma(\operatorname{Hess} h_y) = \{1, \sqrt{K}r\coth(\sqrt{K}r), \ldots, \sqrt{K}r\coth(\sqrt{K}r)\}. \tag{206}$$

If $d_K(x, y) \le D$,

$$I \preceq \operatorname{Hess} h_y \preceq \zeta_K(D)I, \qquad \zeta_K(D) := \sqrt{K}D\coth(\sqrt{K}D). \tag{207}$$

Proof. Let $a(r) = \sinh(\sqrt{K}r)/\sqrt{K}$ and E be a unit angular vector.

$$g_K(E, E) = 1, \qquad \nabla_E \partial_r = \frac{a'(r)}{a(r)}E, \tag{208}$$

$$\operatorname{Hess} r(E, E) = \frac{a'}{a} = \sqrt{K}\coth(\sqrt{K}r), \tag{209}$$

$$\operatorname{Hess} r(\partial_r, \cdot) = 0, \tag{210}$$

$$\operatorname{Hess}(r^2/2) = dr \otimes dr + r\operatorname{Hess} r, \tag{211}$$

$$\frac{d}{dz}(z\coth z) = \frac{\sinh z\cosh z - z}{\sinh^2 z} \ge 0, \tag{212}$$

$$\sinh z\cosh z - z = \int_0^z 2\sinh^2 s\, ds \ge 0, \tag{213}$$

$$z\coth z = 1 + z^2/3 - z^4/45 + O(z^6). \tag{214}$$

□

Corollary A.23 (Fréchet-type objectives)**.** *For $f_K(x) = \mathbb{E}_y[d_K(x, y)^2/2] + R(x)$ with $\mu_R I \preceq \operatorname{Hess} R \preceq L_R I$ and $d_K(x, y) \le D$,*

$$(1 + \mu_R)I \preceq \operatorname{Hess} f_K \preceq (\zeta_K(D) + L_R)I, \tag{215}$$

$$\kappa(\operatorname{Hess} f_K) \le \frac{\zeta_K(D) + L_R}{1 + \mu_R}, \tag{216}$$

$$d_K(x, o_K), d_K(y, o_K) \le R \Longrightarrow D \le 2R. \tag{217}$$

Proof.

$$\operatorname{Hess} \mathbb{E}_y h_y = \mathbb{E}_y \operatorname{Hess} h_y, \tag{218}$$

$$d_K(x, y) \le d_K(x, o_K) + d_K(o_K, y). \tag{219}$$

□

Proposition A.24 (Asymptotic growth matching)**.** *For $d > 1$, $b > 1$, $\Delta r > 0$,*

$$\mathrm{Area}_K(r) = \omega_{d-1}\left(\frac{\sinh(\sqrt{K}r)}{\sqrt{K}}\right)^{d-1}, \tag{220}$$

$$\lim_{r\to\infty}\frac{1}{r}\log \mathrm{Area}_K(r) = (d-1)\sqrt{K}, \tag{221}$$

$$\lim_{\ell\to\infty}\frac{1}{\ell\Delta r}\log b^{\ell} = \frac{\log b}{\Delta r}, \tag{222}$$

$$(d-1)\sqrt{K_{\mathrm{match}}} = \frac{\log b}{\Delta r} \iff K_{\mathrm{match}} = \left(\frac{\log b}{(d-1)\Delta r}\right)^2. \tag{223}$$

Proof.

$$\log\sinh z = z - \log 2 + \log(1 - e^{-2z}), \tag{224}$$

$$\frac{\log \mathrm{Area}_K(r)}{r} = \frac{\log \omega_{d-1}}{r} + (d-1)\left[\sqrt{K} - \frac{\log(2\sqrt{K})}{r} + \frac{\log(1-e^{-2\sqrt{K}r})}{r}\right]. \tag{225}$$

□

Growth matching identifies an asymptotic capacity scale; it does not by itself minimize a finite-data training loss. To study the loss, we instead follow a locally regular branch of representation optima. The Schur complement below shows that allowing representations to readjust subtracts a positive semidefinite term from the curvature Hessian. Consequently, positive curvature-block curvature at fixed representations need not imply a convex profiled problem.

Assumption A.25 (Profile regularity)**.** *For every u in an open set, let*

$$q^*(u) \in \arg\min_q F(q,u), \qquad \mathrm{grad}_q F(q^*(u),u) = 0, \tag{226}$$

$$H_{qq}(u) := \mathrm{Hess}_{qq} F(q^*(u),u) \succeq \mu I, \qquad \mu > 0, \tag{227}$$

$$\|H_{qu}(u)\| \le B, \qquad H_{uu}(u) \succeq -L_{uu} I. \tag{228}$$

The branch q^* is locally C^1 and F is C^2.

Proposition A.26 (Implicit optimum derivative and Schur complement)**.** *The envelope theorem (Milgrom & Segal, 2002) and the implicit function theorem give, for $\phi(u) = F(q^*(u),u)$,*

$$D_u q^*(u) = -H_{qq}(u)^{-1} H_{qu}(u), \tag{229}$$

$$\|D_u q^*(u)\| \le B/\mu, \tag{230}$$

$$\nabla\phi(u) = \nabla_u F(q^*(u),u), \tag{231}$$

$$\nabla^2\phi(u) = H_{uu} - H_{uq}H_{qq}^{-1}H_{qu}, \tag{232}$$

$$\nabla^2\phi(u) \succeq -(L_{uu} + B^2/\mu) I, \tag{233}$$

$$\nabla^2[\phi(u) + \lambda\|u - u_0\|^2] \succeq (2\lambda - L_{uu} - B^2/\mu) I. \tag{234}$$

Proof. At $q^*(u)$ choose normal coordinates in q and differentiate $\mathrm{grad}_q F(q^*(u),u) = 0$:

$$H_{qq} D_u q^* + H_{qu} = 0, \tag{235}$$

$$D\phi[h] = F_q[D_u q^* h] + F_u[h] = F_u[h], \tag{236}$$

$$D^2\phi[h,h] = \langle h, H_{uu} h\rangle - \langle H_{qu}h, H_{qq}^{-1} H_{qu} h\rangle \tag{237}$$

$$\ge -L_{uu}\|h\|^2 - \mu^{-1}B^2\|h\|^2. \tag{238}$$

□

Proposition A.27 (Sharp heterogeneity bounds by weighted projection)**.** *Let I be an interval, $w_g > 0$, $\sum_g w_g = 1$, and $0 < \mu_g \le f_g'' \le L_g$ on I, with $u_g^* \in I$ and $f_g'(u_g^*) = 0$. Define*

$$a_g := w_g \mu_g, \qquad b_g := w_g L_g, \tag{239}$$

$$A := \sum_g a_g, \qquad B := \sum_g b_g, \tag{240}$$

$$u_a := A^{-1} \sum_g a_g u_g^*, \qquad u_b := B^{-1} \sum_g b_g u_g^*, \tag{241}$$

$$V_a := \sum_g a_g (u_g^* - u_a)^2, \qquad V_b := \sum_g b_g (u_g^* - u_b)^2, \tag{242}$$

$$\Delta_{\mathrm{grp}} := \min_{u \in I} \sum_g w_g f_g(u) - \sum_g w_g f_g(u_g^*). \tag{243}$$

Then the sharp bounds are

$$\tfrac{1}{2} V_a \le \Delta_{\mathrm{grp}} \le \tfrac{1}{2} V_b, \tag{244}$$

$$V_a = A^{-1} \sum_{g<h} a_g a_h (u_g^* - u_h^*)^2. \tag{245}$$

Writing $\bar{u} = \sum_g w_g u_g^*$ and $V_* = \sum_g w_g (u_g^* - \bar{u})^2$ recovers

$$\tfrac{1}{2} (\min_g \mu_g) V_* \le \tfrac{1}{2} V_a \le \Delta_{\mathrm{grp}} \le \tfrac{1}{2} V_b \le \tfrac{1}{2} (\max_g L_g) V_*. \tag{246}$$

Both endpoints are attained by admissible quadratic profiles; if $\mu_g = L_g$ for every group, the lower and upper bounds coincide exactly.

Proof. Taylor's integral formula at u_g^* gives

$$\tfrac{1}{2} \mu_g (u - u_g^*)^2 \le f_g(u) - f_g(u_g^*) \le \tfrac{1}{2} L_g (u - u_g^*)^2. \tag{247}$$

The weighted projection identity yields

$$\sum_g a_g (u - u_g^*)^2 = A(u - u_a)^2 + V_a, \tag{248}$$

$$A V_a = A \sum_g a_g (u_g^*)^2 - \left(\sum_g a_g u_g^* \right)^2 = \sum_{g<h} a_g a_h (u_g^* - u_h^*)^2. \tag{249}$$

Minimizing the lower quadratic and evaluating the upper quadratic at $u_b \in I$ prove Eq. (244). Finally,

$$V_a = \min_u \sum_g w_g \mu_g (u - u_g^*)^2 \ge (\min_g \mu_g) V_*, \tag{250}$$

$$V_b \le \sum_g w_g L_g (\bar{u} - u_g^*)^2 \le (\max_g L_g) V_*. \tag{251}$$

The profiles $f_g(u) = \mu_g (u - u_g^*)^2/2 + c_g$ attain the lower endpoint; replacing μ_g by L_g attains the upper endpoint. □

The bound is a weighted distance to the shared-curvature subspace. Keeping the individual Hessian bounds weights disagreement by each group's sensitivity, rather than charging every group the smallest μ_g or largest L_g. The pairwise formula shows which incompatible group preferences contribute to the approximation cost, and the quadratic equality cases establish exactly which constants are sharp.

Corollary A.28 (Exact quadratic profile)**.** *For $f_g(u) = a_g(u - u_g^*)^2/2 + c_g$, $a_g > 0$,*

$$u^*_{\text{shared}} = \frac{\sum_g w_g a_g u_g^*}{\sum_g w_g a_g}, \tag{252}$$

$$\Delta_{\text{grp}} = \frac{1}{2}\sum_g w_g a_g (u_g^* - u^*_{\text{shared}})^2. \tag{253}$$

Proof.

$$\frac{d}{du}\sum_g w_g f_g(u) = u\sum_g w_g a_g - \sum_g w_g a_g u_g^*, \tag{254}$$

$$\frac{d^2}{du^2}\sum_g w_g f_g(u) = \sum_g w_g a_g > 0. \tag{255}$$

□

A.4. Product descent and blockwise stationarity

We first treat the update directions as arbitrary inexact directions. This separates a geometric descent statement from the stochastic and momentum arguments needed to control its errors. The representation step and curvature step enter with separate weights, which must be retained when converting a summed descent budget into stationarity rates.

Assumption A.29 (Smooth objective)**.** *Let $\mathcal{M} = \mathcal{Q} \times \mathbb{R}^G$ with product metric $g_1 \oplus I$. For each update segment,*

$$F \geq F_* > -\infty, \qquad \|\operatorname{Hess} F\|_{\text{op}} \leq L, \tag{256}$$

$$g_t = \operatorname{grad}_q F(q_t, u_t), \qquad b_t = \nabla_u F(q_t, u_t), \tag{257}$$

$$q_{t+1} = \operatorname{Exp}_{q_t}(-\eta_t a_t), \qquad u_{t+1} = u_t - \rho_t h_t, \tag{258}$$

$$e_t = a_t - g_t, \qquad r_t = h_t - b_t, \tag{259}$$

$$\Delta_F = F(q_0, u_0) - F_*. \tag{260}$$

Lemma A.30 (Integral product descent)**.**

$$F(q_{t+1}, u_{t+1}) \leq F(q_t, u_t) - \eta_t\langle g_t, a_t\rangle - \rho_t\langle b_t, h_t\rangle \tag{261}$$

$$+ \tfrac{L}{2}\left(\eta_t^2\|a_t\|^2 + \rho_t^2\|h_t\|^2\right). \tag{262}$$

Proof. For $\Gamma(s) = (\operatorname{Exp}_{q_t}(-s\eta_t a_t), u_t - s\rho_t h_t)$,

$$\varphi(s) = F(\Gamma(s)), \tag{263}$$

$$\varphi'(0) = -\eta_t\langle g_t, a_t\rangle - \rho_t\langle b_t, h_t\rangle, \tag{264}$$

$$\varphi''(s) = \operatorname{Hess} F(\Gamma(s))[\dot\Gamma(s), \dot\Gamma(s)], \tag{265}$$

$$\|\dot\Gamma(s)\|^2 = \eta_t^2\|a_t\|^2 + \rho_t^2\|h_t\|^2, \tag{266}$$

$$\varphi(1) - \varphi(0) = \varphi'(0) + \int_0^1 (1-s)\varphi''(s)\,ds. \tag{267}$$

□

Theorem A.31 (Error-explicit stationarity budget)**.** *For $0 < \eta_t, \rho_t \leq 1/(4L)$,*

$$\sum_{t<T}\left(\frac{\eta_t}{2}\mathbb{E}\|g_t\|^2 + \frac{\rho_t}{2}\mathbb{E}\|b_t\|^2\right) \leq \Delta_F + \frac{5}{4}\sum_{t<T}\left(\eta_t\mathbb{E}\|e_t\|^2 + \rho_t\mathbb{E}\|r_t\|^2\right). \tag{268}$$

If $F(z_{t+1}^{\text{impl}}) \leq F(z_{t+1}^{\text{exact}}) + \varepsilon_t$, add $\sum_t \mathbb{E}\varepsilon_t$ to the right-hand side.

Proof.

$$-\langle g, g+e\rangle = -\|g\|^2 - \langle g, e\rangle \le -\tfrac{3}{4}\|g\|^2 + \|e\|^2, \tag{269}$$

$$\|g+e\|^2 \le 2\|g\|^2 + 2\|e\|^2, \tag{270}$$

$$-\eta\langle g, g+e\rangle + \tfrac{L}{2}\eta^2\|g+e\|^2 \le -\eta(\tfrac{3}{4} - L\eta)\|g\|^2 + \eta(1+L\eta)\|e\|^2 \tag{271}$$

$$\le -\tfrac{\eta}{2}\|g\|^2 + \tfrac{5}{4}\eta\|e\|^2, \tag{272}$$

$$\mathbb{E}F_{t+1} - \mathbb{E}F_t \le -\tfrac{\eta_t}{2}\mathbb{E}\|g_t\|^2 - \tfrac{\rho_t}{2}\mathbb{E}\|b_t\|^2 \tag{273}$$

$$+ \tfrac{5}{4}\eta_t\mathbb{E}\|e_t\|^2 + \tfrac{5}{4}\rho_t\mathbb{E}\|r_t\|^2, \tag{274}$$

$$\sum_{t<T}(\mathbb{E}F_t - \mathbb{E}F_{t+1}) = F_0 - \mathbb{E}F_T \le \Delta_F. \tag{275}$$

□

Corollary A.32 (Distinct weighted output laws)**.** *Let $S_q = \sum_t \eta_t$, $S_u = \sum_t \rho_t$, and let B_T denote the right-hand side of Eq. (268). For*

$$\Pr(R_q = t) = \eta_t/S_q, \qquad \Pr(R_u = t) = \rho_t/S_u, \tag{276}$$

$$\mathbb{E}\|g_{R_q}\|^2 \le 2B_T/S_q, \qquad \mathbb{E}\|b_{R_u}\|^2 \le 2B_T/S_u. \tag{277}$$

For constant horizon steps,

$$T^{-1}\sum_{t<T}\mathbb{E}\|g_t\|^2 \le 2B_T/(T\eta), \tag{278}$$

$$T^{-1}\sum_{t<T}\mathbb{E}\|b_t\|^2 \le 2B_T/(T\rho). \tag{279}$$

Proof.

$$\mathbb{E}\|g_{R_q}\|^2 = S_q^{-1}\sum_t \eta_t\mathbb{E}\|g_t\|^2, \tag{280}$$

$$\mathbb{E}\|b_{R_u}\|^2 = S_u^{-1}\sum_t \rho_t\mathbb{E}\|b_t\|^2. \tag{281}$$

□

The two output laws express what the budget actually controls. Dividing the curvature term by the representation step would give an unsupported rate when the steps differ. The predictable-preconditioner result below is a useful baseline because conditional unbiasedness cancels its linear noise term; the nonlinear normalized-momentum update requires the separate tracking analysis in Section A.5.

Assumption A.33 (Predictable preconditioning)**.** *For a pre-sampling filtration $\mathcal{F}_t$,*

$$A_t, B_t \in \mathcal{F}_t, \qquad a_-I \preceq A_t \preceq a_+I, \qquad b_-I \preceq B_t \preceq b_+I, \tag{282}$$

$$\mathbb{E}_t\widehat{g}_t = g_t, \qquad \mathbb{E}_t\widehat{b}_t = b_t, \tag{283}$$

$$\mathbb{E}_t\|\widehat{g}_t - g_t\|^2 \le \sigma_q^2/\mathfrak{b}_t, \qquad \mathbb{E}_t\|\widehat{b}_t - b_t\|^2 \le \sigma_u^2/\mathfrak{b}_t, \tag{284}$$

$$a_t = A_t\widehat{g}_t, \qquad h_t = B_t\widehat{b}_t. \tag{285}$$

Theorem A.34 (Unbiased preconditioned block budget)**.** *If $L\eta_t a_+^2 \le a_-$ and $L\rho_t b_+^2 \le b_-$, then*

$$\sum_{t<T}\left(\tfrac{a_-\eta_t}{2}\mathbb{E}\|g_t\|^2 + \tfrac{b_-\rho_t}{2}\mathbb{E}\|b_t\|^2\right) \le \Delta_F + \tfrac{L}{2}\sum_{t<T}\frac{a_+^2\eta_t^2\sigma_q^2 + b_+^2\rho_t^2\sigma_u^2}{\mathfrak{b}_t}. \tag{286}$$

Proof.

$$\mathbb{E}_t\langle g_t, A_t\widehat{g}_t\rangle = \langle g_t, A_t g_t\rangle \geq a_-\|g_t\|^2, \tag{287}$$

$$\mathbb{E}_t\|A_t\widehat{g}_t\|^2 \leq a_+^2\mathbb{E}_t\|\widehat{g}_t\|^2 \leq a_+^2(\|g_t\|^2 + \sigma_q^2/\mathfrak{b}_t), \tag{288}$$

$$\mathbb{E}_t\langle b_t, B_t\widehat{b}_t\rangle \geq b_-\|b_t\|^2, \tag{289}$$

$$\mathbb{E}_t F_{t+1} - F_t \leq -\eta_t(a_- - La_+^2\eta_t/2)\|g_t\|^2 \tag{290}$$

$$- \rho_t(b_- - Lb_+^2\rho_t/2)\|b_t\|^2 + \frac{L(a_+^2\eta_t^2\sigma_q^2 + b_+^2\rho_t^2\sigma_u^2)}{2\mathfrak{b}_t}. \tag{291}$$

□

Corollary A.35 (Unequal steps)**.** *For fixed batch size, $\eta = c_\eta T^{-1/2}$, $\rho = c_\rho T^{-3/4}$,*

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\|g_t\|^2 = O(T^{-1/2}), \qquad \frac{1}{T}\sum_{t<T}\mathbb{E}\|b_t\|^2 = O(T^{-1/4}). \tag{292}$$

For $\eta, \rho \asymp T^{-1/2}$, both bounds are $O(T^{-1/2})$.

Proof.

$$\sum_t \eta^2 = c_\eta^2, \qquad \sum_t \rho^2 = c_\rho^2 T^{-1/2}, \tag{293}$$

$$T\eta = c_\eta T^{1/2}, \qquad T\rho = c_\rho T^{1/4}. \tag{294}$$

□

Proposition A.36 (Sharp obstruction for the uniform curvature average)**.** *For $f(u) = (u-1)^2/2$, $u_0 = 0$, $u_{t+1} = u_t - \rho f'(u_t)$, $\rho = T^{-3/4}$,*

$$u_t - 1 = -(1-\rho)^t, \tag{295}$$

$$\frac{1}{T}\sum_{t<T}|f'(u_t)|^2 = \frac{1-(1-\rho)^{2T}}{T(2\rho-\rho^2)} \sim \frac{1}{2T\rho} = \tfrac{1}{2}T^{-1/4}. \tag{296}$$

Proof.

$$(1-\rho)^{2T} \leq e^{-2T\rho} = e^{-2T^{1/4}} \longrightarrow 0, \tag{297}$$

$$(2\rho-\rho^2)/(2\rho) \longrightarrow 1. \tag{298}$$

□

A.5. Full normalized momentum: tracking, descent, and finite-time rates

Momentum estimates a moving preconditioned gradient, rather than a fixed vector. The target changes through both representation motion and curvature adaptation, and successive estimates belong to different tangent spaces. Parallel transport permits an exact subtraction; the resulting recursion separates memory decay, target drift, and fresh sampling noise.

Assumption A.37 (Canonical stochastic dynamics)**.** *For $\mathcal{F}_t$ generated before the current sample,*

$$\mathbb{E}_t\widehat{g}_t = g_t, \qquad \mathbb{E}_t\widehat{b}_t = b_t, \tag{299}$$

$$\mathbb{E}_t\|\widehat{g}_t - g_t\|^2 \leq \sigma_q^2/\mathfrak{b}_t, \qquad \mathbb{E}_t\|\widehat{b}_t - b_t\|^2 \leq \sigma_u^2/\mathfrak{b}_t, \tag{300}$$

$$\|\widehat{g}_t\| \leq G_0, \qquad \|\widehat{b}_t\| \leq H_0 \quad a.s., \tag{301}$$

$$A_t = A(u_t), \qquad a_- I \preceq A_t \preceq a_+ I, \tag{302}$$

$$p_t = \beta P_{t-1}p_{t-1} + (1-\beta)A_t\widehat{g}_t, \qquad 0 \leq \beta < 1, \tag{303}$$

$$q_{t+1} = \mathrm{Exp}_{q_t}[-\eta_t N_\gamma(p_t)], \qquad u_{t+1} = u_t - \rho_t\widehat{b}_t. \tag{304}$$

Here $P_{t-1} = P^1_{q_{t-1}\to q_t}$; compatible cross-curvature transport has already been applied. Assume along the relevant paths

$$\|g(q,u) - P_{q'\to q} g(q',u)\| \le L_{qq} d_1(q,q'), \tag{305}$$
$$\|g(q,u) - g(q,u')\| \le L_{qu}\|u - u'\|. \tag{306}$$

The full product smoothness assumption holds and every update is interior and exact.

Proposition A.38 (Canonical realization of the physical update)**.**

$$\widetilde{m}_{g,t} = \beta m^{\mathrm{carry}}_{g,t} + (1-\beta)\widehat{g}^x_{g,t}, \tag{307}$$
$$p_{g,t} = \sqrt{K_{g,t}}\widetilde{m}_{g,t} = \beta p^{\mathrm{carry}}_{g,t} + (1-\beta)K_{g,t}\widehat{g}^q_{g,t}, \tag{308}$$
$$\sqrt{K_{g,t}}\,\mathrm{Exp}^{K_{g,t}}_{x_{g,t}}\left(\frac{-\eta_t \widetilde{m}_{g,t}}{1+\gamma\sqrt{K_{g,t}}\|\widetilde{m}_{g,t}\|}\right) = \mathrm{Exp}^1_{q_{g,t}}[-\eta_t N_\gamma(p_{g,t})], \tag{309}$$
$$\sqrt{K_{g,t+1}}\, m^{\mathrm{carry}}_{g,t+1} = P^1_{q_{g,t}\to q_{g,t+1}} p_{g,t}. \tag{310}$$

Proof. Use Corollary A.7 and Proposition A.5 in each line. □

Lemma A.39 (Pathwise momentum invariant)**.** *Let $M = \max(\|p_{-1}\|, a_+ G_0)$. Then*

$$\|p_t\| \le M, \qquad \|N_\gamma(p_t)\| \le M_N, \tag{311}$$
$$M_N := \min(M, \sqrt{\nu}/\gamma), \qquad \|m_{g,t}\| \le M/\sqrt{K_-}, \tag{312}$$

where ν is the number of separately normalized tangent blocks ($\nu = 1$ for global normalization).

Proof.

$$\|p_t\| \le \beta\|p_{t-1}\| + (1-\beta)a_+ G_0, \tag{313}$$
$$\|p_t\| \le \beta^{t+1}\|p_{-1}\| + (1-\beta^{t+1})a_+ G_0 \le M, \tag{314}$$
$$\|N_\gamma(p_t)\|^2 = \sum_{j=1}^{\nu} \frac{\|p_{t,j}\|^2}{(1+\gamma\|p_{t,j}\|)^2} \le \min(M^2, \nu/\gamma^2). \tag{315}$$

□

Definition A.40 (Tracking variables)**.**

$$v_t = A_t g_t, \qquad \xi_t = \widehat{g}_t - g_t, \qquad e^m_t = p_t - v_t, \tag{316}$$
$$d_t = v_t - P_{t-1}v_{t-1} \quad (t \ge 1), \qquad d_0 = 0, \tag{317}$$
$$v_{-1} = v_0, \qquad P_{-1} = I, \qquad e^m_{-1} = p_{-1} - v_0. \tag{318}$$

Lemma A.41 (Exact error recursion)**.**

$$e^m_t = \beta P_{t-1} e^m_{t-1} - \beta d_t + (1-\beta)A_t\xi_t, \tag{319}$$
$$\mathbb{E}\|e^m_t\|^2 \le \beta\mathbb{E}\|e^m_{t-1}\|^2 + \frac{\beta^2}{1-\beta}\mathbb{E}\|d_t\|^2 + \frac{(1-\beta)^2 a_+^2\sigma_q^2}{\mathfrak{b}_t}, \tag{320}$$
$$\sum_{t<T}\mathbb{E}\|e^m_t\|^2 \le \frac{\beta}{1-\beta}\mathbb{E}\|e^m_{-1}\|^2 + \frac{\beta^2}{(1-\beta)^2}\sum_{t<T}\mathbb{E}\|d_t\|^2 + (1-\beta)a_+^2\sigma_q^2\sum_{t<T}\mathfrak{b}_t^{-1}. \tag{321}$$

Proof.

$$p_t - v_t = \beta P_{t-1}(p_{t-1} - v_{t-1}) + \beta(P_{t-1}v_{t-1} - v_t) + (1-\beta)A_t\xi_t, \tag{322}$$

$$\mathbb{E}_t\langle \beta P_{t-1}e_{t-1}^m - \beta d_t, A_t\xi_t\rangle = 0, \tag{323}$$

$$\|\beta a - \beta b\|^2 \le \beta\|a\|^2 + \frac{\beta^2}{1-\beta}\|b\|^2, \tag{324}$$

$$(1-\beta)\sum_{t<T}\mathbb{E}\|e_t^m\|^2 \le \beta\mathbb{E}\|e_{-1}^m\|^2 - \beta\mathbb{E}\|e_{T-1}^m\|^2 \tag{325}$$

$$+ \frac{\beta^2}{1-\beta}\sum_{t<T}\mathbb{E}\|d_t\|^2 + (1-\beta)^2 a_+^2\sigma_q^2\sum_{t<T}\mathfrak{b}_t^{-1}. \tag{326}$$

The case $\beta = 0$ follows directly from $e_t^m = A_t\xi_t$. □

The factor $(1-\beta)$ averages fresh noise, whereas target motion persists through the momentum memory. This explains the opposite dependence of variance and drift on $1-\beta$ in the summed tracking bound. Curvature enters twice: through the objective gradient and through $A(u)$, so a bound on $\nabla_u F$ alone is insufficient to control this drift.

Lemma A.42 (Representation–curvature drift)**.** *For $A(u) = \mathrm{diag}(e^{u_g}I)$, set*

$$L_{vq} = K_+L_{qq}, \qquad L_{vu} = K_+(L_{qu} + G_0). \tag{327}$$

Then

$$\|d_t\| \le L_{vq}d_1(q_t, q_{t-1}) + L_{vu}\|u_t - u_{t-1}\| \tag{328}$$

$$\le L_{vq}M_N\eta_{t-1} + L_{vu}H_0\rho_{t-1}, \tag{329}$$

$$D_T := \sum_{t<T}\mathbb{E}\|d_t\|^2 \le \sum_{t<T-1}(L_{vq}M_N\eta_t + L_{vu}H_0\rho_t)^2 =: \overline{D}_T \tag{330}$$

$$\le 2L_{vq}^2M_N^2\sum_{t<T-1}\eta_t^2 + 2L_{vu}^2H_0^2\sum_{t<T-1}\rho_t^2. \tag{331}$$

Proof.

$$v_t - P_{t-1}v_{t-1} = A_t(g_t - P_{t-1}g_{t-1}) + (A_t - A_{t-1})P_{t-1}g_{t-1}, \tag{332}$$

$$\|A_t - A_{t-1}\|_{\mathrm{op}} \le K_+\|u_t - u_{t-1}\|_\infty \le K_+\|u_t - u_{t-1}\|, \tag{333}$$

$$d_1(q_t, q_{t-1}) = \eta_{t-1}\|N_\gamma(p_{t-1})\| \le \eta_{t-1}M_N, \tag{334}$$

$$\|u_t - u_{t-1}\| \le \rho_{t-1}H_0, \tag{335}$$

$$(a+b)^2 \le 2a^2 + 2b^2. \tag{336}$$

□

Theorem A.43 (Finite-horizon martingale filter bound)**.** *Let $E_T = \sum_{t<T}\mathbb{E}\|e_t^m\|^2$, and define*

$$\mathcal{I}_T := \frac{\beta^2(1-\beta^{2T})}{1-\beta^2}\mathbb{E}\|e_{-1}^m\|^2, \tag{337}$$

$$\mathcal{V}_T := \frac{1-\beta}{1+\beta}\sum_{j<T}(1-\beta^{2(T-j)})\frac{a_+^2\sigma_q^2}{\mathfrak{b}_j}, \tag{338}$$

$$(L_{\beta,T})_{tj} := \beta^{t-j+1}\mathbf{1}_{\{1\le j\le t\}}, \quad 0 \le t < T,\ 1 \le j < T, \tag{339}$$

$$h_{\beta,T} := \|L_{\beta,T}\|_{2\to 2} \le \beta\sum_{k=0}^{T-2}\beta^k. \tag{340}$$

For $T = 1$, set $h_{\beta,1} = 0$. Then

$$E_T \le \left(\sqrt{\mathcal{I}_T + \mathcal{V}_T} + h_{\beta,T}\sqrt{D_T}\right)^2. \tag{341}$$

For a stationary target, zero initial error, and a scalar oracle whose variance saturates the stated bound, $E_T = \mathcal{V}_T$. Thus the stationary variance factor $(1-\beta)/(1+\beta)$, including its finite-horizon weights, is exact.

Proof. Use the predictable isometric frame

$$U_0 = I, \qquad U_t = P_{t-1}U_{t-1}, \tag{342}$$

$$\widetilde{e}_t = U_t^{-1}e_t^m, \qquad \widetilde{d}_t = U_t^{-1}d_t, \qquad \zeta_t = U_t^{-1}A_t\xi_t. \tag{343}$$

Because U_t and A_t are $\mathcal{F}_t$-measurable, $\mathbb{E}_t\zeta_t = 0$ in the fixed Hilbert space $T_{q_0}\mathcal{Q}$. The tracking identity becomes

$$\widetilde{e}_t = \underbrace{\beta^{t+1}e_{-1}^m + (1-\beta)\sum_{j\le t}\beta^{t-j}\zeta_j}_{Y_t} - \sum_{1\le j\le t}\beta^{t-j+1}\widetilde{d}_j. \tag{344}$$

Martingale orthogonality (Williams, 1991) gives

$$\sum_{t<T}\mathbb{E}\|Y_t\|^2 = \mathcal{I}_T + (1-\beta)^2\sum_{j<T}\left(\sum_{t=j}^{T-1}\beta^{2(t-j)}\right)\mathbb{E}\|\zeta_j\|^2 \tag{345}$$

$$\le \mathcal{I}_T + \mathcal{V}_T. \tag{346}$$

Minkowski's inequality in $\ell_2(L^2)$ now yields

$$E_T^{1/2} \le \left(\sum_{t<T}\mathbb{E}\|Y_t\|^2\right)^{1/2} + \|L_{\beta,T}\|_{2\to2}\left(\sum_{j=1}^{T-1}\mathbb{E}\|\widetilde{d}_j\|^2\right)^{1/2}. \tag{347}$$

Young's convolution inequality (Grafakos, 2014) bounds the operator norm by the ℓ_1 norm of its truncated kernel, giving the displayed upper bound for $h_{\beta,T}$. If $d_t = 0$ and $e_{-1}^m = 0$, only the orthogonal noise sum remains, proving the equality case. □

The useful step is the predictable frame: it puts all noise increments in one fixed Hilbert space before invoking orthogonality. Direct transport of each past increment to the final random tangent space would make the transporting operator depend on future noise. The filter bound keeps the exact geometric noise weights and the finite-horizon drift gain, instead of applying Young's scalar inequality separately at every time.

Definition A.44 (Explicit tracking bounds)**.**

$$E_T := \sum_{t<T}\mathbb{E}\|e_t^m\|^2, \tag{348}$$

$$\mathscr{E}_T := \frac{\beta}{1-\beta}\mathbb{E}\|e_{-1}^m\|^2 + \frac{\beta^2\overline{D}_T}{(1-\beta)^2} \tag{349}$$

$$+ (1-\beta)a_+^2\sigma_q^2\sum_{t<T}\mathfrak{b}_t^{-1}, \tag{350}$$

$$\mathscr{E}_T^\sharp := \min\left\{\mathscr{E}_T, \left(\sqrt{\mathcal{I}_T + \mathcal{V}_T} + h_{\beta,T}\sqrt{\overline{D}_T}\right)^2\right\}. \tag{351}$$

Thus $E_T \le \mathscr{E}_T^\sharp \le \mathscr{E}_T$.

Theorem A.45 (Normalized two-block momentum budget)**.** *Under Assumption A.37, put*

$$a = \begin{cases} \dfrac{a_-}{1+\gamma a_- G_0}, & \text{scalar preconditioner on each normalization block,} \\ \dfrac{a_-}{1+\gamma a_+ G_0}, & \text{otherwise,} \end{cases} \tag{352}$$

$$C_e = a^{-1} + L\eta_{\max}. \tag{353}$$

If $\eta_t \le a/(4La_+^2)$ and $\rho_t \le L^{-1}$, then

$$\sum_{t<T}\Big(\frac{a\eta_t}{2}\mathbb{E}\|g_t\|^2 + \frac{\rho_t}{2}\mathbb{E}\|b_t\|^2\Big) \le \Delta_F + C_e\sum_{t<T}\eta_t\mathbb{E}\|e_t^m\|^2 + \frac{L\sigma_u^2}{2}\sum_{t<T}\frac{\rho_t^2}{\mathfrak{b}_t}. \tag{354}$$

For constant steps,

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\|g_t\|^2 \le \frac{2\Delta_F}{aT\eta} + \frac{2C_e\mathscr{E}_T^\sharp}{aT} + \frac{L\sigma_u^2\rho^2}{a\eta T}\sum_{t<T}\mathfrak{b}_t^{-1}, \tag{355}$$

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\|b_t\|^2 \le \frac{2\Delta_F}{T\rho} + \frac{2C_e\eta\mathscr{E}_T^\sharp}{T\rho} + \frac{L\sigma_u^2\rho}{T}\sum_{t<T}\mathfrak{b}_t^{-1}. \tag{356}$$

Proof. Let $n_t = N_\gamma(p_t)$, $n_t^* = N_\gamma(v_t)$ and $\delta_t = n_t - n_t^*$.

$$\|\delta_t\| \le \|e_t^m\|, \tag{357}$$

$$\langle g_t, n_t\rangle = \langle g_t, n_t^*\rangle + \langle g_t, \delta_t\rangle \tag{358}$$

$$\ge a\|g_t\|^2 - \|g_t\|\|e_t^m\| \tag{359}$$

$$\ge \tfrac{3a}{4}\|g_t\|^2 - a^{-1}\|e_t^m\|^2, \tag{360}$$

$$\|n_t\|^2 \le 2\|n_t^*\|^2 + 2\|\delta_t\|^2 \le 2a_+^2\|g_t\|^2 + 2\|e_t^m\|^2, \tag{361}$$

$$-\eta_t\langle g_t, n_t\rangle + \tfrac{L}{2}\eta_t^2\|n_t\|^2 \le -\eta_t(3a/4 - La_+^2\eta_t)\|g_t\|^2 + \eta_t(a^{-1} + L\eta_t)\|e_t^m\|^2 \tag{362}$$

$$\le -\tfrac{a\eta_t}{2}\|g_t\|^2 + C_e\eta_t\|e_t^m\|^2, \tag{363}$$

$$\mathbb{E}_t[-\rho_t\langle b_t, \widehat{b}_t\rangle + \tfrac{L}{2}\rho_t^2\|\widehat{b}_t\|^2] \le -\rho_t(1 - L\rho_t/2)\|b_t\|^2 + \frac{L\rho_t^2\sigma_u^2}{2\mathfrak{b}_t} \tag{364}$$

$$\le -\tfrac{\rho_t}{2}\|b_t\|^2 + \frac{L\rho_t^2\sigma_u^2}{2\mathfrak{b}_t}. \tag{365}$$

Combining these inequalities with Lemma A.30,

$$\mathbb{E}F_{t+1} - \mathbb{E}F_t \le -\tfrac{a\eta_t}{2}\mathbb{E}\|g_t\|^2 - \tfrac{\rho_t}{2}\mathbb{E}\|b_t\|^2 + C_e\eta_t\mathbb{E}\|e_t^m\|^2 + \frac{L\rho_t^2\sigma_u^2}{2\mathfrak{b}_t}, \tag{366}$$

$$\sum_{t<T}(\mathbb{E}F_t - \mathbb{E}F_{t+1}) \le \Delta_F, \tag{367}$$

$$\eta\sum_{t<T}\mathbb{E}\|e_t^m\|^2 \le \eta\mathscr{E}_T^\sharp. \tag{368}$$

□

Corollary A.46 (Increasing-batch unequal-step regime)**.** *For fixed $\beta < 1$ and constants satisfying the step restrictions, let*

$$\eta = c_\eta T^{-1/2}, \qquad \rho = c_\rho T^{-3/4}, \qquad \mathfrak{b}_t \ge c_b T^{1/2}. \tag{369}$$

Then

$$\mathscr{E}_T/T \le C_0T^{-1} + C_1T^{-1} + C_2T^{-3/2} + C_3T^{-1/2}, \tag{370}$$

$$T^{-1}\sum_{t<T}\mathbb{E}\|g_t\|^2 = O(T^{-1/2}), \tag{371}$$

$$T^{-1}\sum_{t<T}\mathbb{E}\|b_t\|^2 = O(T^{-1/4}). \tag{372}$$

The same rates hold for deterministic gradients with fixed β.

Proof.

$$\frac{\eta}{\rho} = \frac{c_\eta}{c_\rho} T^{1/4}, \qquad \frac{1}{T\eta} = c_\eta^{-1} T^{-1/2}, \qquad \frac{1}{T\rho} = c_\rho^{-1} T^{-1/4}, \tag{373}$$

$$\frac{\rho^2}{\eta T} \sum_t \mathfrak{b}_t^{-1} = O(T^{-3/2}), \qquad \frac{\rho}{T} \sum_t \mathfrak{b}_t^{-1} = O(T^{-5/4}), \tag{374}$$

$$\frac{\eta \mathscr{E}_T}{T\rho} = O(T^{-1/4}). \tag{375}$$

□

The increasing-batch regime makes the variance term vanish while keeping the memory parameter fixed. With a fixed batch, the same proof requires the memory parameter to change with the horizon. The exponent calculation below displays this trade-off explicitly; its sufficient schedule is not an optimality claim or a statement that a fixed-step benchmark run realizes these asymptotics.

Corollary A.47 (General horizon exponents)**.** *Let $\eta \asymp T^{-\alpha}$, $\rho \asymp T^{-\theta}$, $1-\beta \asymp T^{-\kappa}$ and $\mathfrak{b}_t \asymp T^{\omega}$, with $0 < \alpha, \theta < 1$, $\kappa, \omega \geq 0$. Then*

$$\mathscr{E}_T/T = O\left(T^{-1+\kappa} + T^{-2\alpha+2\kappa} + T^{-2\theta+2\kappa} + T^{-\kappa-\omega}\right), \tag{376}$$

$$\frac{1}{T}\sum_t \mathbb{E}\|g_t\|^2 = O\left(T^{\alpha-1} + \frac{\mathscr{E}_T}{T} + T^{-2\theta+\alpha-\omega}\right), \tag{377}$$

$$\frac{1}{T}\sum_t \mathbb{E}\|b_t\|^2 = O\left(T^{\theta-1} + T^{\theta-\alpha}\frac{\mathscr{E}_T}{T} + T^{-\theta-\omega}\right). \tag{378}$$

Proof. Substitute the four horizon powers into the explicit inequalities of Theorem A.45; C_e and the pathwise bound M remain bounded independently of T. □

Corollary A.48 (A fixed-batch sufficient schedule)**.** *For $\omega = 0$, $\alpha = \theta = 3/5$, $\kappa = 2/5$,*

$$\mathscr{E}_T/T = O(T^{-2/5}), \tag{379}$$

$$\frac{1}{T}\sum_t \mathbb{E}(\|g_t\|^2 + \|b_t\|^2) = O(T^{-2/5}). \tag{380}$$

For fixed β and fixed batch, the preceding tracking argument retains the term

$$(1-\beta)a_+^2\sigma_q^2/\mathfrak{b}, \tag{381}$$

which is not set to zero.

Proof.

$$-1+\kappa = -3/5, \qquad -2\alpha+2\kappa = -2/5, \tag{382}$$

$$-2\theta+2\kappa = -2/5, \qquad -\kappa-\omega = -2/5, \tag{383}$$

$$\alpha - 1 = \theta - 1 = -2/5, \qquad \theta - \alpha = 0. \tag{384}$$

□

Proposition A.49 (PL tracking with a drifting curvature)**.** *Assume $\|g(q,u)\|^2 \geq 2\mu[F(q,u) - F^*(u)]$ and*

$$|F(q,u') - F(q,u)| \leq H\|u' - u\|, \tag{385}$$

$$|F^*(u') - F^*(u)| \leq H_*\|u' - u\|. \tag{386}$$

For $\mathcal{D}_t = F(q_t, u_t) - F^*(u_t)$ and $0 < a\mu\eta_t \le 1$,

$$\mathbb{E}\mathcal{D}_{t+1} \le (1 - a\mu\eta_t)\mathbb{E}\mathcal{D}_t + C_e\eta_t\mathbb{E}\|e_t^m\|^2 + (H + H_*)\mathbb{E}\|u_{t+1} - u_t\|. \tag{387}$$

For constant steps and $\sup_t \mathbb{E}\|e_t^m\|^2 \le Z^2$,

$$\mathbb{E}\mathcal{D}_T \le (1 - a\mu\eta)^T\mathcal{D}_0 + \frac{C_e Z^2}{a\mu} + \frac{(H + H_*)H_0\rho}{a\mu\eta}. \tag{388}$$

Proof.

$$F(q_{t+1}, u_t) - F^*(u_t) \le \mathcal{D}_t - \tfrac{a\eta_t}{2}\|g_t\|^2 + C_e\eta_t\|e_t^m\|^2, \tag{389}$$

$$\mathcal{D}_{t+1} - [F(q_{t+1}, u_t) - F^*(u_t)] \le (H + H_*)\|u_{t+1} - u_t\|, \tag{390}$$

$$\sum_{j=0}^{T-1}(1 - a\mu\eta)^j \le (a\mu\eta)^{-1}. \tag{391}$$

□

A.6. Constrained curvature, coordinate error, and sequential coupling

Curvature bounds change the appropriate optimality condition. At a boundary optimum, the loss can decrease only in an infeasible direction, leaving a nonzero raw derivative. We therefore analyze a projected gradient mapping, and then account for the softplus coordinate, step caps, and the fact that the curvature derivative may be evaluated before the representation moves.

Definition A.50 (Weighted projected residual)**.** *For a positive diagonal B, a box U, and $b = \nabla_u F(q, u)$,*

$$\Pi_U^B(z) := \arg\min_{v \in U}\|v - z\|_{B^{-1}}^2 = \Pi_U(z), \tag{392}$$

$$G_{\rho,B}(q, u) := \rho^{-1}\{u - \Pi_U(u - \rho Bb)\}, \tag{393}$$

$$\|v\|_{B^{-1}}^2 = v^\top B^{-1}v, \qquad \|v\|_B^2 = v^\top Bv. \tag{394}$$

Proposition A.51 (KKT equivalence)**.**

$$G_{\rho,B}(q, u) = 0 \iff \langle b, v - u\rangle \ge 0 \quad (\forall v \in U) \tag{395}$$

$$\iff 0 \in b + N_U(u). \tag{396}$$

Coordinatewise,

$$u_{g,-} < u_g < u_{g,+} \Longrightarrow b_g = 0, \tag{397}$$

$$u_g = u_{g,-} \Longrightarrow b_g \ge 0, \tag{398}$$

$$u_g = u_{g,+} \Longrightarrow b_g \le 0. \tag{399}$$

Proof.

$$u = \Pi_U^B(u - \rho Bb) \iff \langle B^{-1}(u - (u - \rho Bb)), v - u\rangle \ge 0 \tag{400}$$

$$\iff \rho\langle b, v - u\rangle \ge 0. \tag{401}$$

□

Proposition A.52 (Projected curvature descent)**.** *Let $u^+ = \Pi_U(u - \rho B[b + r])$ and $b_- I \preceq B \preceq b_+ I$. If $\rho \le 1/(2Lb_+)$, then*

$$F(q, u^+) \le F(q, u) - \frac{\rho}{4}\|G_{\rho,B}(q, u)\|_{B^{-1}}^2 + \frac{3\rho}{2}\|r\|_B^2. \tag{402}$$

Proof. Set $s = (u - u^+)/\rho$. Projection optimality gives

$$\langle b + r, s\rangle \geq \|s\|^2_{B^{-1}}, \tag{403}$$
$$F(q, u^+) - F(q, u) \leq -\rho\langle b, s\rangle + \tfrac{L}{2}\rho^2\|s\|^2 \tag{404}$$
$$\leq -\rho(1 - L\rho b_+/2)\|s\|^2_{B^{-1}} + \rho\langle r, s\rangle, \tag{405}$$
$$\langle r, s\rangle \leq \tfrac{1}{4}\|s\|^2_{B^{-1}} + \|r\|^2_B, \tag{406}$$
$$F(q, u^+) - F(q, u) \leq -\tfrac{\rho}{2}\|s\|^2_{B^{-1}} + \rho\|r\|^2_B, \tag{407}$$
$$\|s - G_{\rho,B}\|_{B^{-1}} \leq \|Br\|_{B^{-1}} = \|r\|_B, \tag{408}$$
$$\|G_{\rho,B}\|^2_{B^{-1}} \leq 2\|s\|^2_{B^{-1}} + 2\|r\|^2_B. \tag{409}$$

□

Corollary A.53 (Constrained joint budget)**.** *For the representation update of Theorem A.45, followed by the projected curvature update at q_{t+1},*

$$\sum_{t<T}\left[\tfrac{a\eta_t}{2}\mathbb{E}\|g_t\|^2 + \tfrac{\rho_t}{4}\mathbb{E}\|G_{\rho_t,B_t}(q_{t+1}, u_t)\|^2_{B_t^{-1}}\right] \leq \Delta_F + C_e\sum_t \eta_t\mathbb{E}\|z_t\|^2 + \tfrac{3}{2}\sum_t \rho_t\mathbb{E}\|r_t\|^2_{B_t}. \tag{410}$$

If $\widehat{b}_t$ is evaluated at q_t,

$$r_t = \widehat{b}_t - \nabla_u F(q_{t+1}, u_t), \tag{411}$$
$$\mathbb{E}\|r_t\|^2_{B_t} \leq 2b_+\sigma_u^2/\mathfrak{b}_t + 2b_+L_{uq}^2M_N^2\eta_t^2. \tag{412}$$

Proof.

$$r_t = (\widehat{b}_t - b_t) + [b_t - \nabla_u F(q_{t+1}, u_t)], \tag{413}$$
$$\|b_t - \nabla_u F(q_{t+1}, u_t)\| \leq L_{uq}d(q_t, q_{t+1}) \leq L_{uq}\eta_t M_N. \tag{414}$$

Add the representation and curvature descent inequalities and telescope. □

Lemma A.54 (Biased preconditioned curvature step)**.** *For $h_t = B_t\widehat{b}_t + \delta_t$, $\|\delta_t\| \leq d_t$ a.s., $\mathbb{E}_t\widehat{b}_t = b_t$, and $L\rho_t b_+^2 \leq b_-/4$,*

$$\mathbb{E}_t[-\rho_t\langle b_t, h_t\rangle + \tfrac{L}{2}\rho_t^2\|h_t\|^2] \leq -\frac{b_-\rho_t}{2}\|b_t\|^2 + \rho_t(b_-^{-1} + L\rho_t)d_t^2 + \frac{L\rho_t^2 b_+^2\sigma_u^2}{\mathfrak{b}_t}. \tag{415}$$

Proof.

$$-\langle b_t, \mathbb{E}_t h_t\rangle \leq -b_-\|b_t\|^2 + \|b_t\|d_t \leq -\tfrac{3b_-}{4}\|b_t\|^2 + b_-^{-1}d_t^2, \tag{416}$$
$$\mathbb{E}_t\|h_t\|^2 \leq 2b_+^2(\|b_t\|^2 + \sigma_u^2/\mathfrak{b}_t) + 2d_t^2. \tag{417}$$

□

Corollary A.55 (Softplus implementation remainder)**.** *For $u = \psi(c)$ and $c^+ = c - \rho\psi'(c)\widehat{b}$,*

$$B_t = \mathrm{diag}(\psi'(c_{g,t})^2), \tag{418}$$
$$u_{t+1} = u_t - \rho_t B_t\widehat{b}_t + R_t, \tag{419}$$
$$\|R_t\| \leq C_\psi\rho_t^2 H_0^2, \qquad C_\psi = B_\psi b_+/2, \tag{420}$$
$$|F(q, u + R_t) - F(q, u)| \leq H_F C_\psi\rho_t^2 H_0^2 + \tfrac{L}{2}C_\psi^2\rho_t^4 H_0^4 =: \varepsilon_t^c. \tag{421}$$

The unconstrained budget therefore receives the additional explicit term

$$\sum_t \mathbb{E}\varepsilon_t^c \leq H_F C_\psi H_0^2\sum_t \rho_t^2 + \tfrac{L}{2}C_\psi^2 H_0^4\sum_t \rho_t^4. \tag{422}$$

Proof. Apply Proposition A.12 componentwise and use $\|(\widehat{b}_g^2)_g\|_2 \le \sum_g \widehat{b}_g^2 \le H_0^2$. □

Proposition A.56 (Log-step clipping)**.** *For a scalar coordinate and $\Delta > 0$, define*

$$T_{\rho,\Delta}(u,h) = \Pi_U\{u + \mathrm{clip}(-\rho h, -\Delta, \Delta)\}, \tag{423}$$
$$s_\Delta(u,b) = \rho^{-1}[u - T_{\rho,\Delta}(u,b)], \tag{424}$$
$$G_\rho(u,b) = \rho^{-1}[u - \Pi_U(u - \rho b)]. \tag{425}$$

Then

$$s_\Delta(u,b) = \mathrm{sign}(G_\rho)\min(|G_\rho|, \Delta/\rho), \tag{426}$$
$$s_\Delta(u,b) = 0 \Longleftrightarrow G_\rho(u,b) = 0, \tag{427}$$
$$|b| \le H \Longrightarrow |s_\Delta(u,b)| \ge \min(1, \Delta/(\rho H))|G_\rho(u,b)|, \tag{428}$$
$$|T_{\rho,\Delta}(u,h) - T_{\rho,\Delta}(u',h')| \le |u-u'| + \rho|h-h'|, \tag{429}$$
$$\rho|h| \le \Delta \Longrightarrow T_{\rho,\Delta}(u,h) = \Pi_U(u - \rho h). \tag{430}$$

Proof. For $b > 0$, both displacements equal the minimum of the proposed positive step and the distance $u - u_-$; for $b < 0$, replace $u - u_-$ by $u_+ - u$. Moreover,

$$|\Pi_U(a) - \Pi_U(a')| \le |a - a'|, \tag{431}$$
$$|\,\mathrm{clip}(a, -\Delta, \Delta) - \mathrm{clip}(a', -\Delta, \Delta)| \le |a - a'|. \tag{432}$$

□

Proposition A.57 (Summability consequence)**.** *If the right-hand side of Eq. (268) remains bounded as $T \to \infty$ and $\sum_t \eta_t = \sum_t \rho_t = \infty$, then*

$$\sum_t \eta_t \mathbb{E}\|g_t\|^2 < \infty, \qquad \sum_t \rho_t \mathbb{E}\|b_t\|^2 < \infty, \tag{433}$$
$$\liminf_{t\to\infty} \mathbb{E}\|g_t\|^2 = 0, \qquad \liminf_{t\to\infty} \mathbb{E}\|b_t\|^2 = 0. \tag{434}$$

No convergence of every iterate to a single critical point is inferred.

Proof. If $\liminf \mathbb{E}\|g_t\|^2 \ge \epsilon > 0$, then $\sum_t \eta_t \mathbb{E}\|g_t\|^2 \ge (\epsilon/2)\sum_{t \ge t_0} \eta_t = \infty$, a contradiction. The curvature block is identical. □

The combined theorem uses the residual of the admissible curvature step and charges sequential evaluation as an explicit derivative error. Projection controls feasibility, while the Taylor remainder controls the discrepancy between the softplus step and its preconditioned log-curvature approximation. Point-radius clipping is a separate modification of the representation dynamics and must be inactive here or included in the perturbation budget of the next subsection.

Theorem A.58 (Softplus, box constraints and log-step caps together)**.** *Let $U = \psi(C)$ for a coordinate box C and componentwise increasing ψ. Let $U_t = U \cap [u_t - \Delta, u_t + \Delta]$ and*

$$q_{t+1} = \mathrm{Exp}_{q_t}[-\eta_t N_\gamma(p_t)], \tag{435}$$
$$c_t^{\mathrm{raw}} = c_t - \rho_t\,\mathrm{diag}(\psi'(c_t))\widehat{b}_t, \tag{436}$$
$$u_{t+1} = \Pi_{U_t}\psi(c_t^{\mathrm{raw}}), \qquad B_t = \mathrm{diag}(\psi'(c_t)^2), \tag{437}$$
$$b_t^+ = \nabla_u F(q_{t+1}, u_t), \tag{438}$$
$$G_t^\Delta = \rho_t^{-1}\{u_t - \Pi_{U_t}(u_t - \rho_t B_t b_t^+)\}. \tag{439}$$

Assume the oracle, smoothness, compact-coordinate and tangent-momentum assumptions above, with inactive point-radius clipping and no unaccounted gradient-estimator bias. If $\eta_t \le a/(4La_+^2)$, $\rho_t \le 1/(2Lb_+)$, then

$$\sum_{t<T}\left[\tfrac{a\eta_t}{2}\mathbb{E}\|g_t\|^2 + \tfrac{\rho_t}{4}\mathbb{E}\|G_t^{\Delta}\|^2_{B_t^{-1}}\right] \le \Delta_F + C_e\sum_t \eta_t\mathbb{E}\|e_t^m\|^2 \tag{440}$$

$$+ 3b_+\sigma_u^2\sum_t \rho_t/\mathfrak{b}_t + 3b_+L_{uq}^2M_N^2\sum_t \rho_t\eta_t^2 + \sum_t \mathbb{E}\varepsilon_t^c. \tag{441}$$

For fixed $\beta < 1$, $\eta \asymp T^{-1/2}$, $\rho \asymp T^{-3/4}$, $\mathfrak{b}_t \gtrsim T^{1/2}$,

$$T^{-1}\sum_t \mathbb{E}\|g_t\|^2 = O(T^{-1/2}), \tag{442}$$

$$T^{-1}\sum_t \mathbb{E}\|G_t^{\Delta}\|^2_{B_t^{-1}} = O(T^{-1/4}). \tag{443}$$

Proof. For fixed u_t, projection onto U_t is nonexpansive and $u_t \in U_t$. By the softplus remainder bound,

$$\bar{u}_{t+1} = \Pi_{U_t}(u_t - \rho_t B_t\widehat{b}_t), \tag{444}$$

$$\|u_{t+1} - \bar{u}_{t+1}\| \le C_\psi\rho_t^2H_0^2, \tag{445}$$

$$F(q_{t+1}, u_{t+1}) \le F(q_{t+1}, \bar{u}_{t+1}) + \varepsilon_t^c, \tag{446}$$

$$\|u_{t+1} - u_t\| \le \rho_t\widetilde{H}, \qquad \widetilde{H} := b_+H_0 + C_\psi\rho_{\max}H_0^2, \tag{447}$$

$$\sum_t \mathbb{E}\|e_t^m\|^2 \le \frac{\beta\mathbb{E}\|e_{-1}^m\|^2}{1-\beta} + \frac{2\beta^2}{(1-\beta)^2}\left[L_{vq}^2M_N^2\sum_t\eta_t^2 + L_{vu}^2\widetilde{H}^2\sum_t\rho_t^2\right] \tag{448}$$

$$+ (1-\beta)a_+^2\sigma_q^2\sum_t \mathfrak{b}_t^{-1}. \tag{449}$$

Apply projected descent to U_t with error $\widehat{b}_t - b_t^+$; its expected squared B_t-norm is bounded by $2b_+\sigma_u^2/\mathfrak{b}_t + 2b_+L_{uq}^2\eta_t^2M_N^2$. Adding the normalized representation descent gives the budget. Finally,

$$\eta\sum_t \mathbb{E}\|e_t^m\|^2 = O(1), \tag{450}$$

$$\sum_t \rho/\mathfrak{b}_t = O(T^{-1/4}), \qquad \sum_t \rho\eta^2 = O(T^{-3/4}), \tag{451}$$

$$\sum_t \varepsilon_t^c = O(T^{-1/2}), \qquad T\eta \asymp T^{1/2}, \qquad T\rho \asymp T^{1/4}. \tag{452}$$

□

A.7. Geometric perturbations and finite precision

An intrinsic isometry can be poorly conditioned when represented in ambient Euclidean coordinates. We keep these notions separate: Jacobi operators control intrinsic displacement, while ambient operator norms and Lorentz cancellation control numerical sensitivity. The descent budget then determines how an implementation error affects the attainable stationarity level.

Proposition A.59 (One-step intrinsic perturbation)**.** *For $v \in T_xH_K$, $\bar{v} \in T_{\bar{x}}H_K$, with $\max(\|v\|, \|\bar{v}\|) \le R$,*

$$d_K(\mathrm{Exp}_x^K v, \mathrm{Exp}_{\bar{x}}^K \bar{v}) \le \cosh(\sqrt{K}R)d_K(x,\bar{x}) + s(\sqrt{K}R)\|P_{x\to\bar{x}}^K v - \bar{v}\|. \tag{453}$$

For $v = -\eta N_{\gamma\sqrt{K}}(p)$ and $\bar{v} = -\eta N_{\gamma\sqrt{K}}(\bar{p})$,

$$\|P_{x\to\bar{x}}v - \bar{v}\| \le \eta\|P_{x\to\bar{x}}p - \bar{p}\|, \tag{454}$$

$$d_K(x^+, \bar{x}^+) \le \cosh(\sqrt{K}R)d_K(x,\bar{x}) + \eta s(\sqrt{K}R)\|P_{x\to\bar{x}}p - \bar{p}\|. \tag{455}$$

Retaining the normalization–Jacobi cancellation of Theorem A.18 gives the sharper uniform bound

$$d_K(x^+, \bar{x}^+) \le \cosh(\eta/\gamma) d_K(x, \bar{x}) + \eta \Lambda(\eta/\gamma) \|P_{x \to \bar{x}} p - \bar{p}\|. \tag{456}$$

In particular, the velocity coefficient is exactly bounded by η when $\eta/\gamma \le \alpha_c$.

Proof. Insert $\operatorname{Exp}_{\bar{x}}^K(P_{x \to \bar{x}} v)$ and use the triangle inequality and the two finite-increment bounds. Since parallel transport is an isometry, $P N_\gamma(p) = N_\gamma(Pp)$. For the sharper bound, first vary the base point while transporting the same momentum, and then apply Theorem A.18 at the common base point. The step cap gives $\sqrt{K} R \le \eta/\gamma$ for the first part. □

Proposition A.60 (Curvature-coordinate perturbation)**.** *For $a = e^{(u-u')/2}$, perturbed values $u + \delta u$, $u' + \delta u'$ and $x + \delta x$,*

$$\bar{a} = a \exp[(\delta u - \delta u')/2], \tag{457}$$

$$|\bar{a} - a| \le \tfrac{a}{2} e^{(|\delta u| + |\delta u'|)/2} (|\delta u| + |\delta u'|), \tag{458}$$

$$\|\bar{a}(x + \delta x) - a x\|_2 \le \bar{a} \|\delta x\|_2 + |\bar{a} - a| \|x\|_2. \tag{459}$$

The same formula holds for ambient momentum perturbations.

Proof.

$$\bar{a}(x + \delta x) - a x = \bar{a} \delta x + (\bar{a} - a) x, \tag{460}$$

$$e^z - 1 = z \int_0^1 e^{sz} \, ds. \tag{461}$$

□

Proposition A.61 (Ambient sensitivity of transport)**.** *For fixed $x, y \in H_K$,*

$$P_{x \to y}^K = I + \frac{K}{1 - K\langle x, y \rangle_L} (x + y)(J_d y)^\top, \tag{462}$$

$$\|P_{x \to y}^K\|_{2 \to 2} \le 1 + \frac{K \|x + y\|_2 \|y\|_2}{1 - K\langle x, y \rangle_L}, \tag{463}$$

$$d_K(x, o_K), d_K(y, o_K) \le R \Longrightarrow \|P_{x \to y}^K\|_{2 \to 2} \le 1 + \cosh(2\sqrt{K} R). \tag{464}$$

Proof.

$$1 - K\langle x, y \rangle_L \ge 2, \tag{465}$$

$$\|x\|_2^2 = x_0^2 + \|x_s\|_2^2 = K^{-1}[\cosh^2(\sqrt{K} r) + \sinh^2(\sqrt{K} r)] \tag{466}$$

$$= K^{-1} \cosh(2\sqrt{K} r), \tag{467}$$

$$\|x + y\|_2 \le 2K^{-1/2} \sqrt{\cosh(2\sqrt{K} R)}. \tag{468}$$

□

Lemma A.62 (Objective effect of map error)**.** *If $d_{\mathcal{M}}(\bar{z}_{t+1}, z_{t+1}) \le \epsilon_t$, $\|\operatorname{grad} F\| \le G_F$ and $\|\operatorname{Hess} F\| \le L$ along the discrepancy geodesic, then*

$$F(\bar{z}_{t+1}) - F(z_{t+1}) \le G_F \epsilon_t + \tfrac{L}{2} \epsilon_t^2 =: \varepsilon_t. \tag{469}$$

Proof. For $w = \operatorname{Log}_{z_{t+1}} \bar{z}_{t+1}$,

$$\|w\| \le \epsilon_t, \tag{470}$$

$$F(\operatorname{Exp}_z w) - F(z) \le \langle \operatorname{grad} F(z), w \rangle + \tfrac{L}{2} \|w\|^2. \tag{471}$$

□

Corollary A.63 (Error floor in both blocks). *Let $\mathcal{B}_T$ be an exact-arithmetic descent budget. Then*

$$\frac{1}{T}\sum_t \mathbb{E}\|g_t\|^2 \le \frac{2(\mathcal{B}_T + \sum_t \mathbb{E}\varepsilon_t)}{aT\eta}, \tag{472}$$

$$\frac{1}{T}\sum_t \mathbb{E}\|b_t\|^2 \le \frac{2(\mathcal{B}_T + \sum_t \mathbb{E}\varepsilon_t)}{T\rho}. \tag{473}$$

For constant per-step displacement error $\epsilon_t \le \epsilon$,

$$\frac{1}{T\eta}\sum_t \mathbb{E}\varepsilon_t \le \frac{G_F\epsilon + L\epsilon^2/2}{\eta}, \tag{474}$$

$$\frac{1}{T\rho}\sum_t \mathbb{E}\varepsilon_t \le \frac{G_F\epsilon + L\epsilon^2/2}{\rho}. \tag{475}$$

A constant per-step displacement error need not disappear when more steps are taken, and dividing its accumulated objective effect by a smaller learning rate can worsen the bound. Conversely, a small measured manifold residual certifies only approximate feasibility. It cannot certify closeness to the intended trajectory, as the final counterexample shows.

Assumption A.64 (Floating-point model). *Use the standard relative-error and summation model of Higham (2002). For basic arithmetic with no overflow or underflow,*

$$\mathrm{fl}(a \circ b) = (a \circ b)(1+\delta), \qquad |\delta| \le \epsilon_{\mathrm{mach}}, \tag{476}$$

$$\gamma_k := \frac{k\epsilon_{\mathrm{mach}}}{1 - k\epsilon_{\mathrm{mach}}}, \qquad k\epsilon_{\mathrm{mach}} < 1, \tag{477}$$

$$\left|\mathrm{fl}\left(\sum_{i=1}^k a_i\right) - \sum_{i=1}^k a_i\right| \le \gamma_k \sum_{i=1}^k |a_i|. \tag{478}$$

Proposition A.65 (Lorentz constraint measurement). *For exact $x \in H_K$ and $r = d_K(x, o_K)$,*

$$\left|\mathrm{fl}\left(-x_0^2 + \sum_{j=1}^d x_j^2\right) + K^{-1}\right| \le \gamma_{d+2}\left(x_0^2 + \sum_{j=1}^d x_j^2\right) \tag{479}$$

$$= \gamma_{d+2} K^{-1}\cosh(2\sqrt{K}r). \tag{480}$$

For stored $\bar{x} = x + \delta x$,

$$|\bar{x}^\top J_d \bar{x} + K^{-1}| \le 2\|x\|_2\|\delta x\|_2 + \|\delta x\|_2^2. \tag{481}$$

Thus an $O(\epsilon_{\mathrm{mach}})$ relative storage error yields the scale

$$O\Big(\epsilon_{\mathrm{mach}} K^{-1}\cosh(2\sqrt{K}r)\Big). \tag{482}$$

Proof.

$$(x+\delta x)^\top J_d(x+\delta x) - x^\top J_d x = 2x^\top J_d \delta x + \delta x^\top J_d \delta x, \tag{483}$$

$$\|J_d\|_{2\to 2} = 1, \qquad \|x\|_2^2 = K^{-1}\cosh(2\sqrt{K}r). \tag{484}$$

□

Proposition A.66 (Residual alone does not control trajectory error). *For every $D > 0$ there exist $x, y \in H_K$ such that*

$$|x^\top J_d x + K^{-1}| = |y^\top J_d y + K^{-1}| = 0, \tag{485}$$

$$d_K(x, y) = D. \tag{486}$$

Consequently no finite C can satisfy

$$d_K(x,y) \le C\big(|x^\top J_d x + K^{-1}| + |y^\top J_d y + K^{-1}|\big) \tag{487}$$

for all $x, y \in H_K$.

Proof. Take $y = \operatorname{Exp}_x^K(Dv)$ with $v \in T_x H_K$, $\|v\| = 1$. □

A.8. Augmented-state Jacobians and algorithmic stability

For generalization, the relevant question is how replacing one training example changes the entire optimizer state. Position, curvature, and carried momentum influence one another, so a position-only recurrence misses a feedback path. We work in a bounded normal chart, derive the three-state comparison matrix, and only then eliminate momentum with a step-weighted norm.

Definition A.67 (Normal coordinates)**.** *For $z \in \mathbb{R}^d$, $r = \|z\|$, $e = z/r$, define*

$$X(z) = \operatorname{Exp}_{o_1}^1(0, z), \qquad T_z = DX(z), \tag{488}$$

$$s(r) = \sinh(r)/r, \qquad s(0) = 1, \tag{489}$$

$$\mathsf{g}(z) = ee^\top + s(r)^2(I - ee^\top), \qquad \mathsf{g}(0) = I. \tag{490}$$

For products, X, T, g are block diagonal.

Lemma A.68 (Coordinate metric and derivative bounds)**.** *For $\|z\| \le B$,*

$$I \preceq \mathsf{g}(z) \preceq s(B)^2 I, \qquad \|\mathsf{g}(z)^{-1}\| \le 1, \tag{491}$$

$$\|z - z'\| \le d_1(X(z), X(z')) \le s(B)\|z - z'\|, \tag{492}$$

$$\|D\mathsf{g}(z)[h]\| \le L_{\mathsf{g}}(B)\|h\|, \tag{493}$$

$$L_{\mathsf{g}}(B) := \sup_{0<r\le B} \left(2s(r)s'(r) + \frac{2(s(r)^2 - 1)}{r}\right) < \infty, \tag{494}$$

$$\|D(\mathsf{g}^{-1})(z)[h]\| \le L_{\mathsf{g}}(B)\|h\|. \tag{495}$$

Proof.

$$T_z^* T_z = \mathsf{P}_\parallel + s(r)^2 \mathsf{P}_\perp, \tag{496}$$

$$D(ee^\top)[h] = r^{-1}[(I - ee^\top)he^\top + eh^\top(I - ee^\top)], \tag{497}$$

$$\|D(ee^\top)[h]\| \le 2\|h\|/r, \tag{498}$$

$$D\mathsf{g}[h] = 2ss'\langle e, h\rangle(I - ee^\top) + (1 - s^2)D(ee^\top)[h], \tag{499}$$

$$s(r) = 1 + r^2/6 + O(r^4), \qquad s'(r) = r/3 + O(r^3), \tag{500}$$

$$D(\mathsf{g}^{-1})[h] = -\mathsf{g}^{-1} D\mathsf{g}[h] \mathsf{g}^{-1}. \tag{501}$$

For the distance comparisons, integrate DX along a Euclidean segment and $D\operatorname{Log}_{o_1}$ along the hyperbolic geodesic inside the convex ball. □

Proposition A.69 (Connection bound in the chart)**.** *For the coordinate Christoffel bilinear map Γ_z,*

$$2\langle \mathsf{g}(z)\Gamma_z(v, w), a\rangle = D\mathsf{g}(z)[v](w, a) + D\mathsf{g}(z)[w](v, a) - D\mathsf{g}(z)[a](v, w), \tag{502}$$

$$\|\Gamma_z(v, w)\| \le \tfrac{3}{2} L_{\mathsf{g}}(B)\|v\|\|w\|. \tag{503}$$

Proof. Each term on the first line is bounded by $L_{\mathsf{g}}(B)\|v\|\|w\|\|a\|$ and $\|\mathsf{g}^{-1}\| \le 1$. □

Assumption A.70 (Per-example coordinate derivatives)**.** *Let $f(z, u; \zeta) = \ell(X(z), u; \zeta) + R(z, u)$ on a common compact domain. For every example ζ,*

$$\|\nabla_z f\| \le G_f, \qquad \|\nabla_u f\| \le H_f, \tag{504}$$

$$\|\nabla^2_{zz} f\| \le L_{zz}, \qquad \|\nabla^2_{zu} f\| \le L_{zu}, \qquad \|\nabla^2_{uu} f\| \le L_{uu}. \tag{505}$$

All update segments and compared coordinate segments lie in this domain. The loss satisfies

$$|\ell(z, u; \zeta) - \ell(z', u'; \zeta)| \le L_{\ell,z}\|z - z'\| + L_{\ell,u}\|u - u'\|. \tag{506}$$

Lemma A.71 (Coordinate oracle constants)**.** *Let*

$$\mathcal{G}(z,u;\zeta) = A(u)\mathrm{g}(z)^{-1}\nabla_z f(z,u;\zeta), \tag{507}$$
$$\mathcal{H}(z,u;\zeta) = \nabla_u f(z,u;\zeta), \tag{508}$$
$$G_s = a_+ G_f, \qquad H_s = H_f, \tag{509}$$
$$L_z = a_+[L_{zz} + L_{\mathrm{g}}(B)G_f], \qquad L_u = a_+(L_{zu} + G_f), \tag{510}$$
$$L_{uz} = L_{zu}. \tag{511}$$

Then

$$\|\mathcal{G}(z,u;\zeta)\| \le G_s, \qquad \|\mathcal{H}(z,u;\zeta)\| \le H_s, \tag{512}$$
$$\|\mathcal{G}(z,u;\zeta) - \mathcal{G}(z',u';\zeta)\| \le L_z\|z-z'\| + L_u\|u-u'\|, \tag{513}$$
$$\|\mathcal{H}(z,u;\zeta) - \mathcal{H}(z',u';\zeta)\| \le L_{uz}\|z-z'\| + L_{uu}\|u-u'\|. \tag{514}$$

Proof.

$$D_z\mathcal{G}[h] = A\{D\mathrm{g}^{-1}[h]\nabla_z f + \mathrm{g}^{-1}\nabla^2_{zz} f\, h\}, \tag{515}$$
$$D_u\mathcal{G}[k] = DA[k]\mathrm{g}^{-1}\nabla_z f + A\mathrm{g}^{-1}\nabla^2_{zu} f\, k, \tag{516}$$
$$\|DA[k]\| \le a_+\|k\|, \tag{517}$$
$$D_z\mathcal{H}[h] = \nabla^2_{uz} f\, h, \qquad D_u\mathcal{H}[k] = \nabla^2_{uu} f\, k. \tag{518}$$

Integrate the displayed derivatives along the connecting segments. □

Definition A.72 (Exact coordinate step maps)**.** *For $p = T_z w$,*

$$q^+ = \mathrm{Exp}^1_{X(z)}[-\eta N_\gamma(T_z w)], \tag{519}$$
$$\Phi_q(z,w;\eta) = \mathrm{Log}^1_{o_1}(q^+), \tag{520}$$
$$\Phi_p(z,w;\eta) = D\,\mathrm{Log}^1_{o_1}(q^+)P^1_{X(z)\to q^+}T_z w. \tag{521}$$

At $\eta = 0$,

$$\Phi_q(z,w;0) = z, \qquad \Phi_p(z,w;0) = w, \tag{522}$$
$$D_z\Phi_q(z,w;0) = I, \qquad D_w\Phi_q(z,w;0) = 0, \tag{523}$$
$$D_z\Phi_p(z,w;0) = 0, \qquad D_w\Phi_p(z,w;0) = I. \tag{524}$$

Lemma A.73 (Small-step Jacobian constants)**.** *On $\|z\| \le B$, $\|w\| \le W$, $0 \le \eta \le \bar{\eta}$, define finite mixed-derivative bounds*

$$C_q \ge \sup\max\{\|\partial_\eta D_z\Phi_q\|, \|\partial_\eta D_w\Phi_q\|\}, \tag{525}$$
$$C_p \ge \sup\max\{\|\partial_\eta D_z\Phi_p\|, \|\partial_\eta D_w\Phi_p\|\}. \tag{526}$$

Then

$$\|D_z\Phi_q - I\| \le C_q\eta, \qquad \|D_w\Phi_q\| \le C_q\eta, \tag{527}$$
$$\|D_z\Phi_p\| \le C_p\eta, \qquad \|D_w\Phi_p - I\| \le C_p\eta, \tag{528}$$
$$\|\Phi_q(z,w) - \Phi_q(z',w')\| \le (1+C_q\eta)\|z-z'\| + C_q\eta\|w-w'\|, \tag{529}$$
$$\|\Phi_p(z,w) - \Phi_p(z',w')\| \le C_p\eta\|z-z'\| + (1+C_p\eta)\|w-w'\|. \tag{530}$$

Proof. For $\Phi \in \{\Phi_q, \Phi_p\}$,

$$D_z\Phi(\eta) - D_z\Phi(0) = \int_0^\eta \partial_s D_z\Phi(s)\,ds, \tag{531}$$
$$D_w\Phi(\eta) - D_w\Phi(0) = \int_0^\eta \partial_s D_w\Phi(s)\,ds. \tag{532}$$

The mixed derivatives are bounded on the compact update domain by the explicit smooth Exp, Log and transport formulas and the C^1 normalization map. The connecting-segment assumption then gives the finite-increment bounds. □

Lemma A.74 (Geometric refinement of the velocity coefficient)**.** *For pointwise normalization and $\|z\| \le B$, the exact coordinate map satisfies*

$$\|D_w \Phi_q(z, w; \eta)\| \le \eta s(B) \Lambda(\eta/\gamma). \tag{533}$$

Accordingly, set

$$C_{v,t} := \min\{C_q, s(B)\Lambda(\eta_t/\gamma)\} \qquad \textit{for pointwise normalization}, \tag{534}$$

and $C_{v,t} := C_q$ otherwise. The finite-increment bound improves to

$$\|\Phi_q(z, w) - \Phi_q(z', w')\| \le (1 + C_q\eta_t)\|z - z'\| + \eta_t C_{v,t}\|w - w'\|. \tag{535}$$

Proof. By Theorem A.18, the intrinsic momentum-to-point differential has norm at most $\eta\Lambda(\eta/\gamma)$. In the chain rule for Φ_q, the input chart differential has norm at most $s(B)$ and $D\operatorname{Log}_{o_1}$ has norm at most one. Taking the minimum with the preceding C_q bound proves the claim. Integrate first in z at fixed w, then in w at fixed z'. □

Definition A.75 (Coupled neighboring samples)**.** *Let $S = (\zeta_1, \dots, \zeta_n)$ and $S' = (\zeta'_1, \dots, \zeta'_n)$ satisfy*

$$\zeta_i = \zeta'_i \quad (i \ne i_*), \qquad I_t = \mathbf{1}\{i_t = i_*\}, \qquad \mathbb{E}I_t = 1/n. \tag{536}$$

The algorithms share initialization and index sequence. In coordinates,

$$\bar{w}_t = \beta w_t + (1 - \beta)\mathcal{G}(z_t, u_t; \zeta_{i_t}), \tag{537}$$
$$z_{t+1} = \Phi_q(z_t, \bar{w}_t; \eta_t), \tag{538}$$
$$w_{t+1} = \Phi_p(z_t, \bar{w}_t; \eta_t), \tag{539}$$
$$u_{t+1} = \Pi_U[u_t - \rho_t \mathcal{H}(z_t, u_t; \zeta_{i_t})], \tag{540}$$
$$d_t = \|z_t - z'_t\|, \qquad c_t = \|u_t - u'_t\|, \qquad v_t = \|w_t - w'_t\|. \tag{541}$$

Theorem A.76 (Explicit three-state comparison matrix)**.** *Put $\delta_\beta = 1 - \beta$, $\Delta_t = (d_t, c_t, v_t)^\top$. Then pathwise*

$$\Delta_{t+1} \preceq M_t \Delta_t + b_t^{\mathrm{inj}} I_t, \tag{542}$$
$$M_t = \begin{pmatrix} 1 + \eta_t(C_q + C_{v,t}\delta_\beta L_z) & C_{v,t}\eta_t\delta_\beta L_u & C_{v,t}\eta_t\beta \\ \rho_t L_{uz} & 1 + \rho_t L_{uu} & 0 \\ C_p\eta_t + (1 + C_p\eta_t)\delta_\beta L_z & (1 + C_p\eta_t)\delta_\beta L_u & \beta(1 + C_p\eta_t) \end{pmatrix}, \tag{543}$$
$$b_t^{\mathrm{inj}} = 2\begin{pmatrix} C_{v,t}\eta_t\delta_\beta G_s \\ \rho_t H_s \\ (1 + C_p\eta_t)\delta_\beta G_s \end{pmatrix}. \tag{544}$$

Consequently, for identical initial states,

$$\mathbb{E}\Delta_T \preceq \frac{1}{n}\sum_{t<T}\left(\prod_{s=T-1}^{t+1} M_s\right) b_t^{\mathrm{inj}}. \tag{545}$$

Proof.

$$\|\bar{w}_t - \bar{w}'_t\| \le \beta v_t + \delta_\beta L_z d_t + \delta_\beta L_u c_t + 2\delta_\beta G_s I_t, \tag{546}$$
$$d_{t+1} \le (1 + C_q\eta_t)d_t + C_{v,t}\eta_t\|\bar{w}_t - \bar{w}'_t\|, \tag{547}$$
$$c_{t+1} \le c_t + \rho_t(L_{uz}d_t + L_{uu}c_t + 2H_s I_t), \tag{548}$$
$$v_{t+1} \le C_p\eta_t d_t + (1 + C_p\eta_t)\|\bar{w}_t - \bar{w}'_t\|. \tag{549}$$

Substitute the first line into the other three and iterate the nonnegative matrix inequality. Only $\mathbb{E}I_t = 1/n$ is used in the final expectation. □

The injection vector records the direct effect of sampling the replaced example; the matrix propagates that effect through later updates. Before reducing it to a scalar recurrence, we retain the exact matrix transfer bound. The neighboring-sample coupling follows the stability approach of Hardt et al. (2016), with the curvature and memory feedback represented explicitly here.

Theorem A.77 (Collatz–Wielandt certificate for coupled stability)**.** *For constant steps, let $M = M_t$, $b = b_t^{\mathrm{inj}}$ and $c_\ell = (L_{\ell,z}, L_{\ell,u}, 0)^\top$. Define*

$$S_T(M) := \sum_{k=0}^{T-1} M^k, \qquad \Phi_T(r) := \begin{cases} (r^T - 1)/(r-1), & r \neq 1, \\ T, & r = 1. \end{cases} \tag{550}$$

Then

$$\epsilon_{\mathrm{stab}}(T) \le \frac{c_\ell^\top S_T(M) b}{n}. \tag{551}$$

If M is irreducible, let $w > 0$ satisfy $M^\top w = r_* w$, where $r_* := \mathrm{spr}(M)$. With $L_w := \max_i (c_{\ell,i}/w_i)$,

$$\epsilon_{\mathrm{stab}}(T) \le \frac{L_w w^\top b}{n} \Phi_T(r_*), \tag{552}$$

$$r_* = \inf_{v>0} \max_i \frac{(M^\top v)_i}{v_i} = \inf_{v>0} \|M\|_{1,v}. \tag{553}$$

Thus r_* is the smallest scalar growth factor obtainable from a fixed positive weighted ℓ_1 norm of this comparison system. If $0 \le \ell \le M_\ell$, every displayed stability upper bound may additionally be capped at M_ℓ.

Proof. Theorem A.76 gives $\mathbb{E}\Delta_T \preceq S_T(M) b/n$. Since the loss discrepancy is at most $c_\ell^\top \Delta_T$, this proves Eq. (551). Perron–Frobenius and the Collatz–Wielandt formula (Horn & Johnson, 2013, Chapter 8) give the positive left eigenvector and the first variational identity. For a nonnegative matrix,

$$\|M\|_{1,v} = \max_i \frac{(M^\top v)_i}{v_i}, \qquad c_\ell^\top M^k b \le L_w w^\top M^k b = L_w r_*^k w^\top b. \tag{554}$$

Summing proves the Perron bound. Every induced norm bounds the spectral radius from below, while $v = w$ attains it. The matrix expression is exact for the linear comparison recurrence; no optimality over nonlinear learning algorithms is asserted. □

For variable steps we use the following scalar reduction. Although the momentum discrepancy can be order one, its immediate effect on position is multiplied by η_t. Weighting the memory component by that same step retains a factor $1 + C\eta_t$ rather than an order-one expansion at every iteration.

Lemma A.78 (Weighted elimination of momentum)**.** *Suppose $\eta_{t+1} \le \eta_t \le \bar{\eta}$, $\rho_t \le r_\rho \eta_t$, $2C_p\bar{\eta} \le 1 - \beta$. Fix $\kappa > 0$ and define*

$$\bar{\beta} = (1+\beta)/2, \qquad a_m = 1 + \frac{C_q \beta}{1 - \bar{\beta}}, \tag{555}$$

$$V_z = C_p\bar{\eta} + (1 + C_p\bar{\eta})\delta_\beta L_z, \qquad V_u = (1 + C_p\bar{\eta})\delta_\beta L_u, \tag{556}$$

$$C_z = C_q(1 + \delta_\beta L_z) + \kappa r_\rho L_{uz} + a_m V_z, \tag{557}$$

$$C_u = r_\rho L_{uu} + \kappa^{-1}(C_q \delta_\beta L_u + a_m V_u), \tag{558}$$

$$C = \max(C_z, C_u), \tag{559}$$

$$B_s = 2\delta_\beta G_s [C_q + a_m(1 + C_p\bar{\eta})] + 2\kappa r_\rho H_s, \tag{560}$$

$$D_t = d_t + \kappa c_t + a_m \eta_t v_t. \tag{561}$$

Then

$$\mathbb{E} D_{t+1} \le (1 + C\eta_t)\mathbb{E} D_t + B_s \eta_t / n. \tag{562}$$

Proof. From Theorem A.76,

$$\beta(1 + C_p\eta_t) \le \beta + C_p\bar{\eta} \le \bar{\beta}, \tag{563}$$

$$C_q\beta + a_m\bar{\beta} \le a_m, \tag{564}$$

$$D_{t+1} \le (1 + C_z\eta_t)d_t + \kappa(1 + C_u\eta_t)c_t + a_m\eta_t v_t + B_s\eta_t I_t \tag{565}$$

$$\le (1 + C\eta_t)(d_t + \kappa c_t + a_m\eta_t v_t) + B_s\eta_t I_t. \tag{566}$$

Take expectations and use $\mathbb{E}I_t = 1/n$. □

Theorem A.79 (Expected uniform algorithmic stability)**.** *For identical initialization, let* $H_T = \sum_{t<T}\eta_t$ *and* $L_\ell = \max(L_{\ell,z}, L_{\ell,u}/\kappa)$*. Then*

$$\epsilon_{\text{stab}}(T) := \sup_{S\sim S'}\sup_{\zeta}\mathbb{E}|\ell(A_T(S);\zeta) - \ell(A_T(S');\zeta)| \tag{567}$$

$$\le \frac{L_\ell B_s}{n}\sum_{t<T}\eta_t\prod_{s=t+1}^{T-1}(1 + C\eta_s) \tag{568}$$

$$\le \begin{cases} \dfrac{L_\ell B_s}{Cn}(e^{CH_T} - 1), & C > 0, \\ \dfrac{L_\ell B_s H_T}{n}, & C = 0. \end{cases} \tag{569}$$

Proof.

$$\mathbb{E}D_T \le \frac{B_s}{n}\sum_{t<T}\eta_t\prod_{s>t}(1 + C\eta_s), \tag{570}$$

$$|\ell(z_T, u_T;\zeta) - \ell(z'_T, u'_T;\zeta)| \le L_{\ell,z}d_T + L_{\ell,u}c_T \le L_\ell D_T, \tag{571}$$

$$C\sum_{t<T}\eta_t\prod_{s>t}(1 + C\eta_s) = \prod_{s<T}(1 + C\eta_s) - 1 \tag{572}$$

$$\le \exp\left(C\sum_{s<T}\eta_s\right) - 1. \tag{573}$$

□

Corollary A.80 (Schedules and mini-batch replacement)**.** *For* $\eta_t = c/\sqrt{T}$*,*

$$\epsilon_{\text{stab}}(T) \le \frac{L_\ell B_s}{Cn}(e^{Cc\sqrt{T}} - 1). \tag{574}$$

For $\eta_t = c/(t + t_0)$*,* $t_0 > 0$*,*

$$H_T \le c/t_0 + c\log\big((T - 1 + t_0)/t_0\big) \quad (T \ge 1), \tag{575}$$

$$\epsilon_{\text{stab}}(T) \le \frac{L_\ell B_s}{Cn}\left[e^{Cc/t_0}\left(\frac{T - 1 + t_0}{t_0}\right)^{Cc} - 1\right]. \tag{576}$$

For averaged mini-batches of size b *sampled uniformly without replacement,*

$$\Pr(i_* \in \mathcal{B}_t) = b/n, \qquad \|\widehat{G}_S - \widehat{G}_{S'}\| \le (2G_s/b)\mathbf{1}\{i_* \in \mathcal{B}_t\}, \tag{577}$$

$$\mathbb{E}[(2G_s/b)\mathbf{1}\{i_* \in \mathcal{B}_t\}] = 2G_s/n. \tag{578}$$

The exponential upper bound is a worst-case sensitivity estimate and may become uninformative over long training horizons. Mini-batch averaging reduces the size of an individual replacement effect but also increases the probability of including that example, yielding the displayed $1/n$ injection scale. A time-uniform bound follows below only when the full coupled update is contractive; positive curvature-block curvature alone does not establish that property.

Proposition A.81 (Contractive full-state regime)**.** *If the implemented coupled update satisfies*

$$\mathbb{E}D_{t+1} \le (1-\mu\eta_t)\mathbb{E}D_t + B_s\eta_t/n, \qquad 0 \le \mu\eta_t \le 1, \tag{579}$$

then

$$\mathbb{E}D_T \le \frac{B_s}{\mu n}\left[1 - \prod_{t<T}(1-\mu\eta_t)\right] \le \frac{B_s}{\mu n}, \tag{580}$$

$$\epsilon_{\mathrm{stab}}(T) \le \frac{L_\ell B_s}{\mu n}. \tag{581}$$

Proof. Subtract $B_s/(\mu n)$ from both sides of the recursion and iterate from $D_0 = 0$. □

Proposition A.82 (Optimal curvature weight for the exponential coefficient)**.** *This two-state calculation is a closed-form instance of Collatz–Wielandt (Horn & Johnson, 2013). Set*

$$\alpha_s = C_q(1+\delta_\beta L_z) + a_m V_z, \qquad \delta_s = r_\rho L_{uu}, \tag{582}$$

$$b_s = r_\rho L_{uz}, \qquad c_s = C_q\delta_\beta L_u + a_m V_u. \tag{583}$$

For $b_s, c_s > 0$,

$$C(\kappa) = \max(\alpha_s + b_s\kappa, \delta_s + c_s/\kappa), \tag{584}$$

$$\kappa_* = \frac{\delta_s - \alpha_s + \sqrt{(\alpha_s-\delta_s)^2 + 4b_sc_s}}{2b_s}, \tag{585}$$

$$\inf_{\kappa>0} C(\kappa) = \frac{\alpha_s + \delta_s + \sqrt{(\alpha_s-\delta_s)^2 + 4b_sc_s}}{2} \tag{586}$$

$$= \mathrm{spr}\begin{pmatrix} \alpha_s & c_s \\ b_s & \delta_s \end{pmatrix}. \tag{587}$$

Proof.

$$\frac{d}{d\kappa}(\alpha_s + b_s\kappa) = b_s > 0, \qquad \frac{d}{d\kappa}(\delta_s + c_s/\kappa) = -c_s/\kappa^2 < 0, \tag{588}$$

$$\alpha_s + b_s\kappa = \delta_s + c_s/\kappa \iff b_s\kappa^2 + (\alpha_s-\delta_s)\kappa - c_s = 0. \tag{589}$$

The unique positive root is κ_*. The larger root of $(\lambda-\alpha_s)(\lambda-\delta_s) - b_sc_s = 0$ equals the displayed value. □

A.9. Expected risk, second moments, concentration, and metric entropy

We distinguish the signed expected generalization gap, its absolute size, and a high-probability statement. A small signed mean can hide cancellation, so the second-moment argument is needed to control the absolute gap. We also give a separate complexity bound that holds uniformly over the compact hypothesis domain and is independent of the training trajectory.

Definition A.83 (Population and empirical risks)**.**

$$S = (Z_1, \dots, Z_n) \sim P^n, \qquad Z \sim P, \tag{590}$$

$$R(\theta) = \mathbb{E}_Z\ell(\theta; Z), \qquad R_S(\theta) = \frac{1}{n}\sum_{i=1}^n \ell(\theta; Z_i), \tag{591}$$

$$S^{(i)} = (Z_1, \dots, Z_{i-1}, Z_i', Z_{i+1}, \dots, Z_n), \qquad Z_i' \sim P, \tag{592}$$

$$\mathcal{G}(S,\omega) = R(A(S;\omega)) - R_S(A(S;\omega)), \qquad \omega \perp (S, Z, Z_1', \dots, Z_n'). \tag{593}$$

Theorem A.84 (Generalization from expected stability)**.** *Under Theorem A.79,*

$$|\mathbb{E}_{S,\omega}\mathcal{G}(S,\omega)| \le \epsilon_{\mathrm{stab}}(T) \le \frac{L_\ell B_s}{Cn}\left(e^{C\sum_{t<T}\eta_t} - 1\right), \qquad C > 0. \tag{594}$$

For $C = 0$, the right-hand side is $L_\ell B_s \sum_t \eta_t/n$.

Proof. By exchangeability of Z_i, Z_i',

$$\mathbb{E}\mathcal{G} = \frac{1}{n}\sum_i \mathbb{E}[\ell(A(S); Z_i') - \ell(A(S); Z_i)] \tag{595}$$

$$= \frac{1}{n}\sum_i \mathbb{E}[\ell(A(S^{(i)}); Z_i) - \ell(A(S); Z_i)], \tag{596}$$

$$|\mathbb{E}\mathcal{G}| \le \frac{1}{n}\sum_i \mathbb{E}|\ell(A(S^{(i)}); Z_i) - \ell(A(S); Z_i)| \tag{597}$$

$$\le \frac{1}{n}\sum_i \epsilon_{\mathrm{stab}}(T) = \epsilon_{\mathrm{stab}}(T). \tag{598}$$

□

Theorem A.85 (Variance-sensitive second moment by one-coordinate decoupling)**.** *Assume* $0 \le \ell \le M_\ell$ *and*

$$\sup_{S\sim S'}\sup_z \mathbb{E}_\omega|\ell(A(S;\omega); z) - \ell(A(S';\omega); z)| \le \epsilon. \tag{599}$$

Let

$$v_A := \mathbb{E}_{S,\omega}\,\mathrm{Var}_{Z\sim P}\,\ell(A(S;\omega); Z), \qquad \bar{R} := \mathbb{E}_{S,\omega}R(A(S;\omega)). \tag{600}$$

Then

$$\mathbb{E}\mathcal{G}^2 \le \frac{v_A}{n} + 4M_\ell\epsilon \le \frac{\bar{R}(M_\ell - \bar{R})}{n} + 4M_\ell\epsilon \le \frac{M_\ell^2}{4n} + 4M_\ell\epsilon, \tag{601}$$

$$\mathbb{E}|\mathcal{G}| \le \sqrt{v_A/n + 4M_\ell\epsilon}, \tag{602}$$

$$\Pr\left\{|\mathcal{G}| > \sqrt{\frac{v_A/n + 4M_\ell\epsilon}{\delta}}\right\} \le \delta. \tag{603}$$

The coefficient $1/4$ *in the worst-case IID variance term cannot be reduced.*

Proof. Work directly with the population-centered variables, avoiding a ghost-sample Jensen step:

$$c_i := R(A(S;\omega)) - \ell(A(S;\omega); Z_i), \qquad \mathcal{G} = n^{-1}\sum_i c_i. \tag{604}$$

Replace only entry i by an independent Z_i' and define, for every k,

$$c_k^{(i)} := R(A(S^{(i)};\omega)) - \ell(A(S^{(i)};\omega); Z_k). \tag{605}$$

For $j \ne i$, condition on S_{-i}, Z_i', ω. The algorithm output and $c_j^{(i)}$ are then fixed, while $Z_i \sim P$ remains independent. Thus

$$\mathbb{E}[c_i^{(i)} \mid S_{-i}, Z_i', \omega] = 0, \qquad \mathbb{E}[c_i^{(i)}c_j^{(i)}] = 0, \tag{606}$$

$$\mathbb{E}[(c_i^{(i)})^2] = v_A, \qquad \mathbb{E}|c_k - c_k^{(i)}| \le 2\epsilon. \tag{607}$$

The last bound uses stability once for the population loss and once for the evaluation at Z_k. Since $|c_k|, |c_k^{(i)}| \le M_\ell$,

$$\mathbb{E}c_i^2 \le \mathbb{E}(c_i^{(i)})^2 + 2M_\ell\mathbb{E}|c_i - c_i^{(i)}| \le v_A + 4M_\ell\epsilon, \tag{608}$$

$$\mathbb{E}[c_ic_j] = \mathbb{E}[(c_i - c_i^{(i)})c_j + c_i^{(i)}(c_j - c_j^{(i)})] \le 4M_\ell\epsilon \qquad (i \ne j), \tag{609}$$

$$\mathbb{E}\mathcal{G}^2 \le n^{-2}\{n(v_A + 4M_\ell\epsilon) + n(n-1)4M_\ell\epsilon\} = v_A/n + 4M_\ell\epsilon. \tag{610}$$

The scalar Bhatia–Davis bound (Bhatia & Davis, 2000) gives

$$\mathrm{Var}_Z\,\ell(A;Z) \le R(A)(M_\ell - R(A)), \tag{611}$$

$$v_A \le \mathbb{E}[R(A)(M_\ell - R(A))] \le \bar{R}(M_\ell - \bar{R}) \le M_\ell^2/4. \tag{612}$$

Indeed, $0 \le Y \le M_\ell$ implies $\mathbb{E}Y^2 \le M_\ell \mathbb{E}Y$; subtracting $(\mathbb{E}Y)^2$ proves the bound. Cauchy–Schwarz and Markov's inequality give the remaining conclusions. A sample-independent predictor with loss $M_\ell\,\mathrm{Bernoulli}(1/2)$ has $\epsilon = 0$ and $\mathbb{E}\mathcal{G}^2 = M_\ell^2/(4n)$, proving sharpness of the IID term. □

Expected absolute stability suffices for the preceding second-moment bound, including randomness in the training algorithm. The next concentration statement instead assumes a bound for a fixed random seed, uniformly over neighboring samples. Keeping these assumptions distinct prevents the expected coupling argument from being used as an unjustified pathwise guarantee.

Theorem A.86 (Concentration under pathwise uniform stability)**.** *Assume A is deterministic, or fix a seed for which*

$$\sup_{S\sim S'}\sup_z |\ell(A(S);z) - \ell(A(S');z)| \le \epsilon, \tag{613}$$

$$0 \le \ell \le M_\ell. \tag{614}$$

Then, with probability at least $1-\delta$,

$$|\mathcal{G}(S)| \le \epsilon + (2n\epsilon + M_\ell)\sqrt{\frac{\log(2/\delta)}{2n}}. \tag{615}$$

Proof. For $S \sim S'$,

$$|R(A(S)) - R(A(S'))| \le \epsilon, \tag{616}$$

$$|R_S(A(S)) - R_{S'}(A(S'))| \le \frac{n-1}{n}\epsilon + \frac{M_\ell}{n}, \tag{617}$$

$$|\mathcal{G}(S) - \mathcal{G}(S')| \le 2\epsilon + M_\ell/n =: c_n. \tag{618}$$

The bounded-difference martingale inequality gives

$$\Pr(|\mathcal{G} - \mathbb{E}\mathcal{G}| \ge t) \le 2\exp\left(-\frac{2t^2}{nc_n^2}\right), \tag{619}$$

$$|\mathbb{E}\mathcal{G}| \le \epsilon, \tag{620}$$

$$t = c_n\sqrt{n\log(2/\delta)/2}. \tag{621}$$

□

Theorem A.87 (Sharper stability concentration via seed averaging)**.** *Let $n \ge 2$, $0 \le \ell \le M_\ell$, and assume uniform expected absolute stability with parameter ϵ. Define*

$$\bar{\ell}_T(S;z) := \mathbb{E}_\omega \ell(A_T(S;\omega);z), \qquad \overline{\mathcal{G}}_T(S) := \mathbb{E}_\omega \mathcal{G}(S,\omega). \tag{622}$$

There is a universal constant C_{BKW} such that, with probability at least $1-\delta$ over S,

$$|\overline{\mathcal{G}}_T(S)| \le C_{\mathrm{BKW}}\left[\epsilon\log(2n)\log(2/\delta) + M_\ell\sqrt{\frac{\log(2/\delta)}{n}}\right]. \tag{623}$$

Under pathwise uniform stability, the same bound holds for $\mathcal{G}(S,\omega)$ at any fixed seed satisfying that hypothesis.

Proof. The seed-averaged loss is deterministic, bounded in $[0, M_\ell]$, and satisfies

$$\sup_{S\sim S'}\sup_z |\bar{\ell}_T(S;z) - \bar{\ell}_T(S';z)| \le \sup_{S\sim S'}\sup_z \mathbb{E}_\omega |\ell(A_T(S;\omega);z) - \ell(A_T(S';\omega);z)| \le \epsilon. \tag{624}$$

Apply Bousquet et al. (2020, Corollary 8) to this uniformly stable loss. Its generalization gap is precisely $\overline{\mathcal{G}}_T(S)$. The fixed-seed statement applies the same result directly to the pathwise stable loss. □

The stability penalty here is logarithmic in n, replacing the $\sqrt{n}\epsilon$ term introduced by bounded differences. Seed averaging is essential for using the expected stability parameter: the conclusion controls the mean loss over training randomness, rather than asserting concentration for one randomly selected run. The second-moment theorem above remains the applicable bound for the joint sample-and-seed gap without stronger assumptions.

Proposition A.88 (Bounded hyperbolic classification loss)**.** *Let a fixed encoder satisfy $q_{\mathrm{in}}(z) \in B_{H_1}(o, B)$ and prototypes satisfy $q_c \in B_{H_1}(o, B)$, $K_c \in [K_-, K_+]$. For*

$$s_c(z) = -K_c^{-1} d_1(q_{\mathrm{in}}(z), q_c)^2, \qquad \ell(z, y) = -s_y(z) + \log \sum_{c=1}^{C} e^{s_c(z)}, \tag{625}$$

$$-S_B \le s_c \le 0, \qquad S_B := 4B^2/K_-, \tag{626}$$

$$0 \le \ell \le M_\ell, \qquad M_\ell := S_B + \log C. \tag{627}$$

In normal coordinates, one valid joint Lipschitz bound is

$$L_s = \sqrt{2(4Bs(B)/K_-)^2 + (4B^2/K_-)^2}, \tag{628}$$

$$L_\ell \le 2L_s. \tag{629}$$

Proof.

$$d_1(q_{\mathrm{in}}, q_c) \le 2B, \tag{630}$$

$$\ell \le S_B + \log C, \tag{631}$$

$$\|\nabla_{q_{\mathrm{in}}} s_c\|, \|\nabla_{q_c} s_c\| \le 2d_1(q_{\mathrm{in}}, q_c)/K_- \le 4B/K_-, \tag{632}$$

$$|\partial_{u_c} s_c| = K_c^{-1} d_1(q_{\mathrm{in}}, q_c)^2 \le 4B^2/K_-, \tag{633}$$

$$\|D_z X\| \le s(B), \tag{634}$$

$$\nabla_s \ell = \mathrm{softmax}(s) - e_y, \qquad \|\nabla_s \ell\|_1 \le 2, \tag{635}$$

$$\|\nabla_\theta \ell\| = \left\| \sum_c (\mathrm{softmax}(s)_c - \mathbf{1}_{c=y}) \nabla_\theta s_c \right\| \le 2L_s. \tag{636}$$

The bound also allows the input point to vary; freezing it only decreases the parameter set. □

Definition A.89 (Compact hypothesis domain)**.** *Let $P_q = \sum_g n_g$, $D_q = \sum_g n_g d_g$ and*

$$\Theta = \left[\prod_g B_{H_1}(o, B)^{n_g} \right] \times U, \tag{637}$$

$$R_u = \tfrac{1}{2} \left[\sum_g (u_{g,+} - u_{g,-})^2 \right]^{1/2}, \tag{638}$$

$$d_\Theta((q, u), (q', u'))^2 = d_{\mathcal{Q}}(q, q')^2 + \|u - u'\|^2. \tag{639}$$

Assume all trainable prediction parameters belong to Θ, and $\ell(\theta; z) \in [0, M_\ell]$ is L_Θ-Lipschitz in d_Θ.

Lemma A.90 (Product metric entropy)**.** *For $\varepsilon > 0$,*

$$\mathcal{N}(\Theta, d_\Theta, \varepsilon) \le \left(1 + \frac{4B\sqrt{P_q} s(B)}{\varepsilon} \right)^{D_q} \left(1 + \frac{4R_u}{\varepsilon} \right)^{G}. \tag{640}$$

Proof. The canonical coordinate domain $\mathcal{Z}$ is convex and contained in a Euclidean ball of radius $B\sqrt{P_q}$ in $\mathbb{R}^{D_q}$. A maximal

τ-separated set in $\mathcal{Z}$ gives disjoint radius-$\tau/2$ balls, hence

$$\mathcal{N}(\mathcal{Z}, \|\cdot\|, \tau) \le \frac{\operatorname{Vol} B_{D_q}(0, B\sqrt{P_q} + \tau/2)}{\operatorname{Vol} B_{D_q}(0, \tau/2)} = \left(1 + \frac{2B\sqrt{P_q}}{\tau}\right)^{D_q}, \tag{641}$$

$$d_{\mathcal{Q}}(X(z), X(z')) \le s(B)\|z - z'\|, \tag{642}$$

$$\mathcal{N}(U, \|\cdot\|, \varepsilon/2) \le (1 + 4R_u/\varepsilon)^G. \tag{643}$$

Take $\tau = \varepsilon/[2s(B)]$ and the product of the two nets; each joint distance is at most $\sqrt{(\varepsilon/2)^2 + (\varepsilon/2)^2} \le \varepsilon$. □

Theorem A.91 (Uniform generalization from geometry entropy)**.** *For any $\varepsilon > 0$, with probability at least $1 - \delta$,*

$$\mathcal{H}_\varepsilon := D_q \log(1 + 4B\sqrt{P_q} s(B)/\varepsilon) + G \log(1 + 4R_u/\varepsilon), \tag{644}$$

$$\sup_{\theta \in \Theta} |R(\theta) - R_S(\theta)| \le 2L_\Theta \varepsilon + M_\ell \sqrt{\frac{\log(2/\delta) + \mathcal{H}_\varepsilon}{2n}}. \tag{645}$$

The same event holds for any data-dependent $A(S) \in \Theta$.

Proof. Let Θ_ε be a deterministic net with cardinality N_ε. For each $\theta_j \in \Theta_\varepsilon$, Hoeffding's inequality gives

$$\Pr(|R(\theta_j) - R_S(\theta_j)| > t) \le 2e^{-2nt^2/M_\ell^2}, \tag{646}$$

$$\Pr\left(\max_j |R(\theta_j) - R_S(\theta_j)| > t\right) \le 2N_\varepsilon e^{-2nt^2/M_\ell^2}. \tag{647}$$

For $d_\Theta(\theta, \theta_j) \le \varepsilon$,

$$|R(\theta) - R(\theta_j)| \le L_\Theta \varepsilon, \tag{648}$$

$$|R_S(\theta) - R_S(\theta_j)| \le L_\Theta \varepsilon, \tag{649}$$

$$|R(\theta) - R_S(\theta)| \le 2L_\Theta \varepsilon + |R(\theta_j) - R_S(\theta_j)|. \tag{650}$$

Take $t = M_\ell \sqrt{\log(2N_\varepsilon/\delta)/(2n)}$ and apply Eq. (640). □

Theorem A.92 (Dudley chaining without a single-net resolution penalty)**.** *Let $\mathcal{H}_\varepsilon$ be the entropy bound above, and set*

$$D_\Theta := 2\sqrt{P_q B^2 + R_u^2}, \qquad d_0 := \min\{D_\Theta, M_\ell/L_\Theta\}, \tag{651}$$

$$I_\Theta(d) := \int_0^d \sqrt{\mathcal{H}_\varepsilon}\, d\varepsilon. \tag{652}$$

For $L_\Theta > 0$, with probability at least $1 - \delta$,

$$\sup_{\theta \in \Theta} |R(\theta) - R_S(\theta)| \le \mathfrak{s}_n^{\text{chain}}(\delta), \tag{653}$$

$$\mathfrak{s}_n^{\text{chain}}(\delta) := \inf_{0 \le \tau \le d_0/2} \left\{ 8L_\Theta \tau + \frac{24L_\Theta}{\sqrt{n}} \int_\tau^{d_0/2} \sqrt{\mathcal{H}_\varepsilon}\, d\varepsilon \right\} + M_\ell \sqrt{\frac{\log(2/\delta)}{2n}} \tag{654}$$

$$\le \frac{24L_\Theta}{\sqrt{n}} I_\Theta(d_0/2) + M_\ell \sqrt{\frac{\log(2/\delta)}{2n}}. \tag{655}$$

For $L_\Theta = 0$, omit the integral term. A fully explicit upper bound on the integral is

$$I_\Theta(d) \le d\sqrt{D_q} \left(\sqrt{\log(1 + A_q/d)} + \frac{\sqrt{\pi}}{2} \right) \tag{656}$$

$$+ d\sqrt{G} \left(\sqrt{\log(1 + A_u/d)} + \frac{\sqrt{\pi}}{2} \right), \tag{657}$$

$$A_q := 4B\sqrt{P_q} s(B), \qquad A_u := 4R_u. \tag{658}$$

Proof. For a fixed sample, let $f_\theta = (\ell(\theta; Z_i))_{i=1}^n$ and

$$d_{2,S}(f_\theta, f_{\theta'})^2 := n^{-1} \sum_i (\ell(\theta; Z_i) - \ell(\theta'; Z_i))^2 \le L_\Theta^2 d_\Theta(\theta, \theta')^2, \tag{659}$$

$$X_\theta := n^{-1} \sum_i \sigma_i \ell(\theta; Z_i), \tag{660}$$

$$\mathbb{E}_\sigma e^{\lambda(X_\theta - X_{\theta'})} \le \exp\left[\frac{\lambda^2 d_{2,S}(f_\theta, f_{\theta'})^2}{2n}\right]. \tag{661}$$

Center the class at a fixed f_{θ_0}; this does not change $\mathbb{E}_\sigma \sup_\theta X_\theta$. Its radius is at most $L_\Theta d_0$. Dudley's entropy integral (Dudley, 1967), in the Rademacher form of Telgarsky (2021, Theorem 15.2), gives

$$\widehat{\mathfrak{R}}_S := \mathbb{E}_\sigma \sup_\theta X_\theta \le \inf_{0 \le \tau \le d_0/2} \left\{ 4L_\Theta \tau + \frac{12 L_\Theta}{\sqrt{n}} \int_\tau^{d_0/2} \sqrt{\mathcal{H}_\varepsilon}\, d\varepsilon \right\}. \tag{662}$$

Symmetrization (Bartlett & Mendelson, 2002) bounds each of $\mathbb{E}\sup_\theta(R - R_S)$ and $\mathbb{E}\sup_\theta(R_S - R)$ by $2\mathbb{E}\widehat{\mathfrak{R}}_S$. Replacing one sample changes either supremum by at most M_ℓ/n; applying bounded differences to both tails proves Eq. (653). Finally, for $0 < \varepsilon \le d$,

$$\log(1 + A/\varepsilon) \le \log(1 + A/d) + \log(d/\varepsilon), \tag{663}$$

$$\int_0^d \sqrt{\log(d/\varepsilon)}\, d\varepsilon = d\,\Gamma(3/2) = d\sqrt{\pi}/2. \tag{664}$$

Use $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$ to obtain the explicit expression. □

For fixed domain constants, chaining removes the extra $\sqrt{\log n}$ introduced by choosing an $n^{-1/2}$-resolution net. The integral is the primary bound; the final closed form is only a convenient relaxation. To retain the strongest certificate at each sample size, define the deterministic envelope

$$\mathfrak{s}_n^{\mathrm{cov}}(\varepsilon, \delta) := 2L_\Theta \varepsilon + M_\ell \sqrt{\frac{\log(2/\delta) + \mathcal{H}_\varepsilon}{2n}}, \tag{665}$$

$$\mathfrak{s}_n^\sharp(\delta) := \min\left\{ \inf_{\varepsilon > 0} \mathfrak{s}_n^{\mathrm{cov}}(\varepsilon, \delta), \mathfrak{s}_n^{\mathrm{chain}}(\delta) \right\}. \tag{666}$$

The same $1 - \delta$ guarantee holds with this envelope: its alternatives are deterministic bounds on the same scalar uniform deviation, so selecting the smaller one requires no additional union bound. No minimax-optimal numerical constant is claimed for the chaining inequality.

The entropy bound charges representation dimension and the G curvature coordinates separately. Thus grouped curvature increases capacity and also incurs an estimation cost. This bound can be used for any data-dependent output in the domain, even when the trajectory-dependent stability bound is large, but it can itself be loose in high dimension. A trainable encoder must contribute its own parameter complexity as follows.

Corollary A.93 (Additional encoder parameters)**.** *If a trainable encoder belongs to a Euclidean radius-R_e domain in $\mathbb{R}^{D_e}$ and the loss is jointly Lipschitz in the product parameter metric, an additional valid entropy term is*

$$D_e \log(1 + 6R_e/\varepsilon). \tag{667}$$

For this three-factor product, replace the constants 4 in Eq. (640) by 6 and take radius-$\varepsilon/3$ nets.

Proof.

$$\sqrt{3(\varepsilon/3)^2} \le \varepsilon, \qquad \mathcal{N}(B_{D_e}(0, R_e), \|\cdot\|, \varepsilon/3) \le (1 + 6R_e/\varepsilon)^{D_e}. \tag{668}$$

□

Proposition A.94 (Independent validation selection)**.** *For M candidates independent of n_v validation examples and losses in $[0, M_\ell]$, define*

$$a_v = M_\ell \sqrt{\frac{\log(2M/\delta)}{2n_v}}, \qquad \widehat{j} \in \arg\min_{j \le M} \widehat{R}_v(\theta_j). \tag{669}$$

With probability at least $1-\delta$,

$$\max_j |R(\theta_j) - \widehat{R}_v(\theta_j)| \le a_v, \tag{670}$$

$$R(\theta_{\widehat{j}}) \le \min_j R(\theta_j) + 2a_v. \tag{671}$$

Proof.

$$\Pr\left(\max_j |R(\theta_j) - \widehat{R}_v(\theta_j)| > a_v\right) \le 2Me^{-2n_v a_v^2/M_\ell^2} = \delta, \tag{672}$$

$$R(\theta_{\widehat{j}}) \le \widehat{R}_v(\theta_{\widehat{j}}) + a_v \le \widehat{R}_v(\theta_{j^*}) + a_v \le R(\theta_{j^*}) + 2a_v. \tag{673}$$

□

Proposition A.95 (Distribution shift)**.** *For $0 \le \ell \le M_\ell$ and $\|P-Q\|_{\mathrm{TV}} := \sup_A |P(A) - Q(A)|$,*

$$|R_Q(\theta) - R_P(\theta)| \le M_\ell \|P - Q\|_{\mathrm{TV}}, \tag{674}$$

$$|\mathbb{E}[R_Q(A(S)) - R_S(A(S))]| \le \epsilon_{\mathrm{stab}} + M_\ell \|P - Q\|_{\mathrm{TV}}, \qquad S \sim P^n. \tag{675}$$

Proof.

$$R_Q(\theta) - R_P(\theta) = \int_0^{M_\ell} [Q\{\ell(\theta; Z) > t\} - P\{\ell(\theta; Z) > t\}]\, dt, \tag{676}$$

$$|R_Q - R_P| \le \int_0^{M_\ell} \|P - Q\|_{\mathrm{TV}}\, dt. \tag{677}$$

□

A.10. Local spectrum of the coupled momentum–curvature dynamics

Finite-time descent controls averaged stationarity but does not establish attraction to a particular critical point. We now examine the deterministic interior update near such a point. Its augmented Jacobian retains the mixed Hessian blocks, which determine whether representation and curvature updates stabilize or destabilize one another. The spectral calculation is followed by a quantitative nonlinear neighborhood argument.

Assumption A.96 (Interior critical point)**.** *Let (q_*, u_*) satisfy*

$$\mathrm{grad}_q F(q_*, u_*) = 0, \qquad \nabla_u F(q_*, u_*) = 0, \tag{678}$$

$$H = \begin{pmatrix} H_{qq} & H_{qu} \\ H_{uq} & H_{uu} \end{pmatrix}, \qquad A_* = A(u_*) \succ 0. \tag{679}$$

Use normal coordinates ξ centered at q_* and the exact deterministic interior dynamics in Eq. (304).

Proposition A.97 (Linearized augmented update)**.** *For the state $(\xi, u - u_*, m)$ with m the carried canonical momentum,*

$$\mathcal{T}(0, u_*, 0) = (0, u_*, 0), \qquad \delta_\beta = 1 - \beta, \tag{680}$$

$$D\mathcal{T}_* = \begin{pmatrix} I - \eta\delta_\beta A_* H_{qq} & -\eta\delta_\beta A_* H_{qu} & -\eta\beta I \\ -\rho H_{uq} & I - \rho H_{uu} & 0 \\ \delta_\beta A_* H_{qq} & \delta_\beta A_* H_{qu} & \beta I \end{pmatrix}. \tag{681}$$

Proof. At the critical point,

$$\operatorname{grad}_q F(q,u) = H_{qq}\xi + H_{qu}(u-u_*) + o(\|\xi\| + \|u-u_*\|), \tag{682}$$
$$\nabla_u F(q,u) = H_{uq}\xi + H_{uu}(u-u_*) + o(\|\xi\| + \|u-u_*\|), \tag{683}$$
$$D[A(u)\operatorname{grad}_q F]_* = A_* D \operatorname{grad}_q F, \tag{684}$$
$$N_\gamma(0) = 0, \qquad DN_\gamma(0) = I, \tag{685}$$
$$\bar{m} = \beta m + \delta_\beta A_*[H_{qq}\xi + H_{qu}(u-u_*)] + o(\|\text{state}\|), \tag{686}$$
$$\xi^+ = \xi - \eta\bar{m} + o(\|\text{state}\|), \tag{687}$$
$$m^+ = \bar{m} + o(\|\text{state}\|), \tag{688}$$
$$u^+ - u_* = (u-u_*) - \rho[H_{uq}\xi + H_{uu}(u-u_*)] + o(\|\text{state}\|). \tag{689}$$

□

Proposition A.98 (Characteristic block polynomial)**.** *For $\zeta \neq 0$, $\zeta \in \sigma(D\mathcal{T}_*)$ if and only if*

$$\det \begin{pmatrix} (\zeta-\beta)(\zeta-1)I + \eta\zeta\delta_\beta A_* H_{qq} & \eta\zeta\delta_\beta A_* H_{qu} \\ \rho H_{uq} & (\zeta-1)I + \rho H_{uu} \end{pmatrix} = 0. \tag{690}$$

Proof. For an eigenvector (ξ, v, m),

$$\zeta\xi = \xi - \eta\zeta m, \qquad \zeta m = \beta m + \delta_\beta A_*(H_{qq}\xi + H_{qu}v), \tag{691}$$
$$m = \frac{1-\zeta}{\eta\zeta}\xi, \tag{692}$$
$$[(\zeta-\beta)(\zeta-1)I + \eta\zeta\delta_\beta A_* H_{qq}]\xi + \eta\zeta\delta_\beta A_* H_{qu} v = 0, \tag{693}$$
$$\rho H_{uq}\xi + [(\zeta-1)I + \rho H_{uu}]v = 0. \tag{694}$$

Conversely, a nonzero solution (ξ, v) of the last two equations determines m by the third line. □

Corollary A.99 (Uncoupled modal stability)**.** *If $H_{qu} = 0$, let*

$$\lambda_j \in \sigma(A_*^{1/2} H_{qq} A_*^{1/2}), \qquad \mu_k \in \sigma(H_{uu}). \tag{695}$$

The representation modes solve

$$\zeta^2 - [1 + \beta - \eta(1-\beta)\lambda_j]\zeta + \beta = 0, \tag{696}$$

and the curvature modes are $\zeta_k = 1 - \rho\mu_k$. For $0 \leq \beta < 1$, all these modes lie in the open unit disk if and only if

$$0 < \eta(1-\beta)\lambda_j < 2(1+\beta) \quad (\forall j), \tag{697}$$
$$0 < \rho\mu_k < 2 \quad (\forall k). \tag{698}$$

Proof. For $p(\zeta) = \zeta^2 - a\zeta + \beta$, the real quadratic Schur conditions are

$$1 - a + \beta > 0, \qquad 1 + a + \beta > 0, \qquad 1 - \beta > 0. \tag{699}$$

Substitution of $a = 1 + \beta - \eta(1-\beta)\lambda_j$ gives the first condition. The scalar curvature condition is $|1 - \rho\mu_k| < 1$. □

Proposition A.100 (Zero-momentum coupled spectrum)**.** *For $\beta = 0$, let $D = \operatorname{diag}(\eta A_*, \rho I)$. The prediction-state Jacobian is $I - DH$ and the additional momentum eigenvalues are zero:*

$$\sigma(I - DH) = \sigma(I - D^{1/2} H D^{1/2}). \tag{700}$$

Thus local Schur stability is equivalent to

$$0 \prec D^{1/2} H D^{1/2} \prec 2I. \tag{701}$$

Moreover,

$$H \succ 0 \iff H_{qq} \succ 0, \qquad H_{uu} - H_{uq} H_{qq}^{-1} H_{qu} \succ 0. \tag{702}$$

Proof.

$$D^{-1/2}(I - DH)D^{1/2} = I - D^{1/2}HD^{1/2}, \tag{703}$$

$$\begin{pmatrix} I & 0 \\ -H_{uq}H_{qq}^{-1} & I \end{pmatrix} H \begin{pmatrix} I & -H_{qq}^{-1}H_{qu} \\ 0 & I \end{pmatrix} = \operatorname{diag}(H_{qq}, H_{uu} - H_{uq}H_{qq}^{-1}H_{qu}). \tag{704}$$

□

The Schur complement inertia criterion (Horn & Johnson, 2013) connects the profiled objective to the coupled stability test. It accounts for the representation's response to curvature changes. In particular, separate positive diagonal Hessian blocks are insufficient if their coupling makes this complement indefinite. Normalization has a different role locally: its derivative at zero momentum is the identity, so its influence begins beyond first order.

Corollary A.101 (Normalization and linearization)**.** *For every $\gamma \geq 0$,*

$$DN_\gamma(0) = I, \qquad \partial_\gamma D\mathcal{T}_* = 0. \tag{705}$$

Curvature–momentum normalization changes finite-displacement sensitivity, while its first-order linearization at a zero-momentum critical point is independent of γ.

Corollary A.102 (Softplus-coordinate linearization)**.** *For $B_* = \operatorname{diag}(\psi'(c_{g,*})^2)$ and the softplus update of Proposition A.12, replace the curvature row of Eq. (681) by*

$$\big(-\rho B_* H_{uq},\ I - \rho B_* H_{uu},\ 0\big). \tag{706}$$

The zero-momentum spectral criterion holds with $D = \operatorname{diag}(\eta A_*, \rho B_*)$.

Proof.

$$D[B(u)\nabla_u F]_* = B_* D\nabla_u F + DB_*\nabla_u F_* = B_* D\nabla_u F, \tag{707}$$

$$R_c = O(\rho^2 \|\nabla_u F\|^2) = o(\|\text{state}\|). \tag{708}$$

□

Theorem A.103 (Slow-curvature eigenvalues and the profiled Hessian)**.** *Fix $\eta > 0$ and $0 \leq \beta < 1$. Assume $H_{qq} \succ 0$ and that every eigenvalue λ_j of $A_*^{1/2} H_{qq} A_*^{1/2}$ satisfies*

$$0 < \eta(1 - \beta)\lambda_j < 2(1 + \beta). \tag{709}$$

Let the symmetric profiled Hessian

$$S_* := H_{uu} - H_{uq}H_{qq}^{-1}H_{qu} \tag{710}$$

have simple eigenvalues $\mu_1, \ldots, \mu_G$. As $\rho \downarrow 0$, exactly G eigenvalues of the augmented Jacobian approach one, with

$$\zeta_k(\rho) = 1 - \rho\mu_k + O(\rho^2), \qquad k = 1, \ldots, G. \tag{711}$$

All remaining eigenvalues stay strictly inside the unit disk for sufficiently small ρ. Consequently, $S_* \succ 0$ implies local Schur stability for all sufficiently small positive ρ; any $\mu_k < 0$ implies instability for all sufficiently small positive ρ. For a positive softplus preconditioner B_*, the same statement holds with μ_k the simple eigenvalues of $B_*^{1/2} S_* B_*^{1/2}$.

Proof. At $\rho = 0$, the curvature state is fixed. The representation–momentum subsystem is Schur stable by Corollary A.99, so its spectrum is separated from one. The G eigenvalues at one are semisimple: writing the state in representation–momentum and curvature blocks gives

$$J_0 = \begin{pmatrix} J_f & C_f \\ 0 & I \end{pmatrix}, \qquad 1 \notin \sigma(J_f), \tag{712}$$

$$\begin{pmatrix} I & -(I - J_f)^{-1}C_f \\ 0 & I \end{pmatrix} J_0 \begin{pmatrix} I & (I - J_f)^{-1}C_f \\ 0 & I \end{pmatrix} = \operatorname{diag}(J_f, I). \tag{713}$$

For ζ near one, set $\delta_\beta = 1 - \beta$ and

$$Q(\zeta) = (\zeta - \beta)(\zeta - 1)I + \eta\zeta\delta_\beta A_* H_{qq}, \tag{714}$$

$$R(\zeta) = \eta\zeta\delta_\beta A_* H_{qu}, \tag{715}$$

$$Q(1)^{-1}R(1) = H_{qq}^{-1}H_{qu}. \tag{716}$$

Eliminating the representation component in the characteristic block polynomial yields

$$0 = \det\big[(\zeta - 1)I + \rho S(\zeta)\big], \tag{717}$$

$$S(\zeta) := H_{uu} - H_{uq}Q(\zeta)^{-1}R(\zeta), \qquad S(1) = S_*. \tag{718}$$

For $\zeta = 1 + \rho\nu$, divide by ρ^G and use analyticity of $Q^{-1}R$ near one:

$$0 = \det[\nu I + S_* + O(\rho)], \tag{719}$$

$$\nu_k(\rho) = -\mu_k + O(\rho). \tag{720}$$

The simple-root implicit function theorem gives the last line. The remaining eigenvalues remain separated from the unit circle by continuity. If $\mu_k > 0$, then $|1 - \rho\mu_k + O(\rho^2)| < 1$ for small positive ρ; if $\mu_k < 0$, this modulus exceeds one. For the softplus update, elimination replaces $S(\zeta)$ by $B_* S(\zeta)$, and

$$B_* S_* = B_*^{1/2}(B_*^{1/2} S_* B_*^{1/2})B_*^{-1/2}. \tag{721}$$

The same simple-root argument therefore applies. The positive-definiteness and negative-eigenvalue sign conclusions follow by congruence. □

This expansion explains what a slower curvature update can and cannot do. Once the representation–momentum subsystem is stable, the leading slow modes depend on the profiled Hessian, with η and β entering higher-order terms. Reducing ρ can preserve stability when that Hessian is positive, but it slows the corresponding contraction and cannot stabilize a negative profiled direction. A zero eigenvalue requires higher-order analysis and is not covered by the sign conclusion.

Theorem A.104 (Quantitative nonlinear attraction from the augmented spectrum)**.** *Center the full normal-coordinate state at the fixed point, so $s = (\xi, u - u_*, m)$ and $\mathcal{T}(0) = 0$. Suppose the exact deterministic interior map is $C^{1,1}$ near zero and*

$$J_* := D\mathcal{T}(0), \qquad \operatorname{spr}(J_*) < 1. \tag{722}$$

Define

$$W_* := \sum_{k=0}^{\infty} (J_*^k)^\top J_*^k, \qquad \|s\|_{W_*} = (s^\top W_* s)^{1/2}, \tag{723}$$

$$\|B\|_{W_*} := \sup_{s \neq 0} \|Bs\|_{W_*}/\|s\|_{W_*}, \qquad \alpha_* := \sqrt{1 - \lambda_{\max}(W_*)^{-1}}. \tag{724}$$

Let the closed W_*-ball of radius r_0 lie in the chart and in the interior update region, and choose $L_J > 0$ such that

$$\|D\mathcal{T}(s) - J_*\|_{W_*} \le L_J \|s\|_{W_*} \qquad (\|s\|_{W_*} \le r_0). \tag{725}$$

For any radius satisfying

$$0 < r \le r_0, \qquad L_J r < 2(1 - \alpha_*), \qquad \vartheta := \alpha_* + L_J r/2 < 1, \tag{726}$$

the ball is forward invariant and

$$\|s_t\|_{W_*} \le \vartheta^t \|s_0\|_{W_*}, \tag{727}$$

$$\|s_t\|_2 \le \sqrt{\frac{\lambda_{\max}(W_*)}{\lambda_{\min}(W_*)}}\vartheta^t \|s_0\|_2 \qquad (\|s_0\|_{W_*} \le r). \tag{728}$$

Between any two states in this ball, its Lipschitz constant is at most $\alpha_* + L_J r$; hence pairwise contraction holds under the additional restriction $L_J r < 1 - \alpha_*$.

Proof. Stein's discrete Lyapunov theorem (Stein, 1952, Theorem 1) motivates a quadratic metric adapted to the augmented dynamics. Schur stability makes the defining series converge. Shifting its index gives

$$W_* \succeq I, \qquad J_*^\top W_* J_* = W_* - I, \tag{729}$$

$$\|J_* s\|_{W_*}^2 = \|s\|_{W_*}^2 - \|s\|_2^2 \le \big(1 - \lambda_{\max}(W_*)^{-1}\big)\|s\|_{W_*}^2. \tag{730}$$

For attraction to the fixed point, retaining the distance along the integration segment improves the remainder constant:

$$\mathcal{T}(s) = J_* s + \int_0^1 [D\mathcal{T}(as) - J_*]s \, da, \tag{731}$$

$$\|\mathcal{T}(s)\|_{W_*} \le \alpha_* \|s\|_{W_*} + \tfrac{1}{2} L_J \|s\|_{W_*}^2 \le \vartheta \|s\|_{W_*}. \tag{732}$$

This proves forward invariance and permits iteration. For s, s' in the ball, convexity gives the separate pairwise estimate

$$\mathcal{T}(s) - \mathcal{T}(s') = \int_0^1 D\mathcal{T}(s' + a(s - s'))(s - s') \, da, \tag{733}$$

$$\|\mathcal{T}(s) - \mathcal{T}(s')\|_{W_*} \le (\alpha_* + L_J r)\|s - s'\|_{W_*}. \tag{734}$$

The Euclidean bound follows from

$$\sqrt{\lambda_{\min}(W_*)}\|s\|_2 \le \|s\|_{W_*} \le \sqrt{\lambda_{\max}(W_*)}\|s\|_2. \tag{735}$$

□

Corollary A.105 (Best quadratic contraction factor)**.** *For any $\nu \in (\mathrm{spr}(J_*), 1)$, the matrix*

$$W_\nu := \sum_{k \ge 0} \nu^{-2k} (J_*^k)^\top J_*^k, \tag{736}$$

$$J_*^\top W_\nu J_* = \nu^2 (W_\nu - I) \prec \nu^2 W_\nu \tag{737}$$

is a valid Lyapunov metric. In particular,

$$\inf_{W \succ 0} \|J_*\|_W = \mathrm{spr}(J_*). \tag{738}$$

The local theorem applies with α_* replaced by any certified $\|J_*\|_W < 1$ and with the corresponding W-norm derivative constant.

Proof. Apply the convergent Stein series to J_*/ν. Every induced norm dominates the spectral radius; the displayed family approaches that lower bound as $\nu \downarrow \mathrm{spr}(J_*)$. □

Corollary A.106 (Local robustness to state perturbations)**.** *Under Theorem A.104, consider coordinate-state perturbations*

$$s_{t+1} = \mathcal{T}(s_t) + e_t, \qquad \|e_t\|_{W_*} \le \bar{e}, \tag{739}$$

$$\|s_0\|_{W_*} \le r, \qquad \bar{e} \le (1 - \vartheta) r. \tag{740}$$

Then every iterate stays in the ball, and

$$\|s_t\|_{W_*} \le \vartheta^t \|s_0\|_{W_*} + \sum_{j<t} \vartheta^{t-1-j} \|e_j\|_{W_*} \tag{741}$$

$$\le \vartheta^t \|s_0\|_{W_*} + \frac{\bar{e}}{1 - \vartheta}(1 - \vartheta^t). \tag{742}$$

If $\|e_t\|_{W_*} \to 0$, then $s_t \to 0$.

Proof. The recurrence $\|s_{t+1}\|_{W_*} \le \vartheta \|s_t\|_{W_*} + \bar{e}$ preserves the radius r. Iteration gives the convolution. For vanishing perturbations, split this convolution at a fixed index: the finite prefix tends to zero geometrically, while the tail is at most $\sup_{j \ge N} \|e_j\|_{W_*}/(1 - \vartheta)$, which tends to zero as $N \to \infty$. □

The Lyapunov norm handles transient amplification that a Euclidean norm can miss when the augmented Jacobian is nonnormal. Its condition number and the nonlinear constant L_J determine the certified neighborhood. In a scalar coordinate normalized to the chosen quadratic metric, the map $y^+ = \alpha y + L_J y^2/2, 0 \le \alpha < 1$, has a second fixed point at $2(1-\alpha)/L_J$. Thus the factor 2 in the attraction-radius condition cannot be increased using only the stated derivative bound for a general $C^{1,1}$ map. Although γ is absent from J_*, it can affect L_J and therefore the neighborhood and perturbation tolerance. These are local deterministic statements; applying them to neighboring stochastic runs requires a common invariant neighborhood and suitable per-step contraction and injection bounds.

A.11. PL objective error, oracle complexity, and a conditional risk comparison

The Polyak–Łojasiewicz (PL) condition (Karimi et al., 2016) converts stationarity control into an objective-error statement without asserting that every model satisfies it. The contraction is governed by the smaller effective block step, and its forcing terms retain momentum lag and sampling noise. We then combine optimization error with a uniform statistical bound to state a sufficient condition for a grouped-curvature population-risk advantage.

Assumption A.107 (Joint PL condition)**.** *In addition to Assumption A.37, let*

$$\|g_t\|^2 + \|b_t\|^2 \ge 2\mu(F(q_t,u_t) - F_*), \qquad \mu > 0. \tag{743}$$

Use constant $\eta, \rho, \beta, \mathfrak{b}$ and put

$$\chi = \min(a\eta, \rho), \qquad r = 1 - \mu\chi \in [0,1), \tag{744}$$

$$D_* = L_{vq} M_N \eta + L_{vu} H_0 \rho, \tag{745}$$

$$Z_\infty^2 = \frac{\beta^2 D_*^2}{(1-\beta)^2} + \frac{(1-\beta)a_+^2\sigma_q^2}{\mathfrak{b}}, \tag{746}$$

$$Q_* = C_e \eta Z_\infty^2 + \frac{L\rho^2\sigma_u^2}{2\mathfrak{b}}. \tag{747}$$

Theorem A.108 (PL contraction with explicit momentum transient)**.** *Let $v_\xi = a_+^2\sigma_q^2/\mathfrak{b}$. Separate initial memory and filtered noise from the drift envelope:*

$$\mathfrak{v}_t := \beta^{2(t+1)}\mathbb{E}\|e_{-1}^m\|^2 + \frac{1-\beta}{1+\beta}(1-\beta^{2(t+1)})v_\xi, \tag{748}$$

$$\mathfrak{d}_t := \frac{\beta(1-\beta^t)}{1-\beta} D_*, \tag{749}$$

$$\mathfrak{z}_t := \min\left\{\beta^{t+1}\mathbb{E}\|e_{-1}^m\|^2 + Z_\infty^2, (\sqrt{\mathfrak{v}_t} + \mathfrak{d}_t)^2\right\}. \tag{750}$$

Then the finite-time convolution bound is

$$\mathbb{E}(F_T - F_*) \le r^T \Delta_F + C_e\eta \sum_{t<T} r^{T-1-t}\mathfrak{z}_t + \frac{L\rho^2\sigma_u^2}{2\mathfrak{b}} \sum_{t<T} r^{T-1-t}. \tag{751}$$

In particular, the simpler closed form is

$$\mathbb{E}(F_T - F_*) \le r^T \Delta_F + C_e\eta\mathbb{E}\|e_{-1}^m\|^2\, \mathcal{C}_T(r,\beta) + \frac{Q_*}{\mu\chi}(1-r^T), \tag{752}$$

$$\mathcal{C}_T(r,\beta) := \begin{cases} \dfrac{\beta(r^T - \beta^T)}{r-\beta}, & r \ne \beta, \\ T r^T, & r = \beta. \end{cases} \tag{753}$$

Proof. The predictable-frame decomposition of Theorem A.43, now at a fixed time, gives

$$(\mathbb{E}\|e_t^m\|^2)^{1/2} \le \sqrt{\mathfrak{v}_t} + \mathfrak{d}_t. \tag{754}$$

The earlier tracking recursion also gives

$$\mathbb{E}\|e_t^m\|^2 \le \beta^{t+1}\mathbb{E}\|e_{-1}^m\|^2 + Z_\infty^2. \tag{755}$$

Their minimum is $\mathfrak{z}_t$. Unrolling the next descent inequality with that certificate proves Eq. (751); using the first alternative gives the closed form. The normalized descent and PL inequalities imply

$$\mathbb{E}(F_{t+1} - F_*) \le \mathbb{E}(F_t - F_*) - \frac{\chi}{2}\mathbb{E}(\|g_t\|^2 + \|b_t\|^2) + C_e\eta\mathbb{E}\|e_t^m\|^2 + \frac{L\rho^2\sigma_u^2}{2\mathfrak{b}} \tag{756}$$

$$\le r\mathbb{E}(F_t - F_*) + C_e\eta\beta^{t+1}\mathbb{E}\|e_{-1}^m\|^2 + Q_*, \tag{757}$$

$$\sum_{t<T} r^{T-1-t}\beta^{t+1} = \mathcal{C}_T(r,\beta), \tag{758}$$

$$\sum_{t<T} r^{T-1-t} = (1-r^T)/(1-r). \tag{759}$$

□

The transient is a convolution of the objective contraction and the memory decay. When these rates coincide, the factor Tr^T replaces a single geometric term. With constant steps and nonzero noise or drift, the theorem gives an error neighborhood rather than exact convergence; the explicit floor shows which quantities must decrease to shrink it.

Corollary A.109 (Squared-stationarity oracle budgets)**.** *Let $\tau \in (0,1)$ denote a target for an expected squared stationarity measure. With constants independent of T,*

$$\eta \asymp T^{-1/2}, \quad \rho \asymp T^{-3/4}, \quad \mathfrak{b} \asymp T^{1/2} \Longrightarrow \begin{cases} T = O(\tau^{-4}), \\ \textit{sampled-gradient calls} = O(\tau^{-6}), \end{cases} \tag{760}$$

$$\eta, \rho \asymp T^{-3/5}, \quad 1-\beta \asymp T^{-2/5}, \quad \mathfrak{b} \asymp 1 \Longrightarrow \begin{cases} T = O(\tau^{-5/2}), \\ \textit{sampled-gradient calls} = O(\tau^{-5/2}). \end{cases} \tag{761}$$

The first line controls both representation and curvature averages; the second line concerns the interior exact update of Theorem A.45.

Proof.

$$C_qT^{-1/2} + C_uT^{-1/4} \le \tau \Longleftarrow T \ge \max[(2C_q/\tau)^2, (2C_u/\tau)^4], \tag{762}$$

$$CT^{-2/5} \le \tau \Longleftarrow T \ge (C/\tau)^{5/2}, \tag{763}$$

$$T\mathfrak{b} \asymp T^{3/2} \quad \text{or} \quad T. \tag{764}$$

□

Definition A.110 (Nested curvature classes)**.** *Let*

$$\Theta_{\mathrm{grp}} = \mathcal{Q}_B \times U, \qquad \Theta_{\mathrm{shared}} = \{(q,u) \in \Theta_{\mathrm{grp}} : u_1 = \cdots = u_G\}, \tag{765}$$

$$R^*_{\mathrm{grp}} = \inf_{\theta\in\Theta_{\mathrm{grp}}} R(\theta), \qquad R^*_{\mathrm{shared}} = \inf_{\theta\in\Theta_{\mathrm{shared}}} R(\theta), \tag{766}$$

$$\Delta_* = R^*_{\mathrm{shared}} - R^*_{\mathrm{grp}} \ge 0, \tag{767}$$

$$\delta_{\mathrm{opt}}(S,\omega) = R_S(\widehat{\theta}_{\mathrm{grp}}) - \inf_{\theta\in\Theta_{\mathrm{grp}}} R_S(\theta) \ge 0. \tag{768}$$

The approximation gain Δ_* is a population property, whereas the output is obtained by empirical optimization. To turn a larger hypothesis class into a risk advantage, its gain must exceed both the optimization error and the statistical deviation. The following sufficient condition makes that comparison quantitative; heterogeneity alone does not certify an advantage at a given sample size or training horizon.

Theorem A.111 (Approximation–optimization–estimation comparison)**.** *Let $\mathfrak{s}_n(\delta_s) := \mathfrak{s}_n^\sharp(\delta_s)$ be the optimized deterministic envelope in Eq. (666), and suppose*

$$\mathbb{E}\delta_{\mathrm{opt}}(S,\omega) \le \mathfrak{o}_T. \tag{769}$$

For every data-dependent $\widehat{\theta}_{\mathrm{shared}} \in \Theta_{\mathrm{shared}}$*, with probability at least* $1-\delta_s-\delta_o$*,*

$$R(\widehat{\theta}_{\mathrm{grp}}) - R(\widehat{\theta}_{\mathrm{shared}}) \le -\Delta_* + \mathfrak{o}_T/\delta_o + 2\mathfrak{s}_n(\delta_s). \tag{770}$$

If $U = I^G$ *for a common curvature interval* I *and the population profile separates as*

$$\inf_q R(q,u) = \sum_g w_g f_g(u_g), \qquad 0 < \mu_g \le f_g'' \le L_g \text{ on } I, \qquad f_g'(u_g^*) = 0, \quad u_g^* \in I, \tag{771}$$

with $w_g > 0$*,* $\sum_g w_g = 1$ *and* $\Theta_{\mathrm{shared}} \neq \varnothing$*, set* $a_g = w_g\mu_g$ *and* $u_a = (\sum_g a_g u_g^*)/(\sum_g a_g)$*. A sufficient condition for strict population-risk improvement is*

$$\frac{1}{2}\sum_g a_g(u_g^* - u_a)^2 > \frac{\mathfrak{o}_T}{\delta_o} + 2\mathfrak{s}_n(\delta_s). \tag{772}$$

Proof. By Markov's inequality and uniform generalization,

$$\Pr(\delta_{\mathrm{opt}} > \mathfrak{o}_T/\delta_o) \le \delta_o, \tag{773}$$

$$\Pr\left(\sup_{\theta\in\Theta_{\mathrm{grp}}} |R(\theta) - R_S(\theta)| > \mathfrak{s}_n(\delta_s)\right) \le \delta_s. \tag{774}$$

On their joint complement, for $\theta^* \in \Theta_{\mathrm{grp}}$,

$$R(\widehat{\theta}_{\mathrm{grp}}) \le R_S(\widehat{\theta}_{\mathrm{grp}}) + \mathfrak{s}_n \tag{775}$$

$$\le R_S(\theta^*) + \delta_{\mathrm{opt}} + \mathfrak{s}_n \tag{776}$$

$$\le R(\theta^*) + \mathfrak{o}_T/\delta_o + 2\mathfrak{s}_n, \tag{777}$$

$$R(\widehat{\theta}_{\mathrm{shared}}) \ge R^*_{\mathrm{shared}} = R^*_{\mathrm{grp}} + \Delta_*. \tag{778}$$

Take a minimizing sequence $R(\theta^*) \downarrow R^*_{\mathrm{grp}}$ and apply Proposition A.27 to the separated population profile. □

B. Dataset and Metric Details

WN18RR and FB15k-237 are evaluated as link-prediction knowledge graph datasets using Lorentz distance scoring with negative sampling. The WN18RR main comparison uses dimension 20 and 2000 epochs. Because evaluation occurs every 25 epochs, last-5 and last-10 moving averages correspond to the final 125 and 250 epochs respectively. Synthetic hierarchy experiments use four groups unless otherwise stated, with the principal variable-branching setting using branchings $[2, 4, 6, 8]$ and the heterogeneity stress test using branchings $[2, 8, 32, 128]$.

The primary WN18RR table in `tables/table1_kg_wn18rr.csv` records B3 last-5 MRR 0.1332, B4 last-5 MRR 0.1345, B3 last-5 Hits@10 0.3510, B4 last-5 Hits@10 0.3652, B3 loss 2.1274, B4 loss 2.0680, B4 $K_{\mathrm{mean}} = 1.2563$, and B4 $K_{\mathrm{std}} = 0.2948$. The corresponding relation-curvature table records the high-to-low WN18RR range 2.000 to 1.0084. The synthetic convergence table records the exact checkpoints listed in Table 6.

C. Implementation Notes

The revision base methods use pure FP32 Lorentz operations with every-step projection back to the manifold and tangent space; A6 explicitly disables tangent projection. Historical implementation details vary by script, and the KG Adam protocol is described in Section 6. The optimizer applies log-curvature clipping and a finite $K_{\max}$ cap; WN18RR uses the boundary-like high-curvature regime near $K = 2$, whereas synthetic runs often remain interior. B3 freezes every group at $K = 1$, B4 learns one curvature per group or relation, and B5 additionally transports momentum across curvature changes only in grouped-manifold synthetic experiments.

D. Additional Ablation Tables

For the synthetic rho sweep at 1000 epochs, B3 has distortion 0.24440. B4 changes from 0.24427 at $\rho = 10^{-3}$ to 0.23984 at $\rho = 10^{-2}$ and 0.23570 at $\rho = 10^{-1}$, while B4 loss changes from 0.13537 to 0.13452 and then 0.13689. B5 tracks B4 closely:

Table 13. **Illustrative FP32 conditioning proxy.** At unit curvature, $\cosh^2(r)2^{-23}$ illustrates radius amplification. It is not a measured error or a universally attained lower bound; the rigorous roundoff bound is stated in Appendix A.

Radius r	$\cosh^2(r)2^{-23}$
0	1.192e-07
2	1.687e-06
4	8.890e-05
5	6.565e-04
8	2.648e-01
10	1.446e+01

0.24387, 0.23948, and 0.23508 across the same rho values. For the lambda sweep, $\lambda \in \{0, 10^{-5}, 10^{-4}, 10^{-3}\}$ leaves B4 at 0.24427 with $K_{\text{std}} = 0.0077$, while the qualitative $\lambda = 1.0$ 500-epoch run has distortion 0.30139 and $K_{\text{std}} = 0.0019$.

For the 2000-epoch synthetic main table, the base variable-branching runs end at last-valid distortions 0.22604 for B3, 0.22638 for B4, and 0.22642 for B5; these values are effectively tied and support the statement that the synthetic benefit is not a final-value win. In the convergence-tuned runs, B3 reaches 0.22641, B4 reaches 0.22569, and B5 reaches 0.22555 at epoch 1999. These small terminal differences are much smaller than the mid-training gap and should be interpreted together with the FP32 floor.

E. Complete Experimental Scope and Outcome Audit

The main text focuses on the stronger observed conditions and states that selection explicitly. This appendix preserves the full evaluated scope: mixed graph results, historical negative ablations, all six mechanism cases, and every completed prospective real-data task. Registration and successful short path checks are not benchmark completion. Every original run and configuration remains in the execution ledger.

E.1. Ablations and numerical behavior

The rho sweep shows that curvature learning must be strong enough to move K but not so aggressive that fitting loss degrades. At 1000 epochs, increasing ρ from 10^{-3} to 10^{-2} to 10^{-1} increases B4 K_{std} from 0.008 to 0.065 to 0.150 and decreases B4 distortion from 0.24427 to 0.23984 to 0.23570, but the loss at 10^{-1} rises to 0.13689 compared with 0.13452 at 10^{-2}. The chosen interpretation is that $\rho = 10^{-2}$ is the distortion/loss trade-off point, not that the largest curvature learning rate is uniformly best. The lambda sweep shows that $\lambda \in [0, 10^{-3}]$ is inactive in this setting, with B4 distortion 0.24427 and $K_{\text{std}} = 0.008$ across those values; forcing $\lambda = 1.0$ at 500 epochs collapses K_{std} to 0.002 and worsens distortion to 0.30139, although the shorter run length makes it only a qualitative warning. The WN18RR $K_{\max}$ sweep at 500 epochs shows saturation near $K \approx 2$: $K_{\max} = 5$ and 10 match, while $K_{\max} = 2$ is slightly worse in loss, so large caps do not automatically improve the model.

The B5-vs-B4 ablation is negative. On the synthetic 2000-epoch convergence run, B5 final distortion 0.22555 is essentially tied with B4 final distortion 0.22569. In the rho sweep, B5 and B4 differ by less than 0.3% in final distortion across all three rho values. In the heterogeneity sweep, B5 and B4 differ by less than 0.15%. Moreover, B5 has no early advantage at epochs 50–100 in Table 6, consistent with early momentum overshoot. We report this because it defines the architecture scope of the method: B5 is mathematically coherent in grouped-manifold architectures, but this comparison does not isolate transport from momentum and step normalization. The clean transport comparison is B5 versus A5.

E.2. benchmark revision: full-data graph comparison and coverage

The revision suite registers every dataset–model combination in proposal Section 6, including all four synthetic tree types at 1K/5K/10K nodes and dimensions 2/5/10/20. It additionally implements the A1–A12 mechanism checks on Cora/GCN, with separate flags for momentum, shared curvature, normalization, tangency projection, update-rate ratio, log-step clipping, curvature-update frequency, and initialization. A3 and A11 coincide for this prototype-head architecture and are labeled accordingly. Registration is not completion: the machine-readable coverage file in `tables/revision_coverage.csv` distinguishes completed, claimed, pending, deferred-by-scope, unavailable-data, and failed runs. A claim is checked against a live process before being described as running. The subsequent budget amendment in Appendix E.5 changed which

Table 14. benchmark revision: Cora/GCN, seed 42, complete 1000-epoch runs. Final checkpoint is reported consistently; no significance claim is made. E uses a Euclidean linear head; other rows use Lorentz prototypes. The encoder is shared in form and trained with AdamW.

Method	Final val. acc.	Final test acc.	Test macro-F1	Seconds
E	0.7740	**0.7780**	**0.7694**	10.3
B2	**0.7800**	0.7720	0.7680	62.5
B3	0.7780	0.7710	0.7678	63.2
B4	**0.7800**	0.7720	0.7680	66.1
B5	0.7760	0.7720	0.7692	72.0
A5	0.7760	0.7720	0.7692	80.0
B35	0.7760	0.7720	0.7692	92.0
shared	0.7760	0.7720	0.7692	93.6

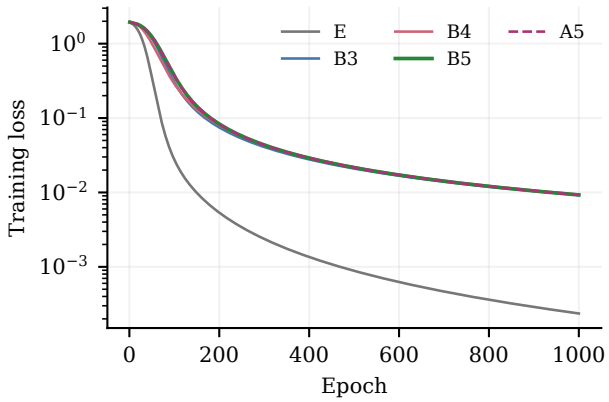


Figure 3. Complete 1000-epoch Cora/GCN training-loss trajectories, with a logarithmic loss axis. Scores are tabulated in Table 14; test data do not select these curves or their run length.

unfinished configurations would execute; it does not alter the original registry.

Table 14 gives completed 1000-epoch Cora/GCN comparisons on compute environment. They use seed 42 and the standard Planetoid split, the same two-layer GCN encoder with AdamW at 10^{-3}, and 32-dimensional output features. E uses a Euclidean linear head; B2 uses fixed-curvature RSGD prototypes; B3 uses fixed-curvature momentum; B4 learns per-class curvature without momentum; B5 adds momentum, normalized steps, and compatible cross-curvature transport; A5 changes only that transport operation. Geometric base step is 0.1, curvature step 0.001, momentum 0.9, curvature interval is bounded above by 5, and the log-step cap is 0.01. These are fixed starting settings, not the result of equal-budget hyperparameter searches, which remain outstanding.

B5 final test accuracy is 0.772 versus 0.771 for B3, and A5 equals B5 at the reported precision. The Euclidean head reaches 0.778. This is a near-tie among the manifold variants, not evidence of a robust graph-family win or an independent transport benefit. Recorded training times are descriptive and include concurrently running jobs; they are not controlled speed benchmarks. The full trajectories and checkpoints remain available for validation-selected analysis.

The new implementation uses actual Lorentz point parameters with closed-form exponential updates and parallel transport. Curvature gradients differentiate through the point homothety, unlike a partial derivative at fixed ambient coordinates. Classification uses one manifold prototype per class; text uses deterministic hashed tokens with an MLP or a two-layer Transformer; graph models use GCN, GAT, or GraphSAGE; recommendation uses a four-factor product distance with MF or origin-chart LightGCN propagation; KG uses grouped product states with a distance or transported-translation score. Synthetic product factors partition the requested total intrinsic dimension. This last architecture differs from the historical independent-tree groups, and the two protocols are not pooled.

KG evaluation in the revision ranks both heads and tails against every entity, filters other known true triples, uses average ranks for ties, and evaluates fixed full splits. MovieLens and Amazon Books use chronological leave-last-two-out per user. Recommendation excludes observed training items from negative draws and evaluates all-item, user-macro

Table 15. Completed graph comparisons on compute environment, seed 42, 1000 epochs. Final test accuracy (%) is the common endpoint; Δ is B5 minus B3 in percentage points. Every dataset–encoder pair with both completed runs is shown, including ties and negative differences. E is an architecture control with a Euclidean linear head. Dashes mean a comparator has not finished.

Dataset	Encoder	E	B3	B5	A5	Δ
arxiv	gat	**68.42**	62.33	62.41	62.45	+0.08
arxiv	gcn	**67.66**	61.34	61.19	61.21	-0.16
arxiv	sage	**68.32**	63.39	63.67	63.50	+0.28
citeseer	gat	**65.80**	64.70	64.60	64.60	-0.10
citeseer	gcn	63.70	**65.60**	**65.60**	**65.60**	+0.00
citeseer	sage	62.30	66.50	**66.60**	**66.60**	+0.10
cora	gat	66.60	**67.90**	67.70	67.70	-0.20
cora	gcn	**77.80**	77.10	77.20	77.20	+0.10
cora	sage	77.80	**78.10**	78.00	78.00	-0.10
pubmed	gat	71.80	**75.90**	**75.90**	**75.90**	+0.00
pubmed	gcn	**76.50**	75.90	75.90	75.90	+0.00
pubmed	sage	74.20	75.10	**75.20**	**75.20**	+0.10

Recall@20/NDCG@20 with observed items masked. Yelp2018 instead preserves its published multi-positive test split, with one seed-42 original training interaction per user held out for validation. All test positives for each user contribute to Recall and normalized DCG. Both Euclidean and Lorentz LightGCN variants use two propagation layers; their geometry differs, while the graph operator is retained in both. Reuters retains all labels and uses binary cross entropy with micro/macro F1. Continual-learning experiments train for 1000 total epochs across five ordered tasks and report task-boundary average accuracy and forgetting; Rotated MNIST uses angles $0, 36, 72, 108, 144$ degrees, while Split CIFAR-100 uses five disjoint blocks of 20 class labels. No replay buffer is added. These protocols are predeclared in the executable configuration rather than adjusted after test scores are observed.

Appendix F contains the generated completion ledger and current coverage counts; all 705 retained configurations are complete, while 188 deferred configurations are excluded from completion claims. Original WN18RR, FB15k-237, and MNIST data were recovered from the hpc project directory. Tiny-ImageNet and YAGO3-10 have been acquired separately. The deterministic Amazon Books 20K-user subset has also been acquired from the official 2018 five-core release. An existing compute environment ImageNet-100 cache has now been located and its manifest verified: 114129 training images, 12560 held-out training images for validation, and 5000 official validation images reserved as test. This is the cached `clane9/imagenet-100` version with shorter-side-160px source images, subsequently resized by the declared input transform; it must not be conflated with an original-resolution protocol. The manifest hash and data source are retained with the execution records. Its 18 formerly blocked configurations were made runnable before the budget amendment and are now explicitly deferred. The Yelp subset is instantiated as the Yelp2018 interaction benchmark released by the LightGCN authors (He et al., 2020), rather than an unspecified raw-review subset. It contains 31668 users, 38048 items, 1205591 retained training interactions, 31668 validation interactions, and 324147 original test interactions. The source commit and file hashes are recorded; the original test split is unchanged. This explicitly named, non-chronological protocol resolved the twelve data-blocked configurations; their execution is now deferred under the same budget amendment.

E.3. Complete paired graph comparisons

Table 15 extends the graph comparison beyond Cora. The generated table contains every dataset–encoder pair for which both B3 and B5 have completed the same 1000-epoch protocol. It uses final test accuracy consistently; validation-selected test metrics are retained separately in the machine-readable evidence file. Positive, tied, and negative differences are all kept. A positive single-seed difference is descriptive evidence for that configuration, not a significance claim or a substitute for equal-budget hyperparameter tuning.

E.4. Long-horizon classification, ranking, and retention audit

We distinguish a complete method block from individual full-length runs. The 20 Newsgroups pooled-token block contains all eight registered methods. Its separate Transformer block has six registered methods; WN18RR translation, MovieLens-1M, and Rotated MNIST each have eight. WN18RR distance has six registered methods. These

tables include only completed runs and state their coverage; missing controls are not assigned scores. Deferred methods are counted separately from unfinished retained controls. All records use seed 42 and the registered stopping length. Raw result and trajectory digests, selected validation epochs, and available per-run source digests are retained in `tables/revision_long_horizon_evidence.json`. These are descriptive comparisons of registered configurations, not a claim that every run has an identical complete source snapshot.

Text classification and checkpoint selection. The registry name `mlp` denotes a pooled-token encoder: deterministic lowercase word hashing into 29999 nonpadding token IDs, truncation to 256 tokens, learned 128-dimensional token embeddings, masked mean pooling, and a 32-dimensional linear projection. The official 20 Newsgroups training split is divided into 10182 training and 1132 stratified validation documents with seed 42; all 7532 official test documents are retained. Headers, footers, and quotes are removed. The encoder uses AdamW with step 10^{-3}; geometric prototypes use base step 0.1, curvature step 0.001, momentum 0.9, and initial curvature 1. These are fixed configurations, not a hyperparameter search. Validation accuracy is checked every epoch; the first maximizing checkpoint is selected after the complete 1000-epoch run, and its test scores are distinguished from the final checkpoint. All eight selected epochs have been checked against the raw validation trajectories.

Table 16. **Complete 20 Newsgroups classification block.** Eight methods, 1000 epochs each, seed 42. The registry labels the pooled-token encoder as `mlp`. The first maximum validation-accuracy checkpoint selects each method's test result; epochs are one-based. All scores are percentages. Final accuracy is a separately labeled endpoint, not an alternative selection rule.

Method	Selected epoch	Val. acc. ↑	Test acc. ↑	Test macro-F1 ↑	Final acc. ↑
E	427	**71.113**	**61.471**	**60.750**	**61.418**
B2	179	68.463	57.554	56.291	56.678
B3	179	68.728	57.820	56.552	56.745
B4	183	68.286	57.754	56.494	56.121
B5	182	68.551	57.674	56.449	56.213
A5	182	68.551	57.687	56.461	56.213
B35	179	68.728	57.820	56.560	56.798
Shared	179	68.728	57.674	56.457	56.028

Separate Transformer comparison. The Transformer retains the same tokenization, data partitions, output dimension, prototype configuration, and validation selection. It adds learned positional embeddings and two Transformer encoder layers, each with hidden dimension 128, four attention heads, feed-forward dimension 256, and zero dropout, before masked mean pooling and projection. Table 17 reports all four retained methods, including the B3–B5 pair. The two runs record identical hashes for the runner, geometry, and structured-task source modules. Their validation maxima occur at epochs 648 and 451, respectively (one-based). Selected test accuracy is 54.673% for B3 and 52.987% for B5, a B5 difference of -1.686 percentage points; selected macro-F1 is also lower for B5. Final accuracy preserves this ordering. The E and A5 controls finish at validation-selected test accuracies 55.709% and 54.169%, respectively. Thus all four retained methods are complete, while B2 and B4 remain deferred. A different encoder on the same dataset is not an independent task family.

Table 17. **20 Newsgroups Transformer: completed method endpoints.** Each listed method finishes 1000 epochs, seed 42. The first maximum validation-accuracy checkpoint selects its test result; epochs are one-based. 4/6 registered methods complete; 2 methods deferred by the budget amendment. Every completed method is retained. All scores are percentages; final accuracy is separately labeled. Bold compares only the displayed rows; B5 is shaded.

Method	Selected epoch	Val. acc. ↑	Test acc. ↑	Test macro-F1 ↑	Final acc. ↑
E	960	**65.724**	**55.709**	**55.006**	**55.616**
B3	648	63.428	54.673	53.539	54.408
B5	451	61.749	52.987	52.485	52.894
A5	592	62.544	54.169	53.213	53.120

Primary endpoints in the other registered families. Tables 18–20 report final endpoints rather than substituting the best observed test checkpoint. WN18RR translation B5 MRR is 0.10230, versus 0.17225 for E, 0.12598 for B3, 0.12596 for B35, and 0.10232 for B4; B5 and A5 agree at the displayed precision. On MovieLens-1M, B5 Recall@20 is slightly

higher than B3, but the primary NDCG@20 is lower (2.1560% versus 2.2072%); E reaches 4.2943%. On Rotated MNIST, all eight methods are complete. B5 mean accuracy is 43.432%, versus 43.388% for B35 and 43.296% for shared curvature: small increases of 0.044 and 0.136 percentage points. However, E, B3, and B2 reach 43.800%, 43.752%, and 44.340%, respectively. Shared curvature gives the highest last-task accuracy, 98.530%, versus 98.470% for B5. A last-task score does not measure retention of earlier tasks. Accordingly, these observations do not establish an additional positive task family or an isolated transport benefit. All completed primary and secondary endpoints are shown, with each deferred method explicitly identified.

The completed shared-curvature WN18RR control has MRR 0.10200. Grouped B5 improves this narrowly by 0.000295 absolute MRR (0.289% relative), while remaining below both fixed-curvature controls; Hits@10 is slightly lower than shared curvature (30.600% versus 30.616%). This mixed profile does not establish a broad grouped-curvature gain. The MovieLens-1M A5 endpoint equals B5 exactly in both stored primary NDCG@20 (0.02156021) and secondary Recall@20 (0.05546358). Equality is an observed ranking-score result for this run, not a claim that the optimizer trajectories are identical. Primary endpoint choice, full run lengths, and the original configurations are unchanged.

Table 18. **Available full-length endpoints.** WN18RR transported-translation model, 2000 epochs. All-entity filtered head-and-tail ranking at the final checkpoint. MRR is primary; Hits scores are percentages. 7/8 registered methods complete; all completed rows are shown. 1 method is deferred by the budget amendment. Bold compares only the displayed completed rows, including rounded ties; B5 is shaded. Seed 42, fixed starting configuration, no equal-budget hyperparameter search.

Method	MRR ↑	Hits@1 (%) ↑	Hits@3 (%) ↑	Hits@10 (%) ↑
E	**0.17225**	0.271	**30.568**	**45.246**
B3	0.12598	**2.106**	18.427	35.354
B4	0.10232	1.133	15.077	30.632
B5	0.10230	1.133	15.061	30.600
A5	0.10230	1.133	15.061	30.600
B35	0.12596	**2.106**	18.427	35.354
Shared	0.10200	1.101	14.853	30.616

Table 19. **Available full-length endpoints.** MovieLens-1M product-distance MF, 1000 epochs. Final all-item user-macro scores after chronological splitting and observed-item masking. NDCG@20 is primary; Recall@20 is secondary. Both are percentages. 5/8 registered methods complete; all completed rows are shown. 3 methods are deferred by the budget amendment. Bold compares only the displayed completed rows, including rounded ties; B5 is shaded. Seed 42, fixed starting configuration, no equal-budget hyperparameter search.

Method	NDCG@20 ↑	Recall@20 ↑
E	**4.2943**	**11.5563**
B3	2.2072	5.4967
B5	2.1560	5.5464
A5	2.1560	5.5464
B35	2.2069	5.4967

Table 20. **Available full-length endpoints.** Rotated MNIST, five ordered tasks and 1000 total epochs without replay. Mean accuracy across all five tasks is primary; forgetting averages the first four tasks, in percentage points. Last-task accuracy is secondary. All rows use the final checkpoint. 8/8 registered methods complete; all completed rows are shown. Bold compares only the displayed completed rows, including rounded ties; B5 is shaded. Seed 42, fixed starting configuration, no equal-budget hyperparameter search.

Method	Mean acc. (%) ↑	Forgetting (pp) ↓	Last-task acc. (%) ↑
E	43.800	68.267	98.380
B2	**44.340**	**67.520**	98.350
B3	43.752	68.320	98.390
B4	42.804	69.535	98.390
B5	43.432	68.715	98.470
A5	43.378	68.782	98.350
B35	43.388	68.807	98.400
Shared	43.296	68.927	**98.530**

E.5. Budget amendment and retained comparisons

On November 4, 2026, the remaining execution budget was reduced after 691 of the 893 registered configurations had completed. The decision used the observed outcomes and compute costs and is therefore exploratory. Of the 202 unfinished configurations, 14 are retained and 188 deferred. Table 21 enumerates every retained unfinished comparison. It preserves ongoing controls, supplies three missing Euclidean anchors, and completes one unopened model block with all six original methods. No completed result is excluded; the registry, raw trajectories, and intermediate checkpoints remain available. The amendment is recorded in `experiments/revision/execution_scope.json` and linked to each deferred record.

Table 21. Remaining execution budget at the amendment. Counts refer to unfinished configurations on November 4, not new trials or completed results. All retained runs keep their registered seed, splits, settings, and full training length.

Dataset and model	Retained unfinished methods	Runs	Epochs
Rotated MNIST / MLP	B35, shared	2	1000
20 Newsgroups / Transformer	A5, E	2	1000
MovieLens-1M / MF	B35, E	2	1000
WN18RR / translation	B4, E	2	2000
WN18RR / distance	E, B2, B3, B4, B5, A5	6	2000
Total retained unfinished		14	
Deferred, never started		187	
Deferred, partially trained	CIFAR-100 / ResNet-18 / B5	1	

The interrupted CIFAR-100 run is not reported as a completed visual benchmark or a numerical failure. Its available checkpoints and loss log are preserved. The broad dataset–encoder expansion and other unstarted ablations are deferred for budget reasons. Within the amended blocks, the retained method counts are four of six for the Transformer, seven of eight for WN18RR translation, five of eight for MovieLens-1M, and all eight for Rotated MNIST. All 705 retained configurations have now completed, including all 14 runs retained by this amendment. The original 893 registrations are not relabeled as completed; the 188 deferred configurations remain separately identified.

Hierarchy-motivated distance comparison. The retained WN18RR distance block replaces transported translation by a relation-weighted product-distance score,

$$s(h,r,t) = -\sum_{g=1}^{4} \big(1 + \|v_{r,g}\|_{K_g}^2\big) d_{K_g}(x_{h,g}, x_{t,g})^2, \qquad v_{r,g} = \mathrm{Log}_{o_{K_g}}(x_{r,g}).$$

It directly tests metric fitting on the same hierarchical graph, with the matched Euclidean architecture replacing each factor distance and relation norm by their Euclidean counterparts. All six originally registered methods are retained together, including the Euclidean, fixed-curvature, and no-transport controls; no hyperparameter or stopping rule is changed to pursue a favorable endpoint. The score is symmetric in its entity arguments, so it does not itself encode edge direction. This is a model-specific test within KG completion, not an additional task family. MRR remains the primary final-checkpoint endpoint under full filtered head-and-tail ranking, and the existing single-seed qualification remains in force. All six methods have now completed 2000 epochs. Table 22 gives B5 MRR 0.13685, compared with 0.15722 for E, 0.14078 for B2, 0.14077 for B3, 0.13688 for B4, and 0.13691 for A5. B5 does not improve the primary endpoint over any of these controls. Its higher Hits@3 than B2 and B3 does not reverse the MRR or Hits@10 comparison. These final results do not establish an additional task-family advantage.

Table 22. **Available full-length endpoints.** WN18RR relation-weighted distance model, 2000 epochs. All six registered methods are retained under the budget amendment. All-entity filtered head-and-tail ranking at the final checkpoint. MRR is primary; Hits scores are percentages. 6/6 registered methods complete; all completed rows are shown. Bold compares only the displayed completed rows, including rounded ties; B5 is shaded. Seed 42, fixed starting configuration, no equal-budget hyperparameter search.

Method	MRR ↑	Hits@1 (%) ↑	Hits@3 (%) ↑	Hits@10 (%) ↑
E	**0.15722**	**0.000**	**30.089**	**38.449**
B2	0.14078	**0.000**	25.973	37.572
B3	0.14077	**0.000**	25.973	37.556
B4	0.13688	**0.000**	27.361	34.923
B5	0.13685	**0.000**	27.393	34.907
A5	0.13691	**0.000**	27.393	34.907

E.6. Complete product-factor reconstruction grid

The four tree constructions are binary, 4-ary, 8-ary, and variable-branching trees. Each uses the full Cartesian grid of three node counts and four total intrinsic dimensions stated in Table 7. The Lorentz product has $\min(4, d)$ factors whose dimensions sum to d; every factor fits the same tree observations. Each update samples 65536 ordered node pairs, removes self-pairs, and minimizes mean squared relative-distance error. A separate generator fixed at seed 1042 draws up to 100000 evaluation pairs once, again removing self-pairs; final distortion is mean absolute relative-distance error on that fixed set. This is a reconstruction diagnostic: evaluation pairs are not withheld from the training-pair distribution. All methods finish 2000 updates. The Euclidean control uses SGD with momentum 0.9 and step 10; the manifold methods use base step 10, curvature step 0.001, and the common geometric safeguards. This is a fixed-configuration comparison, not a validation-selected tuning study. In particular, the B5–B3 contrast combines curvature adaptation and step normalization, while B5–A5 isolates the cross-curvature state map. The full table makes both contrasts and the Euclidean result inspectable.

Table 23. Complete product-factor reconstruction grid. Final fixed-pair distortion (lower is better) after 2000 updates, seed 42. Bold indicates the best value at the displayed precision; B5 is shaded. The last column reports the B5 reduction relative to B3 in percent, calculated from unrounded records. All baselines, including the stronger Euclidean control, are retained.

Tree	n	d	E	B2	B3	B4	B5	A5	Reduction (%)
Binary	1000	2	**0.41637**	0.72640	0.72635	0.72403	0.72399	0.72400	0.3247
Binary	1000	5	**0.32602**	0.66094	0.66133	0.65903	0.65943	0.65945	0.2874
Binary	1000	10	**0.29453**	0.63097	0.63134	0.62930	0.62969	0.62970	0.2620
Binary	1000	20	**0.27958**	0.61316	0.61350	0.61161	0.61197	0.61198	0.2505
Binary	5000	2	**0.69453**	0.95452	0.95471	0.95440	0.95459	0.95459	0.0125
Binary	5000	5	**0.65076**	0.94672	0.94693	0.94662	0.94683	0.94683	0.0099
Binary	5000	10	**0.63689**	0.94239	0.94261	0.94229	0.94250	0.94250	0.0111
Binary	5000	20	**0.62965**	0.93826	0.93848	0.93814	0.93836	0.93836	0.0127
Binary	10000	2	**0.84620**	0.98018	0.98026	0.98016	0.98024	0.98024	0.0026
Binary	10000	5	**0.82480**	0.97607	0.97616	0.97605	0.97614	0.97614	0.0022
Binary	10000	10	**0.81708**	0.97329	0.97338	0.97327	0.97336	0.97336	0.0028
Binary	10000	20	**0.81226**	0.97009	0.97018	0.97005	0.97014	0.97014	0.0036
4-ary	1000	2	**0.36051**	0.50908	0.50823	0.50512	0.50421	0.50422	0.7910
4-ary	1000	5	**0.23455**	0.38139	0.38131	0.37882	0.37876	0.37878	0.6707
4-ary	1000	10	**0.18215**	0.34137	0.34100	0.33942	0.33906	0.33907	0.5688
4-ary	1000	20	**0.15829**	0.32290	0.32242	0.32127	0.32080	0.32081	0.5012
4-ary	5000	2	**0.44106**	0.87753	0.87796	0.87673	0.87717	0.87717	0.0901
4-ary	5000	5	**0.34468**	0.85667	0.85718	0.85606	0.85657	0.85657	0.0713
4-ary	5000	10	**0.31659**	0.84622	0.84675	0.84557	0.84611	0.84611	0.0761
4-ary	5000	20	**0.30328**	0.83768	0.83823	0.83698	0.83754	0.83754	0.0828
4-ary	10000	2	**0.62564**	0.94424	0.94446	0.94405	0.94427	0.94427	0.0195
4-ary	10000	5	**0.57078**	0.93405	0.93430	0.93390	0.93415	0.93415	0.0159
4-ary	10000	10	**0.55361**	0.92822	0.92847	0.92805	0.92830	0.92831	0.0183
4-ary	10000	20	**0.54395**	0.92227	0.92253	0.92207	0.92233	0.92233	0.0218
8-ary	1000	2	**0.34629**	0.39497	0.39423	0.39353	0.39280	0.39281	0.3624
8-ary	1000	5	**0.18629**	0.24986	0.24962	0.24934	0.24911	0.24912	0.2061
8-ary	1000	10	**0.12706**	0.21326	0.21275	0.21273	0.21224	0.21225	0.2371

Tree	n	d	E	B2	B3	B4	B5	A5	Reduction (%)
8-ary	1000	20	**0.09254**	0.19817	0.19757	0.19764	0.19706	0.19706	0.2576
8-ary	5000	2	**0.36754**	0.78506	0.78563	0.78287	0.78345	0.78346	0.2768
8-ary	5000	5	**0.23428**	0.74700	0.74772	0.74530	0.74603	0.74605	0.2255
8-ary	5000	10	**0.19336**	0.72832	0.72909	0.72659	0.72737	0.72738	0.2360
8-ary	5000	20	**0.17304**	0.71503	0.71583	0.71324	0.71405	0.71406	0.2498
8-ary	10000	2	**0.46059**	0.89895	0.89931	0.89837	0.89874	0.89874	0.0634
8-ary	10000	5	**0.37142**	0.88138	0.88180	0.88093	0.88135	0.88135	0.0509
8-ary	10000	10	**0.34542**	0.87208	0.87252	0.87159	0.87203	0.87203	0.0565
8-ary	10000	20	**0.33156**	0.86344	0.86388	0.86288	0.86333	0.86333	0.0646
Variable	1000	2	**0.37559**	0.56828	0.56758	0.56467	0.56395	0.56396	0.6402
Variable	1000	5	**0.24945**	0.45533	0.45542	0.45274	0.45283	0.45285	0.5673
Variable	1000	10	**0.21387**	0.41568	0.41553	0.41362	0.41348	0.41350	0.4926
Variable	1000	20	**0.19139**	0.39677	0.39651	0.39501	0.39476	0.39477	0.4425
Variable	5000	2	**0.51448**	0.90613	0.90648	0.90565	0.90600	0.90600	0.0529
Variable	5000	5	**0.43579**	0.89022	0.89062	0.88984	0.89025	0.89025	0.0416
Variable	5000	10	**0.41323**	0.88205	0.88248	0.88165	0.88208	0.88208	0.0449
Variable	5000	20	**0.40134**	0.87510	0.87553	0.87466	0.87510	0.87510	0.0496
Variable	10000	2	**0.68836**	0.95593	0.95611	0.95582	0.95599	0.95599	0.0123
Variable	10000	5	**0.64379**	0.94772	0.94792	0.94763	0.94782	0.94782	0.0101
Variable	10000	10	**0.62919**	0.94285	0.94305	0.94273	0.94294	0.94294	0.0118
Variable	10000	20	**0.62107**	0.93769	0.93790	0.93755	0.93776	0.93776	0.0144

E.7. All long-horizon mechanism conditions

E.8. New real-data protocols and complete comparisons

The WordNet tasks use the full mammal and vehicle descendant subtrees in WordNet 3.0, including instances, and all internal transitive hypernym relations. The subtrees contain 1182 and 528 synsets, with 6542 and 2311 directed positive pairs, respectively. This is a custom missing-relation protocol and differs from the filtered mammals reconstruction benchmark of Nickel & Kiela (2017). The corpus is accessed through NLTK (NLTK Project, 2026). Ten percent of relations are reserved for validation and ten percent for test; the training split retains at least one incident relation for every nonisolated vertex. Filtered macro query MAP ranks every other synset and masks known positives outside the target split. The score penalizes both product distance and a reversed descendant–ancestor radial order.

Cora and Citeseer links use the Planetoid graphs (Yang et al., 2016), with node identities but no attributes or class labels. Undirected pairs are canonicalized before splitting. Positive and negative pairs are disjoint across train, validation, and test, including reverse-edge equivalence; known positives are excluded from negative draws. The splits contain 5278 and 4552 positive pairs, respectively, with a matched number of negatives and an approximately 80/10/10 positive split that preserves training vertex coverage. Balanced-pair AP is primary and ROC-AUC is secondary; these scores do not estimate precision at the natural all-pairs prevalence.

Optical digits uses the 1797-image dataset distributed by scikit-learn (scikit-learn developers, 2026b), with stratified training/validation/test counts 1078/359/360. Scaling is fit only on training observations. The 20 Newsgroups retrieval task (scikit-learn developers, 2026a) retains its official test set and partitions official training documents into 10182 training and 1132 validation queries, with 7532 test queries. Headers, footers, and quotes are removed. TF-IDF with at most 6000 features, a 128-component truncated SVD, and scaling are all fit exclusively on the training partition. A shared 128-unit tanh encoder produces eight features arranged as four two-dimensional factors; class proxies are optimized on their home manifolds. Retrieval uses observation embeddings, not class proxies, and the gallery contains training observations only. Primary same-class Recall@1 is the fraction of queries with a same-label top-ranked gallery item; secondary NDCG@10 uses same-class relevance. These are compact custom retrieval protocols rather than published leaderboard reproductions.

For each manifold method, the six configurations cross $K_0 \in \{0.1, 1, 3\}$ with two base steps: $\{10, 100\}$ for relation embeddings and $\{0.1, 1\}$ for retrieval proxies. The Euclidean control receives the same two steps crossed with point-decay coefficients $\{0, 10^{-4}, 10^{-2}\}$, also six runs. Other shared settings are $\beta = 0.9$, $\rho = 0.01$, $K \in [0.001, 5]$, the canonical radius cap 4, the normalized trust cap 0.25, and the log-curvature cap 0.01. Auxiliary parameters use AdamW with learning

Table 24. Long-horizon hierarchy mechanism tests: final held-out pair distortion after 2000 updates (lower is better). All methods use seed 42 and identical pair splits. Only cases with completed B5, B3, and A5 are included. The best-fixed column requires all five B35 curvature settings and selects using final validation distortion, never test distortion. Missing comparisons remain dashes. These controlled synthetic cases are not additional real-data task families.

Case	B3	B5	A5	Shared	B4	Best fixed
abrupt8	0.7417	0.6176	0.6176	0.6143	0.5914	**0.1637**
heterogeneous	0.2510	0.2470	0.2463	0.2543	0.2415	**0.2182**
smooth2	0.3328	0.3295	0.3296	0.3260	0.3202	**0.1823**
smooth5	0.6225	0.4347	0.4347	0.4479	0.4262	**0.2058**
smooth8	0.7411	0.4813	0.4813	0.5078	0.4719	**0.2757**
static	0.2734	0.2727	0.2726	0.2785	0.2604	**0.2011**

Table 25. Matched five-initialization comparison, 2000 updates per run. B35, B5, and A5 each use $K_0 \in \{0.03, 0.1, 0.3, 1, 3\}$, with configurations selected by final validation distortion. Test distortion is reported after selection. A case appears only when all 15 runs finish. The last column compares B5 with A5 at B5's selected initialization, retaining the clean transport contrast.

Case	Best fixed	B5	Best A5	B5 K_0	Matched A5
abrupt8	**0.1637**	0.1655	0.1663	0.03	0.1663
heterogeneous	0.2182	0.2177	**0.2174**	0.1	**0.2174**
smooth2	0.1823	**0.1762**	0.1763	0.1	0.1763
smooth5	**0.2058**	**0.2058**	0.2061	0.03	0.2061
smooth8	0.2757	**0.2619**	0.2624	0.03	0.2624
static	0.2011	0.1952	**0.1951**	0.1	**0.1951**

rate 10^{-3} and decay 10^{-4}. Temperature is restricted to $[e^{-3}, e^{3}]$. Full-batch training spans all training examples each epoch. Validation loss is saved every epoch and the primary ranking metric every 25 epochs and at the final checkpoint. Final-checkpoint validation, not the best test score or best test epoch, selects the configuration. Training data, transformations, model capacity, and stopping length are identical across the relevant manifold comparisons. The canonical-radius constraint still makes physical-distance capacity curvature-dependent.

Registered new-task completion counts: WordNet mammals: 30/30, Cora links: 30/30, Digits retrieval: 30/30, WordNet vehicles: 30/30, Citeseer links: 30/30, 20News retrieval: 30/30. Test comparison requires the full 30-run block.

Table 26. Final-checkpoint test scores in percent; higher is better. Taxonomy: filtered query MAP; links: balanced-pair AP; retrieval: same-class Recall@1. Each method selects from six runs by final validation metric; all 30 configurations finish before any test evaluation. Matched A5 uses B5's selected learning rate and initial curvature. Seed 42; 1000 epochs. Every completed task is retained, including ties and negative differences.

Task	E	B3	B35	B5	A5	Matched A5
WordNet mammals	8.26	8.49	8.31	8.29	**8.96**	8.38
Cora links	**93.46**	**93.46**	**93.46**	**93.46**	**93.46**	**93.46**
Digits retrieval	96.39	96.67	96.67	**96.94**	**96.94**	**96.94**
WordNet vehicles	8.42	8.40	7.80	**8.50**	8.27	8.27
Citeseer links	91.91	91.91	**91.92**	**91.92**	**91.92**	**91.92**
20News retrieval	42.31	**43.49**	**43.49**	**43.49**	43.45	43.45

E.9. Recorded non-loss trajectory checkpoints

Table 27. **Historical KG metric checkpoints.** Recorded MRR, Hits@10 and learned curvature spread. Loss alone is plotted in Figure 1.

Epoch	B3 MRR	B4 MRR	B3 Hits@10	B4 Hits@10	B4 K_{std}
100	0.1331	0.1334	0.3500	0.3500	0.0765
300	0.1386	0.1361	0.3638	0.3688	0.2530
500	0.1217	0.1250	0.3463	0.3575	0.2823
800	0.1353	0.1384	0.3688	0.3750	0.2867
1000	0.1356	0.1422	0.3588	0.3825	0.2886
1500	0.1323	0.1321	0.3463	0.3538	0.2931
1900	0.1337	0.1326	0.3337	0.3613	0.2946
1999	0.1223	0.1355	0.3500	0.3563	0.2948

Table 28. **Smooth-drift validation distortion and curvature states.** All methods start at $K_0 = 1$. $K_1, \ldots, K_4$ are B5 states. These are fixed checkpoints of the complete recorded trajectories, distinct from validation-selected initialization comparisons.

Drift	Update	B3	B5	A5	K_1	K_2	K_3	K_4
2×	1	**0.9852**	**0.9852**	**0.9852**	1.000	1.000	1.000	1.000
2×	100	0.3601	**0.3568**	0.3570	1.007	1.034	1.062	1.066
2×	500	0.3286	0.3277	**0.3276**	0.978	1.052	1.082	1.096
2×	1000	0.3230	0.3217	**0.3216**	0.909	1.035	1.104	1.127
2×	2000	0.3383	0.3362	**0.3361**	0.743	0.879	1.019	1.119
5×	1	**0.9852**	**0.9852**	**0.9852**	1.000	1.000	1.000	1.000
5×	100	0.3440	**0.3425**	**0.3425**	1.004	1.026	1.054	1.064
5×	500	**0.3506**	0.3529	0.3529	0.951	0.983	1.035	1.063
5×	1000	0.4660	**0.4307**	**0.4307**	0.836	0.793	0.812	0.834
5×	2000	0.6231	**0.4382**	**0.4382**	0.569	0.385	0.329	0.299
8×	1	**0.9852**	**0.9852**	**0.9852**	1.000	1.000	1.000	1.000
8×	100	0.3317	0.3316	**0.3314**	1.001	1.019	1.044	1.061
8×	500	0.4446	**0.4329**	**0.4329**	0.929	0.912	0.932	0.958
8×	1000	0.5917	**0.5123**	**0.5123**	0.784	0.651	0.613	0.601
8×	2000	0.7411	**0.4831**	**0.4831**	0.444	0.223	0.165	0.138

F. Generated Completed-Run Ledger

The generated execution snapshot contains 893 registered configurations: 705 complete, 0 claimed, 0 pending, 0 blocked on data, 0 failed, and 188 deferred by scope amendment. The amended execution budget retains 705 configurations, of which 0 are unfinished. A claim alone does not establish that a process is live. Only full-length completed runs appear in the generated result table.

Table 29. Complete runs only. Score is final test accuracy for classification, average accuracy for continual learning, MRR for KG, NDCG@20 for recommendation, fixed-pair distortion for the original synthetic suite, and held-out pair distortion for the new mechanism suite (lower is better only for distortion). For new tasks, unselected rows report the final validation primary score; selected test results are reported separately after the complete search. Rows across different datasets are not compared.

Dataset	Model	Method	Ablation	Epochs	Score	Seconds
cora	gcn	E	–	1000	0.7780	10.3
cora	gcn	B2	–	1000	0.7720	62.5
cora	gcn	B3	–	1000	0.7710	63.2
cora	gcn	B4	–	1000	0.7720	66.1
cora	gcn	B5	–	1000	0.7720	72.0
cora	gcn	B5	A1	1000	0.7730	69.6
cora	gcn	B5	A2	1000	0.7720	90.5
cora	gcn	B5	A4	1000	0.7710	91.2
cora	gcn	B5	A6	1000	0.7720	92.0
cora	gcn	B5	A7	1000	0.7740	98.1
cora	gcn	B5	A8	1000	0.7720	107.9
cora	gcn	B5	A9	1000	0.7720	91.4
cora	gcn	B5	A10	1000	0.7720	108.2
cora	gcn	B5	A12_K0.01	1000	0.7710	127.7
cora	gcn	B5	A12_K0.1	1000	0.7770	141.4
cora	gcn	B5	A12_K0.1001	1000	0.7770	102.1
cora	gcn	B5	A12_K0.5	1000	0.7740	123.2
cora	gcn	B5	A12_K2.0	1000	0.7700	115.6
cora	gcn	B5	A12_K5.0	1000	0.7710	108.6
cora	gcn	A5	–	1000	0.7720	80.0
cora	gcn	A5	A5	1000	0.7720	125.6
rotmnist	mlp	E	–	1000	0.4380	6261.3
rotmnist	mlp	B2	–	1000	0.4434	72081.4
rotmnist	mlp	B3	–	1000	0.4375	53846.6
rotmnist	mlp	B4	–	1000	0.4280	42271.5
rotmnist	mlp	B5	–	1000	0.4343	102629.8
rotmnist	mlp	A5	–	1000	0.4338	56237.5
rotmnist	mlp	B35	–	1000	0.4339	99424.0
rotmnist	mlp	shared	–	1000	0.4330	90981.8
cora	gcn	B35	–	1000	0.7720	92.0
cora	gcn	shared	–	1000	0.7720	93.6
cora	gcn	shared	A3	1000	0.7720	101.3
cora	gcn	shared	A11	1000	0.7720	136.2
wn18rr	lorentz_translation	E	–	2000	0.1723	2535.2
wn18rr	lorentz_translation	B3	–	2000	0.1260	25676.4
wn18rr	lorentz_translation	B4	–	2000	0.1023	36423.5
wn18rr	lorentz_translation	B5	–	2000	0.1023	31008.4
wn18rr	lorentz_translation	A5	–	2000	0.1023	31133.4
wn18rr	lorentz_translation	B35	–	2000	0.1260	30545.4
wn18rr	lorentz_translation	shared	–	2000	0.1020	36423.3
ml1m	mf	E	–	1000	0.0429	6962.5
ml1m	mf	B3	–	1000	0.0221	49850.1
ml1m	mf	B5	–	1000	0.0216	55356.9
ml1m	mf	A5	–	1000	0.0216	57283.4
ml1m	mf	B35	–	1000	0.0221	47567.4
20news	mlp	E	–	1000	0.6142	1043.4
20news	mlp	B2	–	1000	0.5668	18764.6
20news	mlp	B3	–	1000	0.5674	13301.6
20news	mlp	B4	–	1000	0.5612	19079.8
20news	mlp	B5	–	1000	0.5621	16128.6
20news	mlp	A5	–	1000	0.5621	13208.3
20news	mlp	B35	–	1000	0.5680	12493.8
20news	mlp	shared	–	1000	0.5603	13657.4
arxiv	gat	E	–	1000	0.6842	51.5
arxiv	gat	B2	–	1000	0.6247	411.3
arxiv	gat	B3	–	1000	0.6233	384.6

Dataset	Model	Method	Ablation	Epochs	Score	Seconds
arxiv	gat	B4	–	1000	0.6266	412.0
arxiv	gat	B5	–	1000	0.6241	405.8
arxiv	gat	A5	–	1000	0.6245	419.8
arxiv	gcn	E	–	1000	0.6766	40.4
arxiv	gcn	B2	–	1000	0.6141	408.4
arxiv	gcn	B3	–	1000	0.6134	438.7
arxiv	gcn	B4	–	1000	0.6143	449.5
arxiv	gcn	B5	–	1000	0.6119	454.1
arxiv	gcn	A5	–	1000	0.6121	457.7
arxiv	sage	E	–	1000	0.6832	45.4
arxiv	sage	B2	–	1000	0.6389	434.7
arxiv	sage	B3	–	1000	0.6339	426.9
arxiv	sage	B4	–	1000	0.6396	454.8
arxiv	sage	B5	–	1000	0.6367	444.9
arxiv	sage	A5	–	1000	0.6350	466.3
citeseer	gat	E	–	1000	0.6580	19.1
citeseer	gat	B2	–	1000	0.6510	83.9
citeseer	gat	B3	–	1000	0.6470	83.2
citeseer	gat	B4	–	1000	0.6510	86.4
citeseer	gat	B5	–	1000	0.6460	88.5
citeseer	gat	A5	–	1000	0.6460	90.5
citeseer	gcn	E	–	1000	0.6370	23.9
citeseer	gcn	B2	–	1000	0.6570	81.2
citeseer	gcn	B3	–	1000	0.6560	76.6
citeseer	gcn	B4	–	1000	0.6570	74.5
citeseer	gcn	B5	–	1000	0.6560	71.8
citeseer	gcn	A5	–	1000	0.6560	76.4
citeseer	sage	E	–	1000	0.6230	12.9
citeseer	sage	B2	–	1000	0.6630	73.9
citeseer	sage	B3	–	1000	0.6650	91.5
citeseer	sage	B4	–	1000	0.6630	86.6
citeseer	sage	B5	–	1000	0.6660	96.6
citeseer	sage	A5	–	1000	0.6660	84.6
cora	gat	E	–	1000	0.6660	29.2
cora	gat	B2	–	1000	0.6760	86.7
cora	gat	B3	–	1000	0.6790	98.1
cora	gat	B4	–	1000	0.6760	89.2
cora	gat	B5	–	1000	0.6770	99.5
cora	gat	A5	–	1000	0.6770	100.0
cora	sage	E	–	1000	0.7780	40.5
cora	sage	B2	–	1000	0.7820	88.2
cora	sage	B3	–	1000	0.7810	88.9
cora	sage	B4	–	1000	0.7820	93.3
cora	sage	B5	–	1000	0.7800	115.7
cora	sage	A5	–	1000	0.7800	103.1
pubmed	gat	E	–	1000	0.7180	18.4
pubmed	gat	B2	–	1000	0.7590	55.7
pubmed	gat	B3	–	1000	0.7590	52.0
pubmed	gat	B4	–	1000	0.7590	55.2
pubmed	gat	B5	–	1000	0.7590	56.2
pubmed	gat	A5	–	1000	0.7590	57.5
pubmed	gcn	E	–	1000	0.7650	27.2
pubmed	gcn	B2	–	1000	0.7590	63.8
pubmed	gcn	B3	–	1000	0.7590	58.5
pubmed	gcn	B4	–	1000	0.7590	67.3
pubmed	gcn	B5	–	1000	0.7590	72.3
pubmed	gcn	A5	–	1000	0.7590	70.2
pubmed	sage	E	–	1000	0.7420	13.0
pubmed	sage	B2	–	1000	0.7470	43.0
pubmed	sage	B3	–	1000	0.7510	46.3
pubmed	sage	B4	–	1000	0.7470	49.1
pubmed	sage	B5	–	1000	0.7520	48.6
pubmed	sage	A5	–	1000	0.7520	51.2
wn18rr	lorentz_distance	E	–	2000	0.1572	1795.1
wn18rr	lorentz_distance	B2	–	2000	0.1408	19011.7
wn18rr	lorentz_distance	B3	–	2000	0.1408	21320.8
wn18rr	lorentz_distance	B4	–	2000	0.1369	20077.4
wn18rr	lorentz_distance	B5	–	2000	0.1369	26756.9
wn18rr	lorentz_distance	A5	–	2000	0.1369	20677.9
abrupt8	independent_lorentz	B3	–	2000	0.7417	150.2

Dataset	Model	Method	Ablation	Epochs	Score	Seconds
abrupt8	independent_lorentz	B4	–	2000	0.5914	151.8
abrupt8	independent_lorentz	B5	–	2000	0.6176	147.3
abrupt8	independent_lorentz	B5	initial_K0.03	2000	0.1655	74.7
abrupt8	independent_lorentz	B5	initial_K0.1	2000	0.1791	79.5
abrupt8	independent_lorentz	B5	initial_K0.3	2000	0.2841	78.2
abrupt8	independent_lorentz	B5	initial_K3.0	2000	0.8308	78.4
abrupt8	independent_lorentz	A5	–	2000	0.6176	151.6
abrupt8	independent_lorentz	A5	initial_K0.03	2000	0.1663	75.2
abrupt8	independent_lorentz	A5	initial_K0.1	2000	0.1787	77.8
abrupt8	independent_lorentz	A5	initial_K0.3	2000	0.2842	75.3
abrupt8	independent_lorentz	A5	initial_K3.0	2000	0.8308	99.3
abrupt8	independent_lorentz	B35	–	2000	0.7417	145.9
abrupt8	independent_lorentz	shared	–	2000	0.6143	166.3
abrupt8	independent_lorentz	B35	fixed_K0.03	2000	0.1637	73.6
abrupt8	independent_lorentz	B35	fixed_K0.1	2000	0.2978	72.2
abrupt8	independent_lorentz	B35	fixed_K0.3	2000	0.5531	72.5
abrupt8	independent_lorentz	B35	fixed_K3.0	2000	0.8471	74.7
heterogeneous	independent_lorentz	B3	–	2000	0.2510	138.2
heterogeneous	independent_lorentz	B4	–	2000	0.2415	167.5
heterogeneous	independent_lorentz	B5	–	2000	0.2470	153.4
heterogeneous	independent_lorentz	B5	initial_K0.03	2000	0.2227	92.4
heterogeneous	independent_lorentz	B5	initial_K0.1	2000	0.2177	91.7
heterogeneous	independent_lorentz	B5	initial_K0.3	2000	0.2206	90.7
heterogeneous	independent_lorentz	B5	initial_K3.0	2000	0.3115	90.8
heterogeneous	independent_lorentz	A5	–	2000	0.2463	149.5
heterogeneous	independent_lorentz	A5	initial_K0.03	2000	0.2223	90.8
heterogeneous	independent_lorentz	A5	initial_K0.1	2000	0.2174	90.6
heterogeneous	independent_lorentz	A5	initial_K0.3	2000	0.2210	91.0
heterogeneous	independent_lorentz	A5	initial_K3.0	2000	0.3115	92.1
heterogeneous	independent_lorentz	B35	–	2000	0.2516	148.0
heterogeneous	independent_lorentz	shared	–	2000	0.2543	166.1
heterogeneous	independent_lorentz	B35	fixed_K0.03	2000	0.2235	69.8
heterogeneous	independent_lorentz	B35	fixed_K0.1	2000	0.2182	68.9
heterogeneous	independent_lorentz	B35	fixed_K0.3	2000	0.2268	70.7
heterogeneous	independent_lorentz	B35	fixed_K3.0	2000	0.3189	81.5
smooth2	independent_lorentz	B3	–	2000	0.3328	127.8
smooth2	independent_lorentz	B4	–	2000	0.3202	148.2
smooth2	independent_lorentz	B5	–	2000	0.3295	133.7
smooth2	independent_lorentz	B5	initial_K0.03	2000	0.1972	89.5
smooth2	independent_lorentz	B5	initial_K0.1	2000	0.1762	89.2
smooth2	independent_lorentz	B5	initial_K0.3	2000	0.2217	88.6
smooth2	independent_lorentz	B5	initial_K3.0	2000	0.4941	98.5
smooth2	independent_lorentz	A5	–	2000	0.3296	136.9
smooth2	independent_lorentz	A5	initial_K0.03	2000	0.1974	95.4
smooth2	independent_lorentz	A5	initial_K0.1	2000	0.1763	97.1
smooth2	independent_lorentz	A5	initial_K0.3	2000	0.2216	94.6
smooth2	independent_lorentz	A5	initial_K3.0	2000	0.4941	101.8
smooth2	independent_lorentz	B35	–	2000	0.3328	151.2
smooth2	independent_lorentz	shared	–	2000	0.3260	171.3
smooth2	independent_lorentz	B35	fixed_K0.03	2000	0.2000	78.0
smooth2	independent_lorentz	B35	fixed_K0.1	2000	0.1823	75.3
smooth2	independent_lorentz	B35	fixed_K0.3	2000	0.2185	64.3
smooth2	independent_lorentz	B35	fixed_K3.0	2000	0.5503	74.9
smooth5	independent_lorentz	B3	–	2000	0.6225	130.7
smooth5	independent_lorentz	B4	–	2000	0.4262	154.2
smooth5	independent_lorentz	B5	–	2000	0.4347	145.9
smooth5	independent_lorentz	B5	initial_K0.03	2000	0.2058	87.8
smooth5	independent_lorentz	B5	initial_K0.1	2000	0.2445	94.0
smooth5	independent_lorentz	B5	initial_K0.3	2000	0.3372	98.4
smooth5	independent_lorentz	B5	initial_K3.0	2000	0.7136	99.8
smooth5	independent_lorentz	A5	–	2000	0.4347	138.1
smooth5	independent_lorentz	A5	initial_K0.03	2000	0.2061	121.0
smooth5	independent_lorentz	A5	initial_K0.1	2000	0.2444	103.0
smooth5	independent_lorentz	A5	initial_K0.3	2000	0.3372	110.0
smooth5	independent_lorentz	A5	initial_K3.0	2000	0.7136	113.1
smooth5	independent_lorentz	B35	–	2000	0.6225	142.3
smooth5	independent_lorentz	shared	–	2000	0.4479	165.3
smooth5	independent_lorentz	B35	fixed_K0.03	2000	0.2058	64.3
smooth5	independent_lorentz	B35	fixed_K0.1	2000	0.2471	78.5
smooth5	independent_lorentz	B35	fixed_K0.3	2000	0.3968	74.3

Dataset	Model	Method	Ablation	Epochs	Score	Seconds
smooth5	independent_lorentz	B35	fixed_K3.0	2000	0.7724	74.4
smooth8	independent_lorentz	B3	–	2000	0.7411	179.3
smooth8	independent_lorentz	B4	–	2000	0.4719	160.0
smooth8	independent_lorentz	B5	–	2000	0.4813	158.5
smooth8	independent_lorentz	B5	initial_K0.03	2000	0.2619	103.1
smooth8	independent_lorentz	B5	initial_K0.1	2000	0.2921	103.7
smooth8	independent_lorentz	B5	initial_K0.3	2000	0.3770	110.5
smooth8	independent_lorentz	B5	initial_K3.0	2000	0.8090	108.1
smooth8	independent_lorentz	A5	–	2000	0.4813	165.4
smooth8	independent_lorentz	A5	initial_K0.03	2000	0.2624	102.3
smooth8	independent_lorentz	A5	initial_K0.1	2000	0.2922	108.8
smooth8	independent_lorentz	A5	initial_K0.3	2000	0.3769	111.0
smooth8	independent_lorentz	A5	initial_K3.0	2000	0.8090	110.4
smooth8	independent_lorentz	B35	–	2000	0.7411	174.4
smooth8	independent_lorentz	shared	–	2000	0.5078	167.5
smooth8	independent_lorentz	B35	fixed_K0.03	2000	0.2757	73.8
smooth8	independent_lorentz	B35	fixed_K0.1	2000	0.3505	70.7
smooth8	independent_lorentz	B35	fixed_K0.3	2000	0.5561	75.6
smooth8	independent_lorentz	B35	fixed_K3.0	2000	0.8469	73.1
static	independent_lorentz	B3	–	2000	0.2734	119.0
static	independent_lorentz	B4	–	2000	0.2604	158.6
static	independent_lorentz	B5	–	2000	0.2727	124.0
static	independent_lorentz	B5	initial_K0.03	2000	0.2112	102.5
static	independent_lorentz	B5	initial_K0.1	2000	0.1952	102.9
static	independent_lorentz	B5	initial_K0.3	2000	0.2046	159.2
static	independent_lorentz	B5	initial_K3.0	2000	0.3541	109.3
static	independent_lorentz	A5	–	2000	0.2726	129.6
static	independent_lorentz	A5	initial_K0.03	2000	0.2111	154.8
static	independent_lorentz	A5	initial_K0.1	2000	0.1951	102.2
static	independent_lorentz	A5	initial_K0.3	2000	0.2046	103.7
static	independent_lorentz	A5	initial_K3.0	2000	0.3541	155.9
static	independent_lorentz	B35	–	2000	0.2734	129.0
static	independent_lorentz	shared	–	2000	0.2785	177.8
static	independent_lorentz	B35	fixed_K0.03	2000	0.2129	150.4
static	independent_lorentz	B35	fixed_K0.1	2000	0.2011	139.8
static	independent_lorentz	B35	fixed_K0.3	2000	0.2106	76.3
static	independent_lorentz	B35	fixed_K3.0	2000	0.3697	68.2
4ary n1000 d2	product_lorentz	E	–	2000	0.3605	16.9
4ary n1000 d5	product_lorentz	E	–	2000	0.2345	12.0
4ary n1000 d10	product_lorentz	E	–	2000	0.1822	17.2
4ary n1000 d20	product_lorentz	E	–	2000	0.1583	13.6
4ary n5000 d2	product_lorentz	E	–	2000	0.4411	16.8
4ary n5000 d5	product_lorentz	E	–	2000	0.3447	17.6
4ary n5000 d10	product_lorentz	E	–	2000	0.3166	12.6
4ary n5000 d20	product_lorentz	E	–	2000	0.3033	17.5
4ary n10000 d2	product_lorentz	E	–	2000	0.6256	17.9
4ary n10000 d5	product_lorentz	E	–	2000	0.5708	9.5
4ary n10000 d10	product_lorentz	E	–	2000	0.5536	16.1
4ary n10000 d20	product_lorentz	E	–	2000	0.5440	17.1
4ary n1000 d2	product_lorentz	B2	–	2000	0.5091	71.9
4ary n1000 d5	product_lorentz	B2	–	2000	0.3814	134.8
4ary n1000 d10	product_lorentz	B2	–	2000	0.3414	134.6
4ary n1000 d20	product_lorentz	B2	–	2000	0.3229	138.7
4ary n5000 d2	product_lorentz	B2	–	2000	0.8775	70.9
4ary n5000 d5	product_lorentz	B2	–	2000	0.8567	128.7
4ary n5000 d10	product_lorentz	B2	–	2000	0.8462	120.0
4ary n5000 d20	product_lorentz	B2	–	2000	0.8377	130.9
4ary n10000 d2	product_lorentz	B2	–	2000	0.9442	63.5
4ary n10000 d5	product_lorentz	B2	–	2000	0.9341	135.3
4ary n10000 d10	product_lorentz	B2	–	2000	0.9282	142.1
4ary n10000 d20	product_lorentz	B2	–	2000	0.9223	138.6
4ary n1000 d2	product_lorentz	B3	–	2000	0.5082	79.6
4ary n1000 d5	product_lorentz	B3	–	2000	0.3813	133.7
4ary n1000 d10	product_lorentz	B3	–	2000	0.3410	154.9
4ary n1000 d20	product_lorentz	B3	–	2000	0.3224	151.9
4ary n5000 d2	product_lorentz	B3	–	2000	0.8780	81.1
4ary n5000 d5	product_lorentz	B3	–	2000	0.8572	138.6
4ary n5000 d10	product_lorentz	B3	–	2000	0.8468	143.8
4ary n5000 d20	product_lorentz	B3	–	2000	0.8382	136.1
4ary n10000 d2	product_lorentz	B3	–	2000	0.9445	83.2

Dataset	Model	Method	Ablation	Epochs	Score	Seconds
4ary n10000 d5	product_lorentz	B3	–	2000	0.9343	133.1
4ary n10000 d10	product_lorentz	B3	–	2000	0.9285	131.5
4ary n10000 d20	product_lorentz	B3	–	2000	0.9225	154.9
4ary n1000 d2	product_lorentz	B4	–	2000	0.5051	78.3
4ary n1000 d5	product_lorentz	B4	–	2000	0.3788	129.0
4ary n1000 d10	product_lorentz	B4	–	2000	0.3394	133.2
4ary n1000 d20	product_lorentz	B4	–	2000	0.3213	127.7
4ary n5000 d2	product_lorentz	B4	–	2000	0.8767	74.4
4ary n5000 d5	product_lorentz	B4	–	2000	0.8561	139.1
4ary n5000 d10	product_lorentz	B4	–	2000	0.8456	143.8
4ary n5000 d20	product_lorentz	B4	–	2000	0.8370	138.1
4ary n10000 d2	product_lorentz	B4	–	2000	0.9441	69.4
4ary n10000 d5	product_lorentz	B4	–	2000	0.9339	136.3
4ary n10000 d10	product_lorentz	B4	–	2000	0.9280	137.8
4ary n10000 d20	product_lorentz	B4	–	2000	0.9221	134.4
4ary n1000 d2	product_lorentz	B5	–	2000	0.5042	74.7
4ary n1000 d5	product_lorentz	B5	–	2000	0.3788	126.5
4ary n1000 d10	product_lorentz	B5	–	2000	0.3391	137.6
4ary n1000 d20	product_lorentz	B5	–	2000	0.3208	125.8
4ary n5000 d2	product_lorentz	B5	–	2000	0.8772	80.3
4ary n5000 d5	product_lorentz	B5	–	2000	0.8566	132.9
4ary n5000 d10	product_lorentz	B5	–	2000	0.8461	141.5
4ary n5000 d20	product_lorentz	B5	–	2000	0.8375	130.4
4ary n10000 d2	product_lorentz	B5	–	2000	0.9443	80.2
4ary n10000 d5	product_lorentz	B5	–	2000	0.9341	133.6
4ary n10000 d10	product_lorentz	B5	–	2000	0.9283	136.2
4ary n10000 d20	product_lorentz	B5	–	2000	0.9223	132.9
4ary n1000 d2	product_lorentz	A5	–	2000	0.5042	97.3
4ary n1000 d5	product_lorentz	A5	–	2000	0.3788	143.5
4ary n1000 d10	product_lorentz	A5	–	2000	0.3391	149.6
4ary n1000 d20	product_lorentz	A5	–	2000	0.3208	144.1
4ary n5000 d2	product_lorentz	A5	–	2000	0.8772	85.8
4ary n5000 d5	product_lorentz	A5	–	2000	0.8566	143.2
4ary n5000 d10	product_lorentz	A5	–	2000	0.8461	149.1
4ary n5000 d20	product_lorentz	A5	–	2000	0.8375	148.7
4ary n10000 d2	product_lorentz	A5	–	2000	0.9443	87.1
4ary n10000 d5	product_lorentz	A5	–	2000	0.9341	156.5
4ary n10000 d10	product_lorentz	A5	–	2000	0.9283	142.1
4ary n10000 d20	product_lorentz	A5	–	2000	0.9223	131.5
8ary n1000 d2	product_lorentz	E	–	2000	0.3463	14.6
8ary n1000 d5	product_lorentz	E	–	2000	0.1863	17.3
8ary n1000 d10	product_lorentz	E	–	2000	0.1271	17.4
8ary n1000 d20	product_lorentz	E	–	2000	0.0925	17.0
8ary n5000 d2	product_lorentz	E	–	2000	0.3675	17.6
8ary n5000 d5	product_lorentz	E	–	2000	0.2343	11.9
8ary n5000 d10	product_lorentz	E	–	2000	0.1934	17.1
8ary n5000 d20	product_lorentz	E	–	2000	0.1730	12.4
8ary n10000 d2	product_lorentz	E	–	2000	0.4606	16.3
8ary n10000 d5	product_lorentz	E	–	2000	0.3714	17.4
8ary n10000 d10	product_lorentz	E	–	2000	0.3454	17.7
8ary n10000 d20	product_lorentz	E	–	2000	0.3316	13.2
8ary n1000 d2	product_lorentz	B2	–	2000	0.3950	84.7
8ary n1000 d5	product_lorentz	B2	–	2000	0.2499	132.4
8ary n1000 d10	product_lorentz	B2	–	2000	0.2133	133.6
8ary n1000 d20	product_lorentz	B2	–	2000	0.1982	129.1
8ary n5000 d2	product_lorentz	B2	–	2000	0.7851	77.4
8ary n5000 d5	product_lorentz	B2	–	2000	0.7470	125.7
8ary n5000 d10	product_lorentz	B2	–	2000	0.7283	130.2
8ary n5000 d20	product_lorentz	B2	–	2000	0.7150	127.6
8ary n10000 d2	product_lorentz	B2	–	2000	0.8989	82.1
8ary n10000 d5	product_lorentz	B2	–	2000	0.8814	142.6
8ary n10000 d10	product_lorentz	B2	–	2000	0.8721	118.5
8ary n10000 d20	product_lorentz	B2	–	2000	0.8634	131.2
8ary n1000 d2	product_lorentz	B3	–	2000	0.3942	62.5
8ary n1000 d5	product_lorentz	B3	–	2000	0.2496	141.2
8ary n1000 d10	product_lorentz	B3	–	2000	0.2127	136.9
8ary n1000 d20	product_lorentz	B3	–	2000	0.1976	134.3
8ary n5000 d2	product_lorentz	B3	–	2000	0.7856	72.5
8ary n5000 d5	product_lorentz	B3	–	2000	0.7477	127.5
8ary n5000 d10	product_lorentz	B3	–	2000	0.7291	123.9

Dataset	Model	Method	Ablation	Epochs	Score	Seconds
8ary n5000 d20	product_lorentz	B3	–	2000	0.7158	131.0
8ary n10000 d2	product_lorentz	B3	–	2000	0.8993	78.1
8ary n10000 d5	product_lorentz	B3	–	2000	0.8818	128.2
8ary n10000 d10	product_lorentz	B3	–	2000	0.8725	140.5
8ary n10000 d20	product_lorentz	B3	–	2000	0.8639	141.6
8ary n1000 d2	product_lorentz	B4	–	2000	0.3935	77.0
8ary n1000 d5	product_lorentz	B4	–	2000	0.2493	139.5
8ary n1000 d10	product_lorentz	B4	–	2000	0.2127	142.8
8ary n1000 d20	product_lorentz	B4	–	2000	0.1976	150.1
8ary n5000 d2	product_lorentz	B4	–	2000	0.7829	82.4
8ary n5000 d5	product_lorentz	B4	–	2000	0.7453	141.9
8ary n5000 d10	product_lorentz	B4	–	2000	0.7266	137.1
8ary n5000 d20	product_lorentz	B4	–	2000	0.7132	132.7
8ary n10000 d2	product_lorentz	B4	–	2000	0.8984	81.2
8ary n10000 d5	product_lorentz	B4	–	2000	0.8809	135.7
8ary n10000 d10	product_lorentz	B4	–	2000	0.8716	131.6
8ary n10000 d20	product_lorentz	B4	–	2000	0.8629	139.8
8ary n1000 d2	product_lorentz	B5	–	2000	0.3928	76.8
8ary n1000 d5	product_lorentz	B5	–	2000	0.2491	132.7
8ary n1000 d10	product_lorentz	B5	–	2000	0.2122	131.1
8ary n1000 d20	product_lorentz	B5	–	2000	0.1971	138.2
8ary n5000 d2	product_lorentz	B5	–	2000	0.7835	74.6
8ary n5000 d5	product_lorentz	B5	–	2000	0.7460	147.0
8ary n5000 d10	product_lorentz	B5	–	2000	0.7274	136.7
8ary n5000 d20	product_lorentz	B5	–	2000	0.7140	136.2
8ary n10000 d2	product_lorentz	B5	–	2000	0.8987	73.1
8ary n10000 d5	product_lorentz	B5	–	2000	0.8813	132.0
8ary n10000 d10	product_lorentz	B5	–	2000	0.8720	137.4
8ary n10000 d20	product_lorentz	B5	–	2000	0.8633	134.8
8ary n1000 d2	product_lorentz	A5	–	2000	0.3928	76.3
8ary n1000 d5	product_lorentz	A5	–	2000	0.2491	137.4
8ary n1000 d10	product_lorentz	A5	–	2000	0.2122	134.0
8ary n1000 d20	product_lorentz	A5	–	2000	0.1971	132.9
8ary n5000 d2	product_lorentz	A5	–	2000	0.7835	73.1
8ary n5000 d5	product_lorentz	A5	–	2000	0.7460	132.1
8ary n5000 d10	product_lorentz	A5	–	2000	0.7274	139.9
8ary n5000 d20	product_lorentz	A5	–	2000	0.7141	145.1
8ary n10000 d2	product_lorentz	A5	–	2000	0.8987	78.9
8ary n10000 d5	product_lorentz	A5	–	2000	0.8814	124.3
8ary n10000 d10	product_lorentz	A5	–	2000	0.8720	130.7
8ary n10000 d20	product_lorentz	A5	–	2000	0.8633	136.9
binary n1000 d2	product_lorentz	E	–	2000	0.4164	36.1
binary n1000 d5	product_lorentz	E	–	2000	0.3260	11.9
binary n1000 d10	product_lorentz	E	–	2000	0.2945	23.6
binary n1000 d20	product_lorentz	E	–	2000	0.2796	31.8
binary n5000 d2	product_lorentz	E	–	2000	0.6945	29.7
binary n5000 d5	product_lorentz	E	–	2000	0.6508	17.2
binary n5000 d10	product_lorentz	E	–	2000	0.6369	29.7
binary n5000 d20	product_lorentz	E	–	2000	0.6297	37.7
binary n10000 d2	product_lorentz	E	–	2000	0.8462	25.1
binary n10000 d5	product_lorentz	E	–	2000	0.8248	23.8
binary n10000 d10	product_lorentz	E	–	2000	0.8171	25.4
binary n10000 d20	product_lorentz	E	–	2000	0.8123	32.2
binary n1000 d2	product_lorentz	B2	–	2000	0.7264	131.9
binary n1000 d5	product_lorentz	B2	–	2000	0.6609	211.4
binary n1000 d10	product_lorentz	B2	–	2000	0.6310	224.3
binary n1000 d20	product_lorentz	B2	–	2000	0.6132	253.2
binary n5000 d2	product_lorentz	B2	–	2000	0.9545	114.6
binary n5000 d5	product_lorentz	B2	–	2000	0.9467	225.7
binary n5000 d10	product_lorentz	B2	–	2000	0.9424	218.5
binary n5000 d20	product_lorentz	B2	–	2000	0.9383	234.7
binary n10000 d2	product_lorentz	B2	–	2000	0.9802	123.2
binary n10000 d5	product_lorentz	B2	–	2000	0.9761	202.5
binary n10000 d10	product_lorentz	B2	–	2000	0.9733	257.3
binary n10000 d20	product_lorentz	B2	–	2000	0.9701	204.6
binary n1000 d2	product_lorentz	B3	–	2000	0.7263	79.1
binary n1000 d5	product_lorentz	B3	–	2000	0.6613	132.4
binary n1000 d10	product_lorentz	B3	–	2000	0.6313	134.4
binary n1000 d20	product_lorentz	B3	–	2000	0.6135	126.6
binary n5000 d2	product_lorentz	B3	–	2000	0.9547	71.1

Dataset	Model	Method	Ablation	Epochs	Score	Seconds
binary n5000 d5	product_lorentz	B3	–	2000	0.9469	125.5
binary n5000 d10	product_lorentz	B3	–	2000	0.9426	129.3
binary n5000 d20	product_lorentz	B3	–	2000	0.9385	125.6
binary n10000 d2	product_lorentz	B3	–	2000	0.9803	71.7
binary n10000 d5	product_lorentz	B3	–	2000	0.9762	140.0
binary n10000 d10	product_lorentz	B3	–	2000	0.9734	132.2
binary n10000 d20	product_lorentz	B3	–	2000	0.9702	128.6
binary n1000 d2	product_lorentz	B4	–	2000	0.7240	127.9
binary n1000 d5	product_lorentz	B4	–	2000	0.6590	229.0
binary n1000 d10	product_lorentz	B4	–	2000	0.6293	214.7
binary n1000 d20	product_lorentz	B4	–	2000	0.6116	190.1
binary n5000 d2	product_lorentz	B4	–	2000	0.9544	110.3
binary n5000 d5	product_lorentz	B4	–	2000	0.9466	200.8
binary n5000 d10	product_lorentz	B4	–	2000	0.9423	209.1
binary n5000 d20	product_lorentz	B4	–	2000	0.9381	211.9
binary n10000 d2	product_lorentz	B4	–	2000	0.9802	105.6
binary n10000 d5	product_lorentz	B4	–	2000	0.9760	196.9
binary n10000 d10	product_lorentz	B4	–	2000	0.9733	216.1
binary n10000 d20	product_lorentz	B4	–	2000	0.9701	241.8
binary n1000 d2	product_lorentz	B5	–	2000	0.7240	83.3
binary n1000 d5	product_lorentz	B5	–	2000	0.6594	146.4
binary n1000 d10	product_lorentz	B5	–	2000	0.6297	156.2
binary n1000 d20	product_lorentz	B5	–	2000	0.6120	140.7
binary n5000 d2	product_lorentz	B5	–	2000	0.9546	81.4
binary n5000 d5	product_lorentz	B5	–	2000	0.9468	134.6
binary n5000 d10	product_lorentz	B5	–	2000	0.9425	139.8
binary n5000 d20	product_lorentz	B5	–	2000	0.9384	129.9
binary n10000 d2	product_lorentz	B5	–	2000	0.9802	74.9
binary n10000 d5	product_lorentz	B5	–	2000	0.9761	136.3
binary n10000 d10	product_lorentz	B5	–	2000	0.9734	143.9
binary n10000 d20	product_lorentz	B5	–	2000	0.9701	150.5
binary n1000 d2	product_lorentz	A5	–	2000	0.7240	77.6
binary n1000 d5	product_lorentz	A5	–	2000	0.6594	132.9
binary n1000 d10	product_lorentz	A5	–	2000	0.6297	144.9
binary n1000 d20	product_lorentz	A5	–	2000	0.6120	177.5
binary n5000 d2	product_lorentz	A5	–	2000	0.9546	102.0
binary n5000 d5	product_lorentz	A5	–	2000	0.9468	213.7
binary n5000 d10	product_lorentz	A5	–	2000	0.9425	233.8
binary n5000 d20	product_lorentz	A5	–	2000	0.9384	185.6
binary n10000 d2	product_lorentz	A5	–	2000	0.9802	111.0
binary n10000 d5	product_lorentz	A5	–	2000	0.9761	197.3
binary n10000 d10	product_lorentz	A5	–	2000	0.9734	180.0
binary n10000 d20	product_lorentz	A5	–	2000	0.9701	217.3
variable n1000 d2	product_lorentz	E	–	2000	0.3756	22.4
variable n1000 d5	product_lorentz	E	–	2000	0.2494	24.4
variable n1000 d10	product_lorentz	E	–	2000	0.2139	22.8
variable n1000 d20	product_lorentz	E	–	2000	0.1914	22.9
variable n5000 d2	product_lorentz	E	–	2000	0.5145	10.0
variable n5000 d5	product_lorentz	E	–	2000	0.4358	14.0
variable n5000 d10	product_lorentz	E	–	2000	0.4132	21.5
variable n5000 d20	product_lorentz	E	–	2000	0.4013	24.0
variable n10000 d2	product_lorentz	E	–	2000	0.6884	22.0
variable n10000 d5	product_lorentz	E	–	2000	0.6438	19.3
variable n10000 d10	product_lorentz	E	–	2000	0.6292	11.1
variable n10000 d20	product_lorentz	E	–	2000	0.6211	22.0
variable n1000 d2	product_lorentz	B2	–	2000	0.5683	103.7
variable n1000 d5	product_lorentz	B2	–	2000	0.4553	187.9
variable n1000 d10	product_lorentz	B2	–	2000	0.4157	164.4
variable n1000 d20	product_lorentz	B2	–	2000	0.3968	181.8
variable n5000 d2	product_lorentz	B2	–	2000	0.9061	88.3
variable n5000 d5	product_lorentz	B2	–	2000	0.8902	172.4
variable n5000 d10	product_lorentz	B2	–	2000	0.8821	174.7
variable n5000 d20	product_lorentz	B2	–	2000	0.8751	205.7
variable n10000 d2	product_lorentz	B2	–	2000	0.9559	116.2
variable n10000 d5	product_lorentz	B2	–	2000	0.9477	170.4
variable n10000 d10	product_lorentz	B2	–	2000	0.9428	105.2
variable n10000 d20	product_lorentz	B2	–	2000	0.9377	129.9
variable n1000 d2	product_lorentz	B3	–	2000	0.5676	138.3
variable n1000 d5	product_lorentz	B3	–	2000	0.4554	203.3
variable n1000 d10	product_lorentz	B3	–	2000	0.4155	206.6

Dataset	Model	Method	Ablation	Epochs	Score	Seconds
variable n1000 d20	product_lorentz	B3	–	2000	0.3965	213.2
variable n5000 d2	product_lorentz	B3	–	2000	0.9065	122.8
variable n5000 d5	product_lorentz	B3	–	2000	0.8906	230.6
variable n5000 d10	product_lorentz	B3	–	2000	0.8825	200.1
variable n5000 d20	product_lorentz	B3	–	2000	0.8755	252.2
variable n10000 d2	product_lorentz	B3	–	2000	0.9561	96.3
variable n10000 d5	product_lorentz	B3	–	2000	0.9479	225.2
variable n10000 d10	product_lorentz	B3	–	2000	0.9430	197.8
variable n10000 d20	product_lorentz	B3	–	2000	0.9379	177.0
variable n1000 d2	product_lorentz	B4	–	2000	0.5647	86.6
variable n1000 d5	product_lorentz	B4	–	2000	0.4527	166.8
variable n1000 d10	product_lorentz	B4	–	2000	0.4136	196.0
variable n1000 d20	product_lorentz	B4	–	2000	0.3950	212.7
variable n5000 d2	product_lorentz	B4	–	2000	0.9056	72.9
variable n5000 d5	product_lorentz	B4	–	2000	0.8898	190.0
variable n5000 d10	product_lorentz	B4	–	2000	0.8816	193.3
variable n5000 d20	product_lorentz	B4	–	2000	0.8747	191.2
variable n10000 d2	product_lorentz	B4	–	2000	0.9558	116.0
variable n10000 d5	product_lorentz	B4	–	2000	0.9476	160.4
variable n10000 d10	product_lorentz	B4	–	2000	0.9427	206.6
variable n10000 d20	product_lorentz	B4	–	2000	0.9376	215.8
variable n1000 d2	product_lorentz	B5	–	2000	0.5639	127.0
variable n1000 d5	product_lorentz	B5	–	2000	0.4528	240.5
variable n1000 d10	product_lorentz	B5	–	2000	0.4135	241.5
variable n1000 d20	product_lorentz	B5	–	2000	0.3948	223.3
variable n5000 d2	product_lorentz	B5	–	2000	0.9060	136.4
variable n5000 d5	product_lorentz	B5	–	2000	0.8902	227.0
variable n5000 d10	product_lorentz	B5	–	2000	0.8821	229.9
variable n5000 d20	product_lorentz	B5	–	2000	0.8751	223.6
variable n10000 d2	product_lorentz	B5	–	2000	0.9560	123.7
variable n10000 d5	product_lorentz	B5	–	2000	0.9478	251.1
variable n10000 d10	product_lorentz	B5	–	2000	0.9429	224.0
variable n10000 d20	product_lorentz	B5	–	2000	0.9378	214.1
variable n1000 d2	product_lorentz	A5	–	2000	0.5640	135.7
variable n1000 d5	product_lorentz	A5	–	2000	0.4529	184.6
variable n1000 d10	product_lorentz	A5	–	2000	0.4135	179.0
variable n1000 d20	product_lorentz	A5	–	2000	0.3948	181.7
variable n5000 d2	product_lorentz	A5	–	2000	0.9060	108.5
variable n5000 d5	product_lorentz	A5	–	2000	0.8902	167.0
variable n5000 d10	product_lorentz	A5	–	2000	0.8821	193.1
variable n5000 d20	product_lorentz	A5	–	2000	0.8751	165.4
variable n10000 d2	product_lorentz	A5	–	2000	0.9560	92.3
variable n10000 d5	product_lorentz	A5	–	2000	0.9478	183.4
variable n10000 d10	product_lorentz	A5	–	2000	0.9429	185.1
variable n10000 d20	product_lorentz	A5	–	2000	0.9378	177.7
20news	transformer	E	–	1000	0.5562	1702.8
20news	transformer	B3	–	1000	0.5441	38574.0
20news	transformer	B5	–	1000	0.5289	20438.0
20news	transformer	A5	–	1000	0.5312	40850.1
wordnet_mammal	product_embedding	E	search_s0	1000	0.0528	24.0
wordnet_mammal	product_embedding	E	search_s1	1000	0.0533	26.5
wordnet_mammal	product_embedding	E	search_s2	1000	0.0111	29.6
wordnet_mammal	product_embedding	E	search_s3	1000	0.0814	28.5
wordnet_mammal	product_embedding	E	search_s4	1000	0.0816	29.1
wordnet_mammal	product_embedding	E	search_s5	1000	0.0053	24.5
wordnet_mammal	product_embedding	B3	search_s0	1000	0.0541	77.7
wordnet_mammal	product_embedding	B3	search_s1	1000	0.0546	76.7
wordnet_mammal	product_embedding	B3	search_s2	1000	0.0548	79.1
wordnet_mammal	product_embedding	B3	search_s3	1000	0.0783	75.7
wordnet_mammal	product_embedding	B3	search_s4	1000	0.0814	79.0
wordnet_mammal	product_embedding	B3	search_s5	1000	0.0804	73.6
wordnet_mammal	product_embedding	B35	search_s0	1000	0.0547	75.9
wordnet_mammal	product_embedding	B35	search_s1	1000	0.0563	69.9
wordnet_mammal	product_embedding	B35	search_s2	1000	0.0556	67.7
wordnet_mammal	product_embedding	B35	search_s3	1000	0.0769	70.6
wordnet_mammal	product_embedding	B35	search_s4	1000	0.0795	68.4
wordnet_mammal	product_embedding	B35	search_s5	1000	0.0786	70.8
wordnet_mammal	product_embedding	B5	search_s0	1000	0.0546	79.4
wordnet_mammal	product_embedding	B5	search_s1	1000	0.0534	78.8
wordnet_mammal	product_embedding	B5	search_s2	1000	0.0541	79.0

Dataset	Model	Method	Ablation	Epochs	Score	Seconds
wordnet_mammal	product_embedding	B5	search_s3	1000	0.0798	80.1
wordnet_mammal	product_embedding	B5	search_s4	1000	0.0807	85.2
wordnet_mammal	product_embedding	B5	search_s5	1000	0.0776	82.9
wordnet_mammal	product_embedding	A5	search_s0	1000	0.0537	79.9
wordnet_mammal	product_embedding	A5	search_s1	1000	0.0544	78.6
wordnet_mammal	product_embedding	A5	search_s2	1000	0.0548	84.9
wordnet_mammal	product_embedding	A5	search_s3	1000	0.0774	86.5
wordnet_mammal	product_embedding	A5	search_s4	1000	0.0798	83.9
wordnet_mammal	product_embedding	A5	search_s5	1000	0.0804	86.6
cora_links	product_embedding	E	search_s0	1000	0.8057	11.6
cora_links	product_embedding	E	search_s1	1000	0.8029	10.9
cora_links	product_embedding	E	search_s2	1000	0.5000	13.0
cora_links	product_embedding	E	search_s3	1000	0.9307	8.4
cora_links	product_embedding	E	search_s4	1000	0.9135	8.7
cora_links	product_embedding	E	search_s5	1000	0.5000	13.1
cora_links	product_embedding	B3	search_s0	1000	0.8056	69.4
cora_links	product_embedding	B3	search_s1	1000	0.8051	71.3
cora_links	product_embedding	B3	search_s2	1000	0.8029	79.6
cora_links	product_embedding	B3	search_s3	1000	0.9306	86.3
cora_links	product_embedding	B3	search_s4	1000	0.9296	77.8
cora_links	product_embedding	B3	search_s5	1000	0.9279	75.0
cora_links	product_embedding	B35	search_s0	1000	0.8056	70.3
cora_links	product_embedding	B35	search_s1	1000	0.8051	77.9
cora_links	product_embedding	B35	search_s2	1000	0.8029	64.3
cora_links	product_embedding	B35	search_s3	1000	0.9306	77.6
cora_links	product_embedding	B35	search_s4	1000	0.9296	72.0
cora_links	product_embedding	B35	search_s5	1000	0.9279	77.3
cora_links	product_embedding	B5	search_s0	1000	0.8058	69.5
cora_links	product_embedding	B5	search_s1	1000	0.8051	81.5
cora_links	product_embedding	B5	search_s2	1000	0.8029	70.0
cora_links	product_embedding	B5	search_s3	1000	0.9306	76.4
cora_links	product_embedding	B5	search_s4	1000	0.9296	67.0
cora_links	product_embedding	B5	search_s5	1000	0.9279	76.5
cora_links	product_embedding	A5	search_s0	1000	0.8058	82.7
cora_links	product_embedding	A5	search_s1	1000	0.8050	74.6
cora_links	product_embedding	A5	search_s2	1000	0.8029	78.5
cora_links	product_embedding	A5	search_s3	1000	0.9306	77.0
cora_links	product_embedding	A5	search_s4	1000	0.9296	74.9
cora_links	product_embedding	A5	search_s5	1000	0.9279	77.3
digits_retrieval	product_proxy	E	search_s0	1000	0.9666	20.7
digits_retrieval	product_proxy	E	search_s1	1000	0.9666	20.7
digits_retrieval	product_proxy	E	search_s2	1000	0.9694	19.2
digits_retrieval	product_proxy	E	search_s3	1000	0.9610	18.6
digits_retrieval	product_proxy	E	search_s4	1000	0.9638	19.3
digits_retrieval	product_proxy	E	search_s5	1000	0.9749	18.8
digits_retrieval	product_proxy	B3	search_s0	1000	0.9721	60.4
digits_retrieval	product_proxy	B3	search_s1	1000	0.9638	64.6
digits_retrieval	product_proxy	B3	search_s2	1000	0.9638	65.1
digits_retrieval	product_proxy	B3	search_s3	1000	0.9582	59.7
digits_retrieval	product_proxy	B3	search_s4	1000	0.9582	61.6
digits_retrieval	product_proxy	B3	search_s5	1000	0.9582	66.7
digits_retrieval	product_proxy	B35	search_s0	1000	0.9721	64.9
digits_retrieval	product_proxy	B35	search_s1	1000	0.9638	67.0
digits_retrieval	product_proxy	B35	search_s2	1000	0.9638	65.8
digits_retrieval	product_proxy	B35	search_s3	1000	0.9582	61.9
digits_retrieval	product_proxy	B35	search_s4	1000	0.9582	62.9
digits_retrieval	product_proxy	B35	search_s5	1000	0.9554	68.0
digits_retrieval	product_proxy	B5	search_s0	1000	0.9721	69.2
digits_retrieval	product_proxy	B5	search_s1	1000	0.9554	63.6
digits_retrieval	product_proxy	B5	search_s2	1000	0.9638	71.3
digits_retrieval	product_proxy	B5	search_s3	1000	0.9554	64.5
digits_retrieval	product_proxy	B5	search_s4	1000	0.9582	70.7
digits_retrieval	product_proxy	B5	search_s5	1000	0.9554	62.9
digits_retrieval	product_proxy	A5	search_s0	1000	0.9721	69.5
digits_retrieval	product_proxy	A5	search_s1	1000	0.9554	64.9
digits_retrieval	product_proxy	A5	search_s2	1000	0.9638	66.3
digits_retrieval	product_proxy	A5	search_s3	1000	0.9554	69.8
digits_retrieval	product_proxy	A5	search_s4	1000	0.9582	69.2
digits_retrieval	product_proxy	A5	search_s5	1000	0.9554	67.7
wordnet_vehicle	product_embedding	E	search_s0	1000	0.0896	16.1

Dataset	Model	Method	Ablation	Epochs	Score	Seconds
wordnet_vehicle	product_embedding	E	search_s1	1000	0.0959	15.8
wordnet_vehicle	product_embedding	E	search_s2	1000	0.0445	16.3
wordnet_vehicle	product_embedding	E	search_s3	1000	0.0967	17.8
wordnet_vehicle	product_embedding	E	search_s4	1000	0.0961	18.1
wordnet_vehicle	product_embedding	E	search_s5	1000	0.0077	16.6
wordnet_vehicle	product_embedding	B3	search_s0	1000	0.0940	63.4
wordnet_vehicle	product_embedding	B3	search_s1	1000	0.0945	57.9
wordnet_vehicle	product_embedding	B3	search_s2	1000	0.0976	60.0
wordnet_vehicle	product_embedding	B3	search_s3	1000	0.0958	60.3
wordnet_vehicle	product_embedding	B3	search_s4	1000	0.0976	59.1
wordnet_vehicle	product_embedding	B3	search_s5	1000	0.0952	60.7
wordnet_vehicle	product_embedding	B35	search_s0	1000	0.0942	59.7
wordnet_vehicle	product_embedding	B35	search_s1	1000	0.0939	64.3
wordnet_vehicle	product_embedding	B35	search_s2	1000	0.0981	59.7
wordnet_vehicle	product_embedding	B35	search_s3	1000	0.0957	62.8
wordnet_vehicle	product_embedding	B35	search_s4	1000	0.0991	61.2
wordnet_vehicle	product_embedding	B35	search_s5	1000	0.0956	62.8
wordnet_vehicle	product_embedding	B5	search_s0	1000	0.0930	64.8
wordnet_vehicle	product_embedding	B5	search_s1	1000	0.0942	65.4
wordnet_vehicle	product_embedding	B5	search_s2	1000	0.0991	65.2
wordnet_vehicle	product_embedding	B5	search_s3	1000	0.0940	64.7
wordnet_vehicle	product_embedding	B5	search_s4	1000	0.0986	68.9
wordnet_vehicle	product_embedding	B5	search_s5	1000	0.0968	64.1
wordnet_vehicle	product_embedding	A5	search_s0	1000	0.0910	64.7
wordnet_vehicle	product_embedding	A5	search_s1	1000	0.0947	66.0
wordnet_vehicle	product_embedding	A5	search_s2	1000	0.0993	64.9
wordnet_vehicle	product_embedding	A5	search_s3	1000	0.0954	66.8
wordnet_vehicle	product_embedding	A5	search_s4	1000	0.0983	67.6
wordnet_vehicle	product_embedding	A5	search_s5	1000	0.0985	69.8
citeseer_links	product_embedding	E	search_s0	1000	0.8800	13.1
citeseer_links	product_embedding	E	search_s1	1000	0.8639	14.4
citeseer_links	product_embedding	E	search_s2	1000	0.5000	13.7
citeseer_links	product_embedding	E	search_s3	1000	0.9425	14.4
citeseer_links	product_embedding	E	search_s4	1000	0.9240	11.7
citeseer_links	product_embedding	E	search_s5	1000	0.5000	12.1
citeseer_links	product_embedding	B3	search_s0	1000	0.8801	77.3
citeseer_links	product_embedding	B3	search_s1	1000	0.8805	74.4
citeseer_links	product_embedding	B3	search_s2	1000	0.8809	82.3
citeseer_links	product_embedding	B3	search_s3	1000	0.9426	63.7
citeseer_links	product_embedding	B3	search_s4	1000	0.9428	77.5
citeseer_links	product_embedding	B3	search_s5	1000	0.9433	71.6
citeseer_links	product_embedding	B35	search_s0	1000	0.8801	83.1
citeseer_links	product_embedding	B35	search_s1	1000	0.8805	72.4
citeseer_links	product_embedding	B35	search_s2	1000	0.8809	85.6
citeseer_links	product_embedding	B35	search_s3	1000	0.9426	72.4
citeseer_links	product_embedding	B35	search_s4	1000	0.9428	73.4
citeseer_links	product_embedding	B35	search_s5	1000	0.9433	72.5
citeseer_links	product_embedding	B5	search_s0	1000	0.8808	84.0
citeseer_links	product_embedding	B5	search_s1	1000	0.8807	86.9
citeseer_links	product_embedding	B5	search_s2	1000	0.8809	76.2
citeseer_links	product_embedding	B5	search_s3	1000	0.9426	89.6
citeseer_links	product_embedding	B5	search_s4	1000	0.9428	74.2
citeseer_links	product_embedding	B5	search_s5	1000	0.9433	77.4
citeseer_links	product_embedding	A5	search_s0	1000	0.8808	72.4
citeseer_links	product_embedding	A5	search_s1	1000	0.8807	78.7
citeseer_links	product_embedding	A5	search_s2	1000	0.8809	75.3
citeseer_links	product_embedding	A5	search_s3	1000	0.9426	80.9
citeseer_links	product_embedding	A5	search_s4	1000	0.9428	75.3
citeseer_links	product_embedding	A5	search_s5	1000	0.9433	87.7
20news_retrieval	product_proxy	E	search_s0	1000	0.4673	38.6
20news_retrieval	product_proxy	E	search_s1	1000	0.4620	41.9
20news_retrieval	product_proxy	E	search_s2	1000	0.4647	37.5
20news_retrieval	product_proxy	E	search_s3	1000	0.4735	36.1
20news_retrieval	product_proxy	E	search_s4	1000	0.4744	35.8
20news_retrieval	product_proxy	E	search_s5	1000	0.4797	35.3
20news_retrieval	product_proxy	B3	search_s0	1000	0.4532	87.2
20news_retrieval	product_proxy	B3	search_s1	1000	0.4585	81.2
20news_retrieval	product_proxy	B3	search_s2	1000	0.4673	85.1
20news_retrieval	product_proxy	B3	search_s3	1000	0.4832	86.4
20news_retrieval	product_proxy	B3	search_s4	1000	0.4655	81.1

Dataset	Model	Method	Ablation	Epochs	Score	Seconds
20news_retrieval	product_proxy	B3	search_s5	1000	0.4638	88.8
20news_retrieval	product_proxy	B35	search_s0	1000	0.4629	81.7
20news_retrieval	product_proxy	B35	search_s1	1000	0.4514	81.9
20news_retrieval	product_proxy	B35	search_s2	1000	0.4567	81.3
20news_retrieval	product_proxy	B35	search_s3	1000	0.4859	86.7
20news_retrieval	product_proxy	B35	search_s4	1000	0.4691	86.9
20news_retrieval	product_proxy	B35	search_s5	1000	0.4673	84.3
20news_retrieval	product_proxy	B5	search_s0	1000	0.4585	92.4
20news_retrieval	product_proxy	B5	search_s1	1000	0.4638	85.4
20news_retrieval	product_proxy	B5	search_s2	1000	0.4611	87.0
20news_retrieval	product_proxy	B5	search_s3	1000	0.4859	90.8
20news_retrieval	product_proxy	B5	search_s4	1000	0.4682	85.6
20news_retrieval	product_proxy	B5	search_s5	1000	0.4726	92.3
20news_retrieval	product_proxy	A5	search_s0	1000	0.4611	89.5
20news_retrieval	product_proxy	A5	search_s1	1000	0.4638	83.2
20news_retrieval	product_proxy	A5	search_s2	1000	0.4523	89.2
20news_retrieval	product_proxy	A5	search_s3	1000	0.4850	88.9
20news_retrieval	product_proxy	A5	search_s4	1000	0.4691	84.1
20news_retrieval	product_proxy	A5	search_s5	1000	0.4700	91.0